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(54) **SELECTIVE MODULATION OF
PROTEIN-PROTEIN INTERACTIONS**

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C12N 15/81 (2006.01)
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CPC **C12Q 1/6897** (2013.01); **C12N 15/1055** (2013.01); **C12N 15/1093** (2013.01); **C12N 15/80** (2013.01); **C12N 15/81** (2013.01); **C12N 15/815** (2013.01); **C12Q 2521/50** (2013.01); **C12Q 2600/158** (2013.01)

(58) **Field of Classification Search**
CPC C12N 15/10; C12N 15/81
See application file for complete search history.

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(57) **ABSTRACT**

The present disclosure provides methods to identify peptides and small molecule moieties that are able to modulate protein-protein interactions (PPIs). Some moieties can disrupt specific PPIs within a complex, or disrupt variant-specific PPIs. Some moieties can alternatively bridge between two proteins in a protein-specific or a variant-specific manner. The methods described enable generation of compounds able to modulate PPI networks within cells with implications for drug development for pathological conditions.

16 Claims, 17 Drawing Sheets

Specification includes a Sequence Listing.

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Disruption of specific
complex components

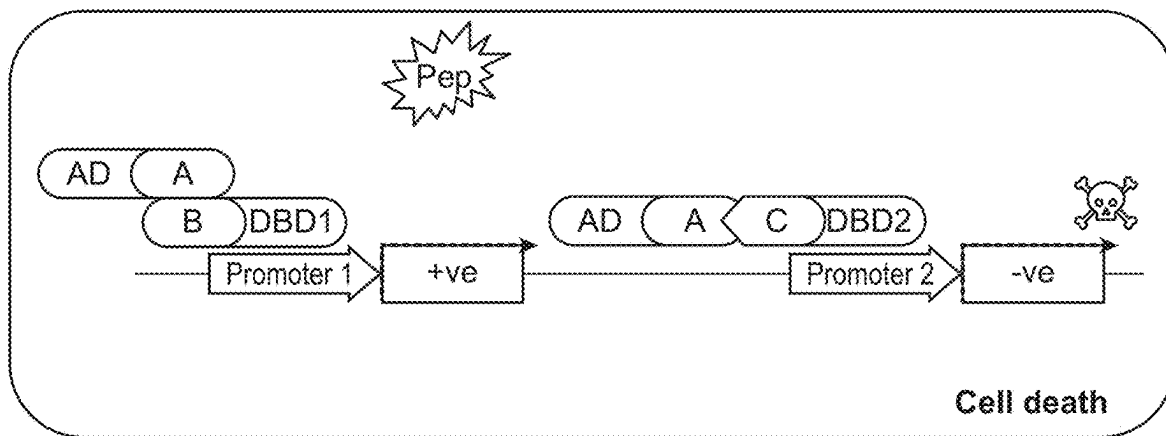
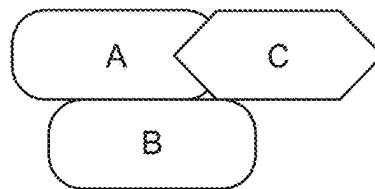


FIG. 1A

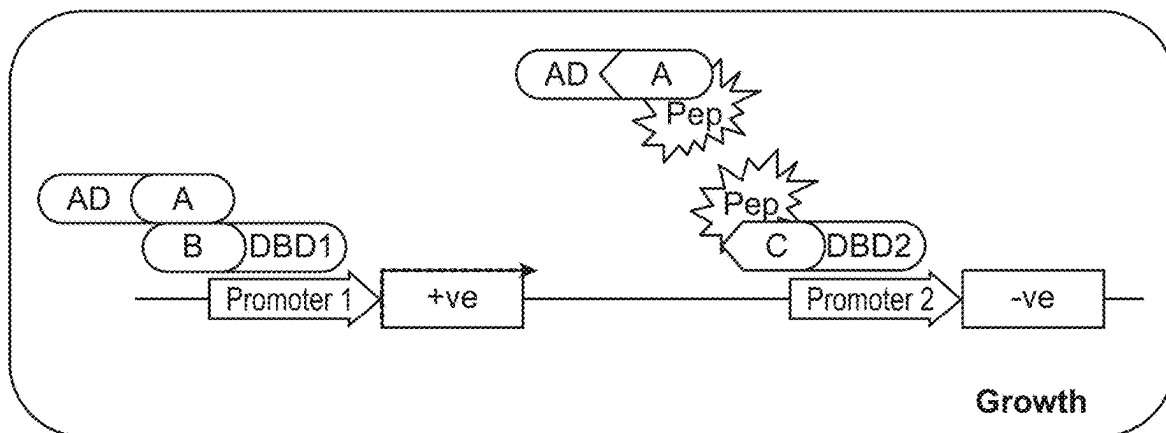
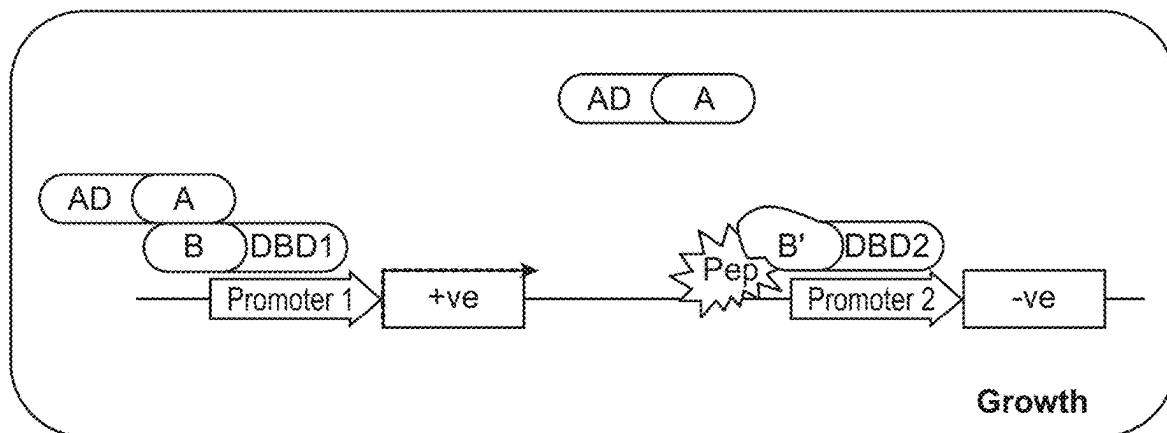
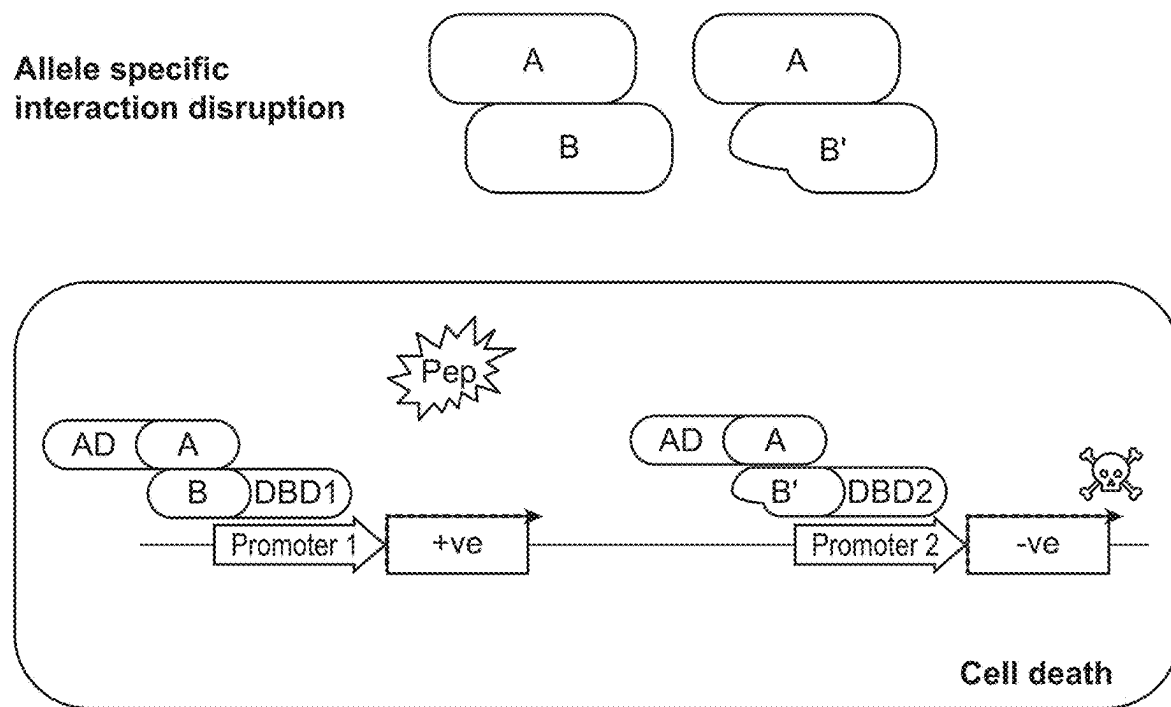


FIG. 1B



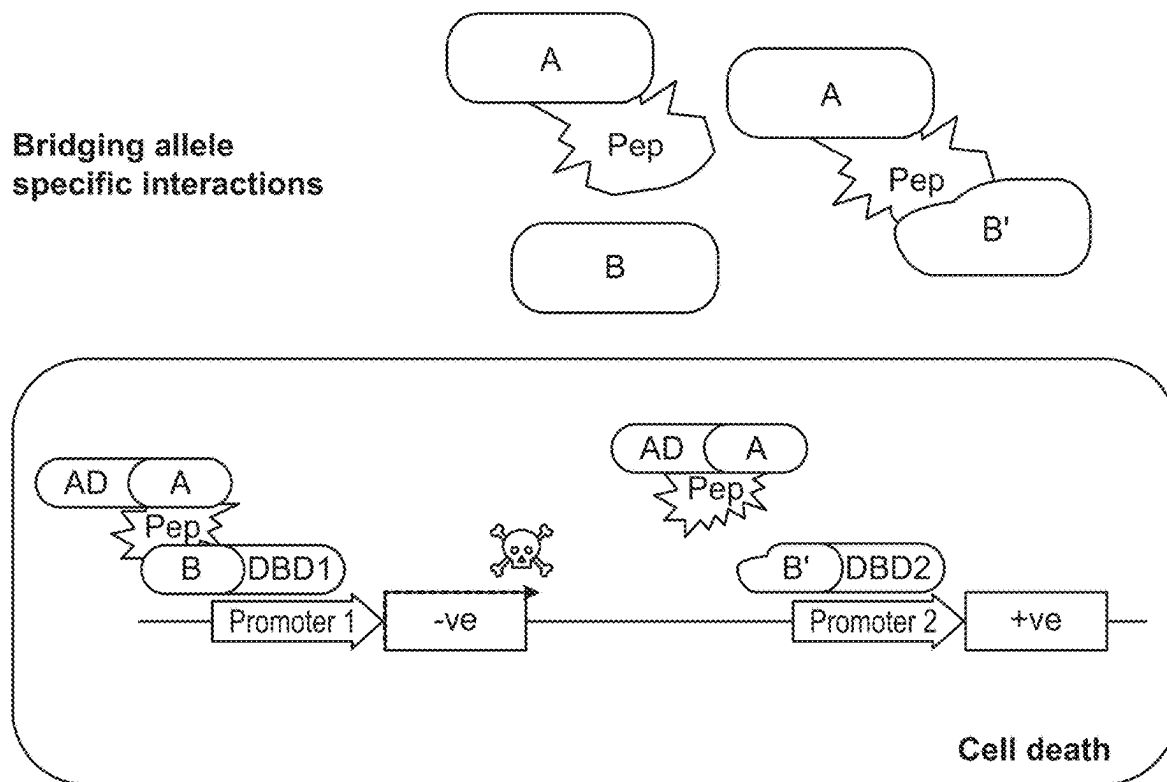


FIG. 3A

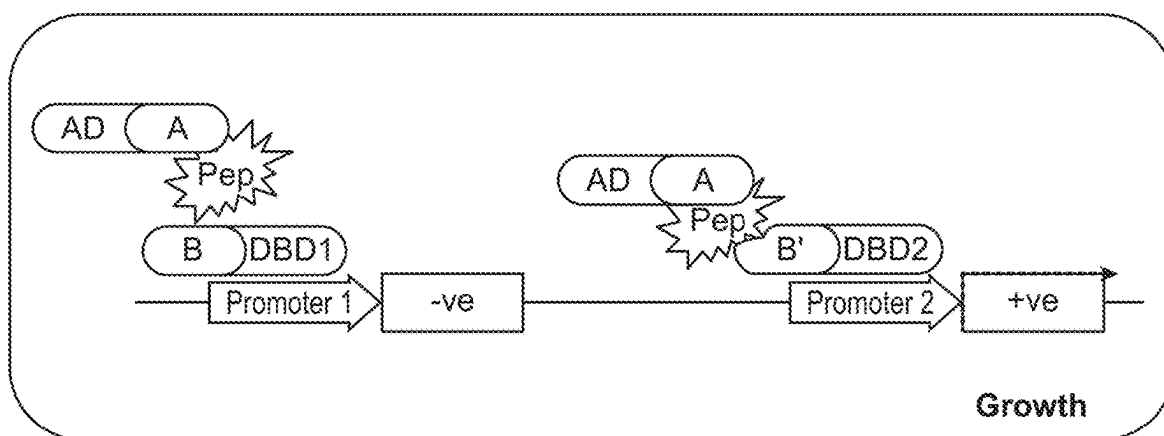


FIG. 3B

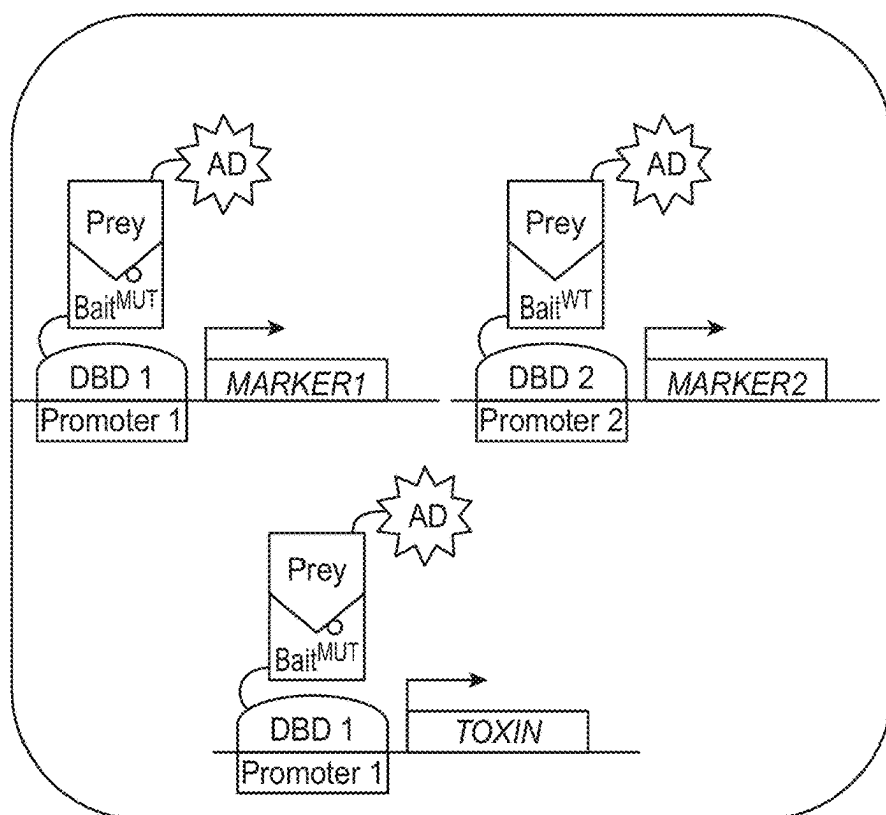


FIG. 4A

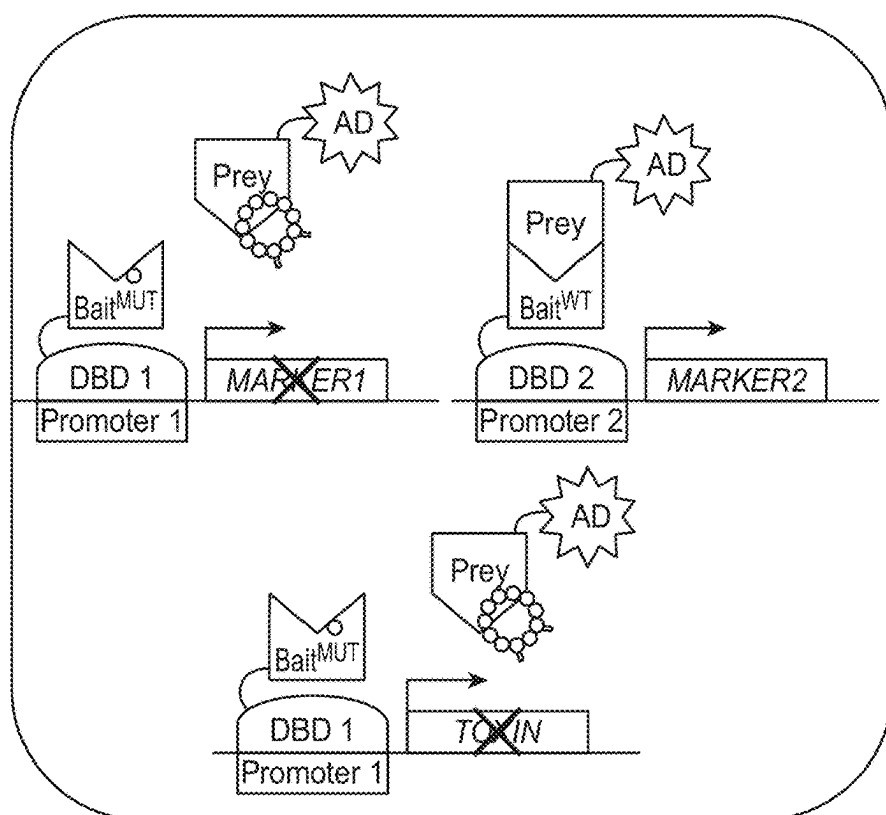


FIG. 4B

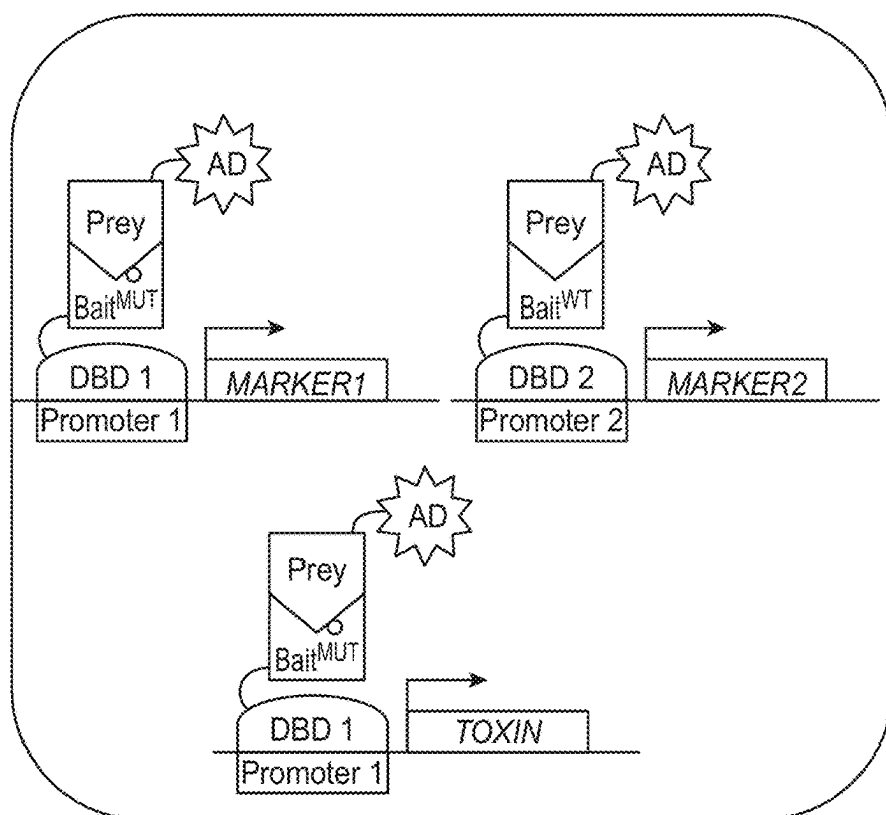


FIG. 5A

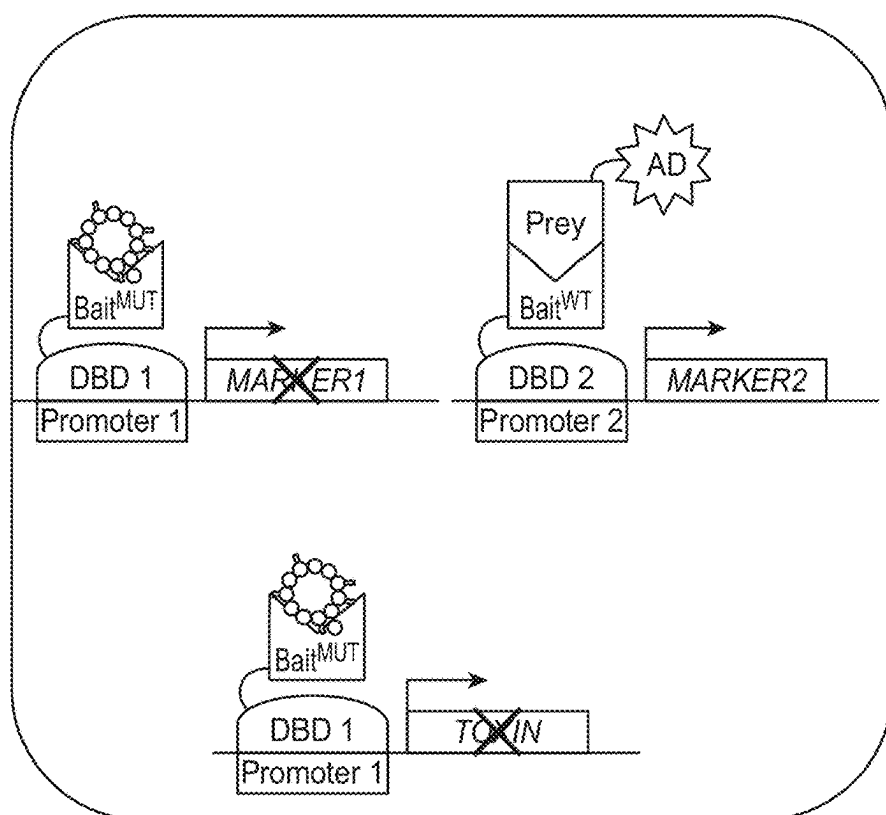


FIG. 5B

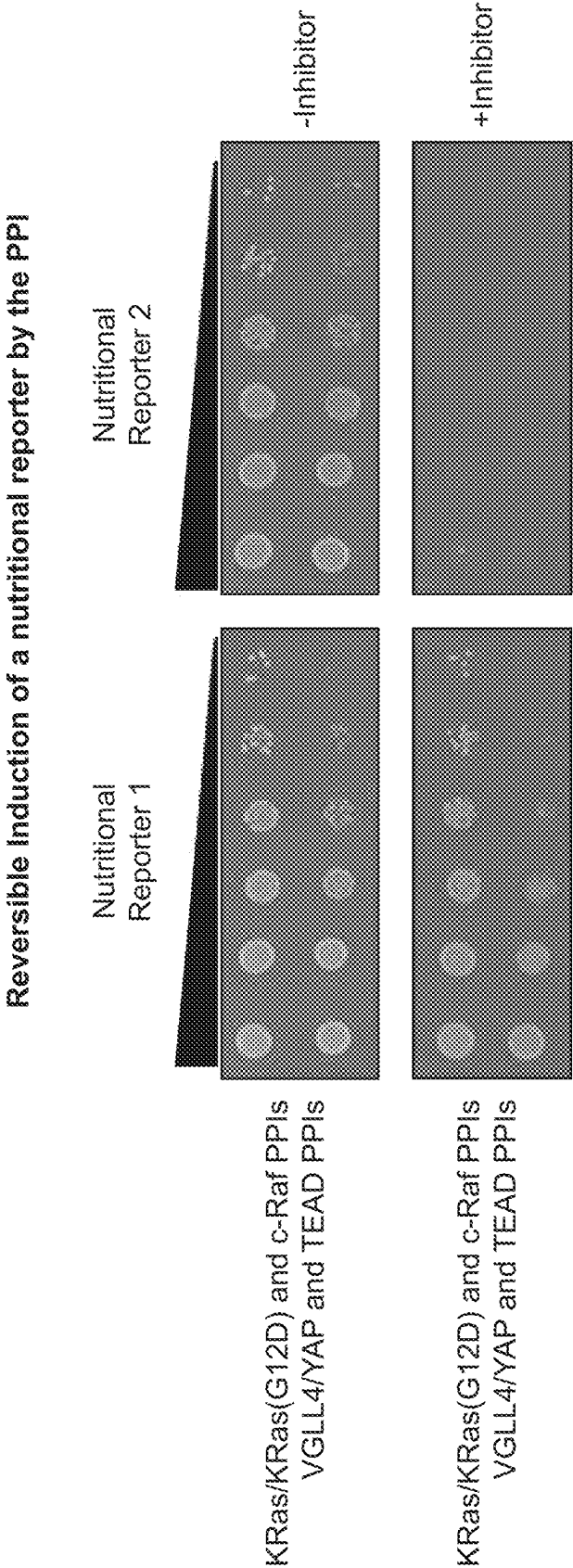
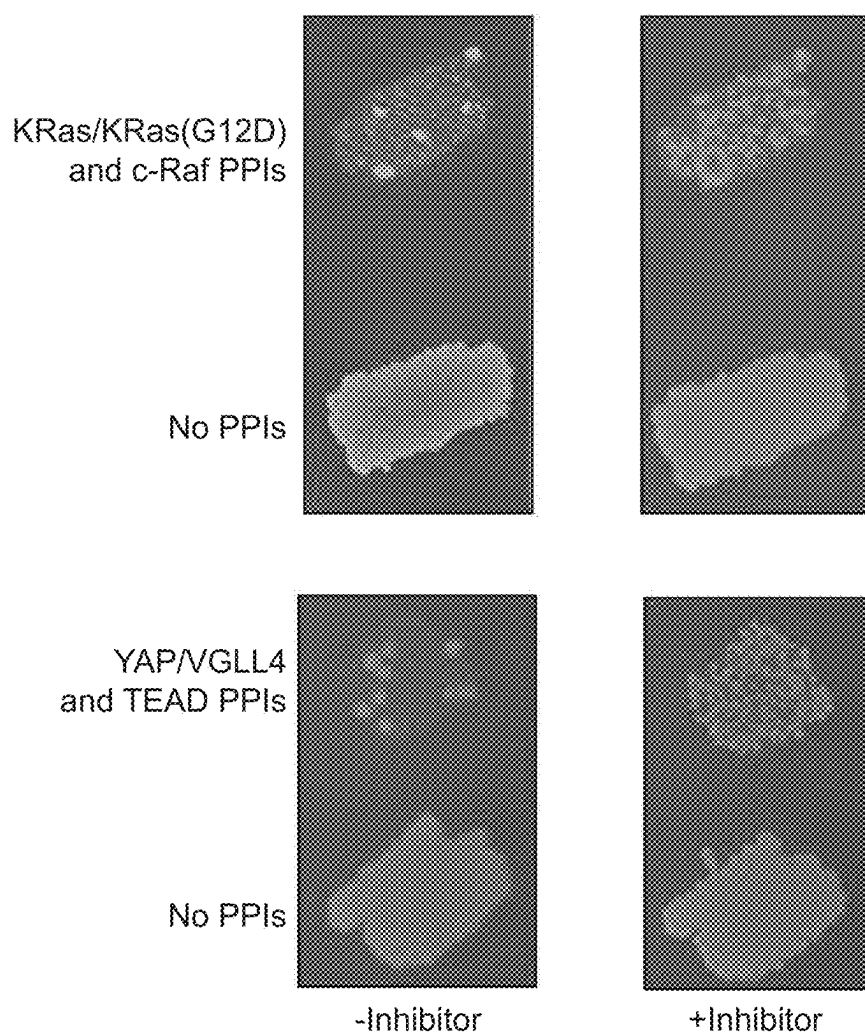


FIG. 6

Reversible induction of a cytotoxic reporter by the PPI**FIG. 7**

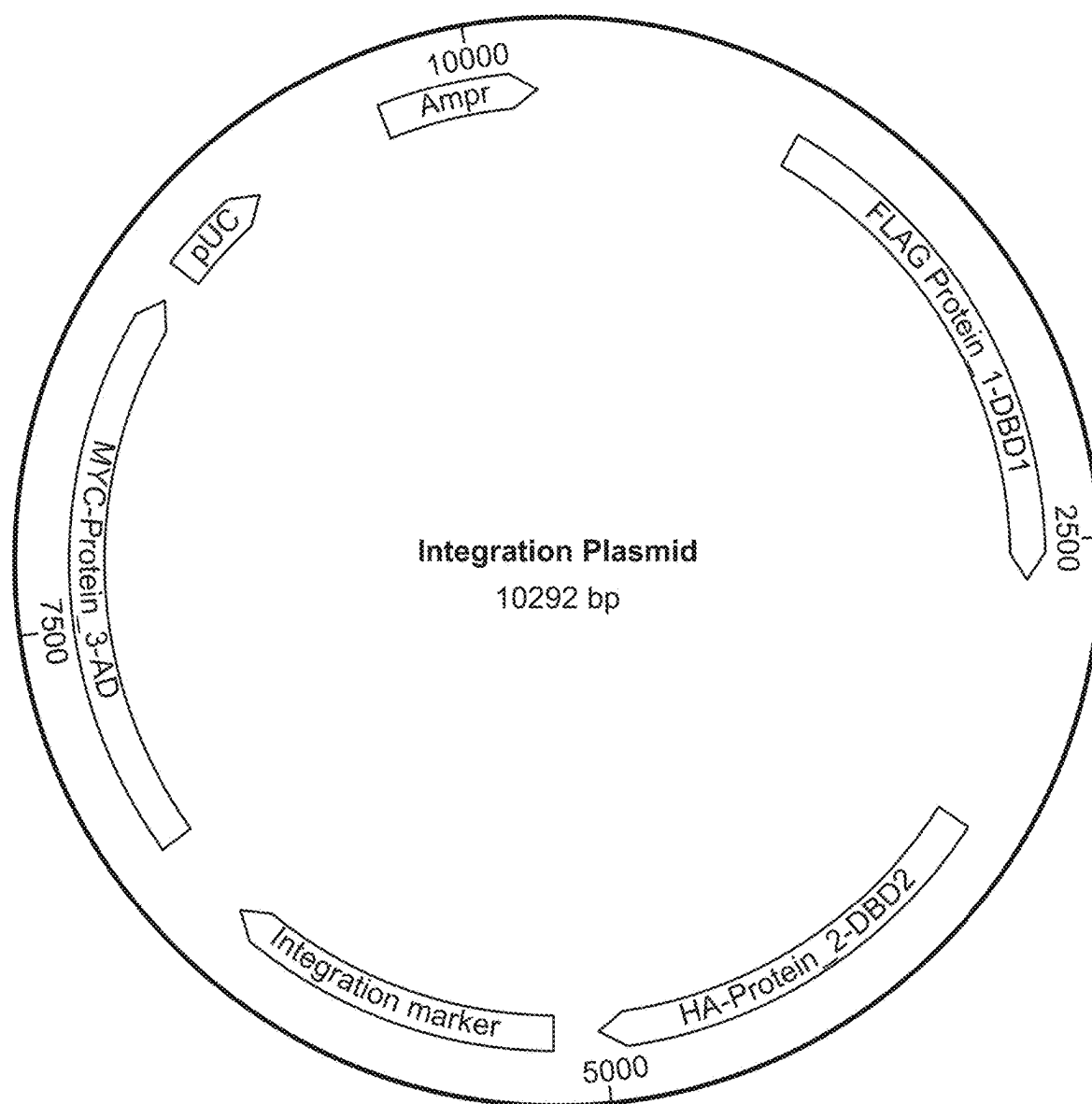


FIG. 8

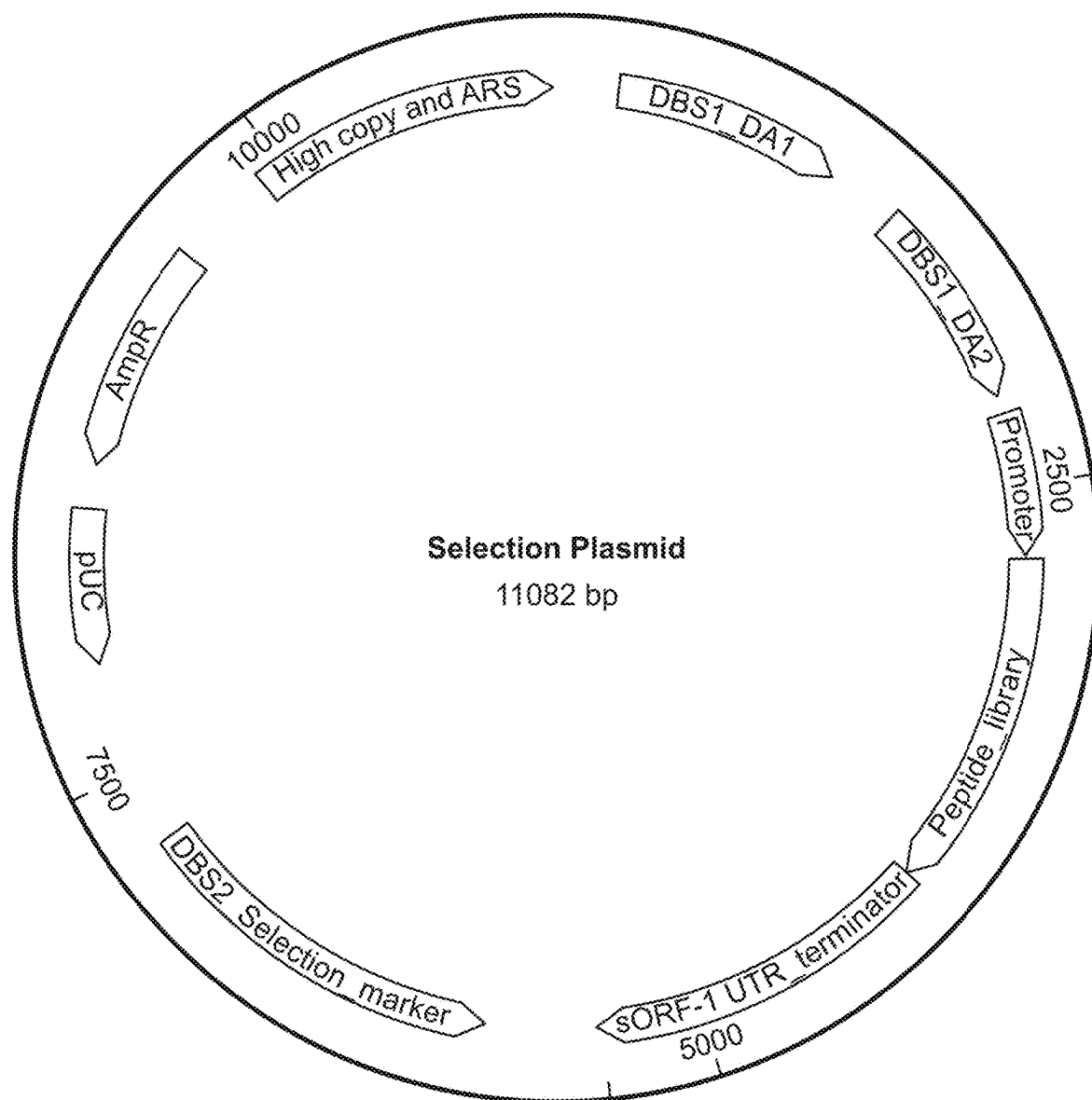


FIG. 9

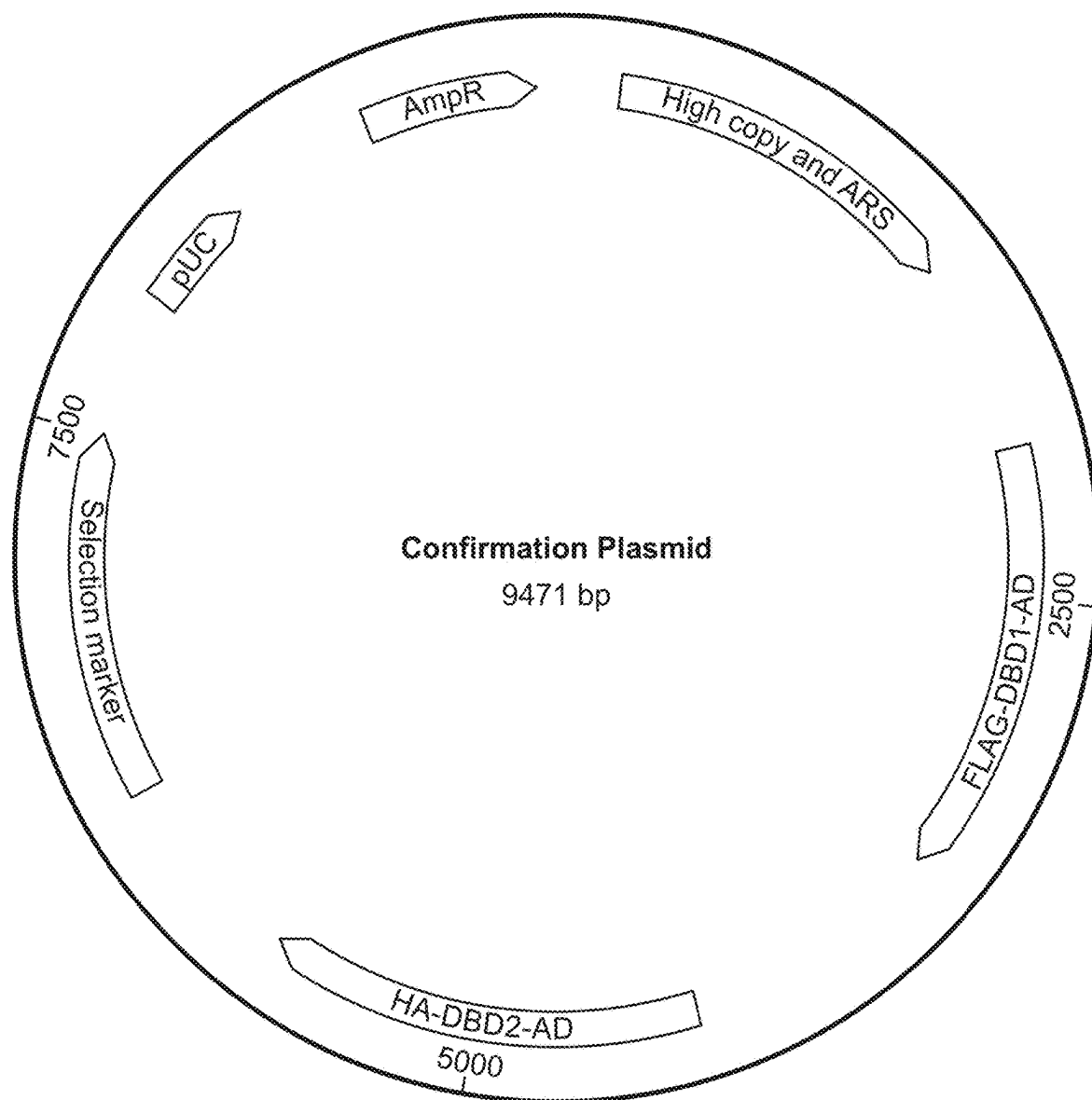
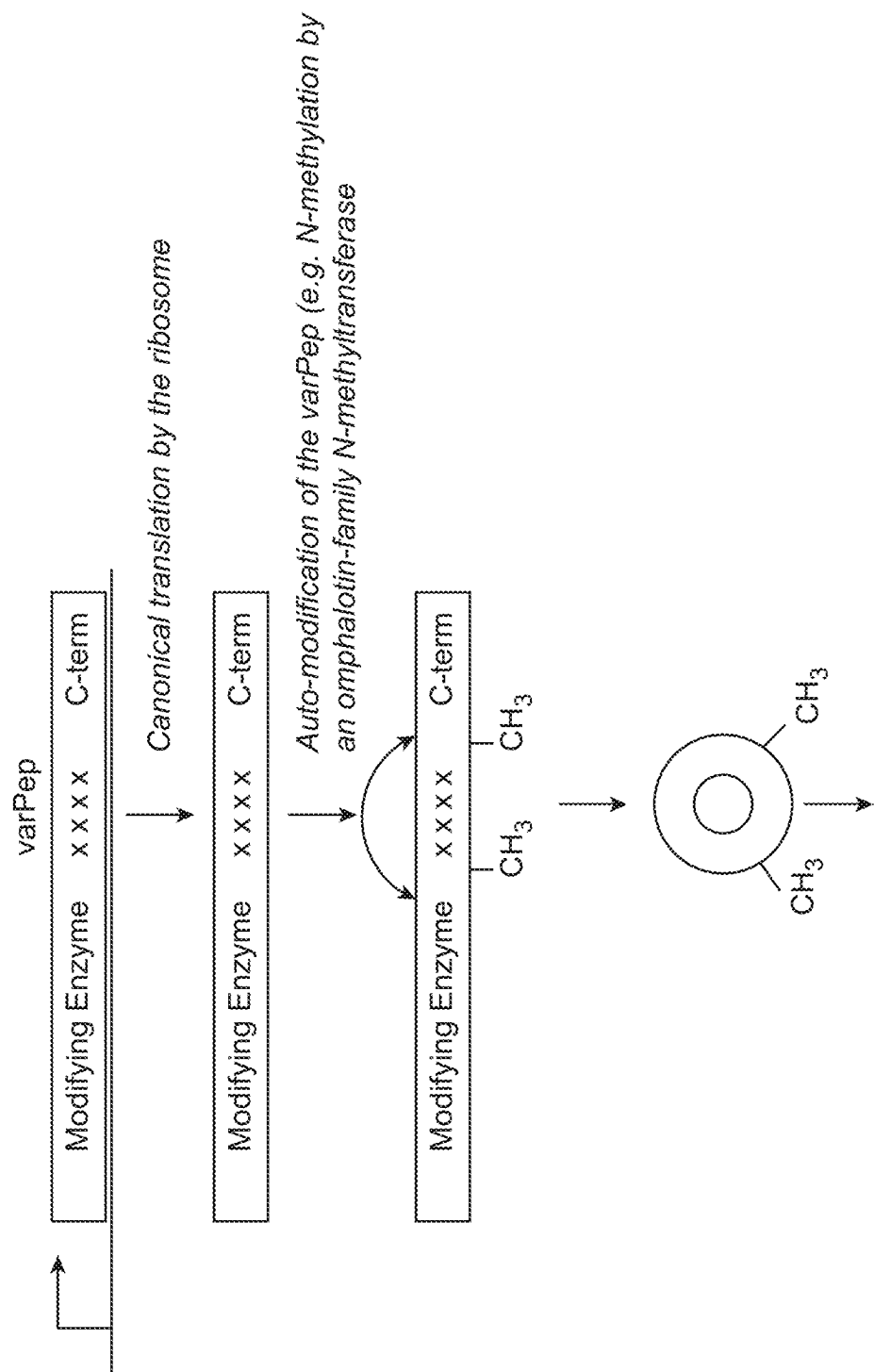


FIG. 10



Further steps of processing include side-chain modification (e.g. hydroxylation, isomerization, etc.)

FIG. 11

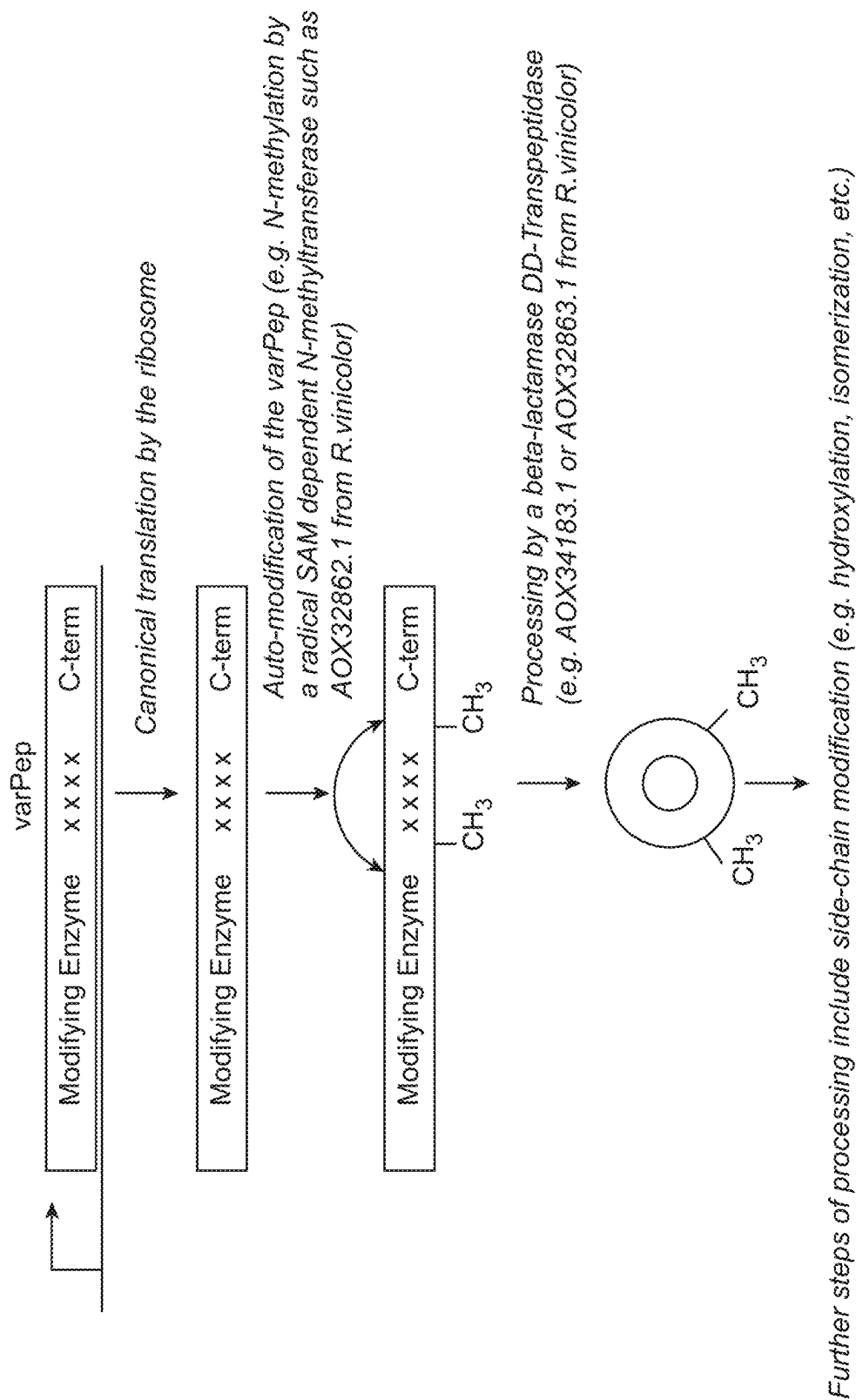


FIG. 12

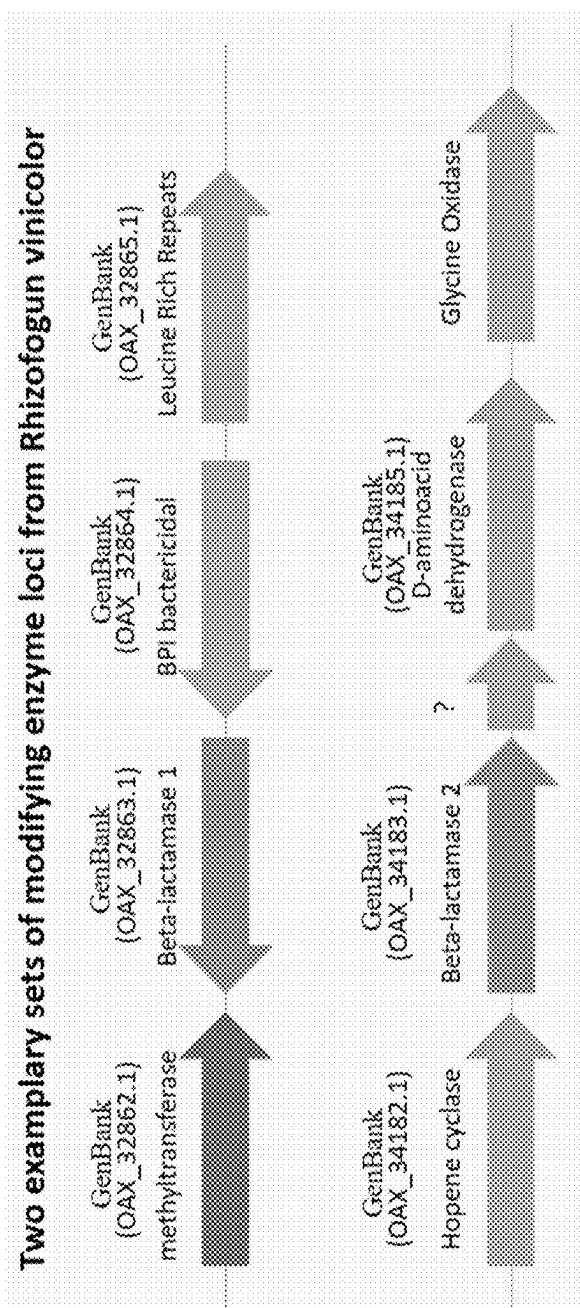
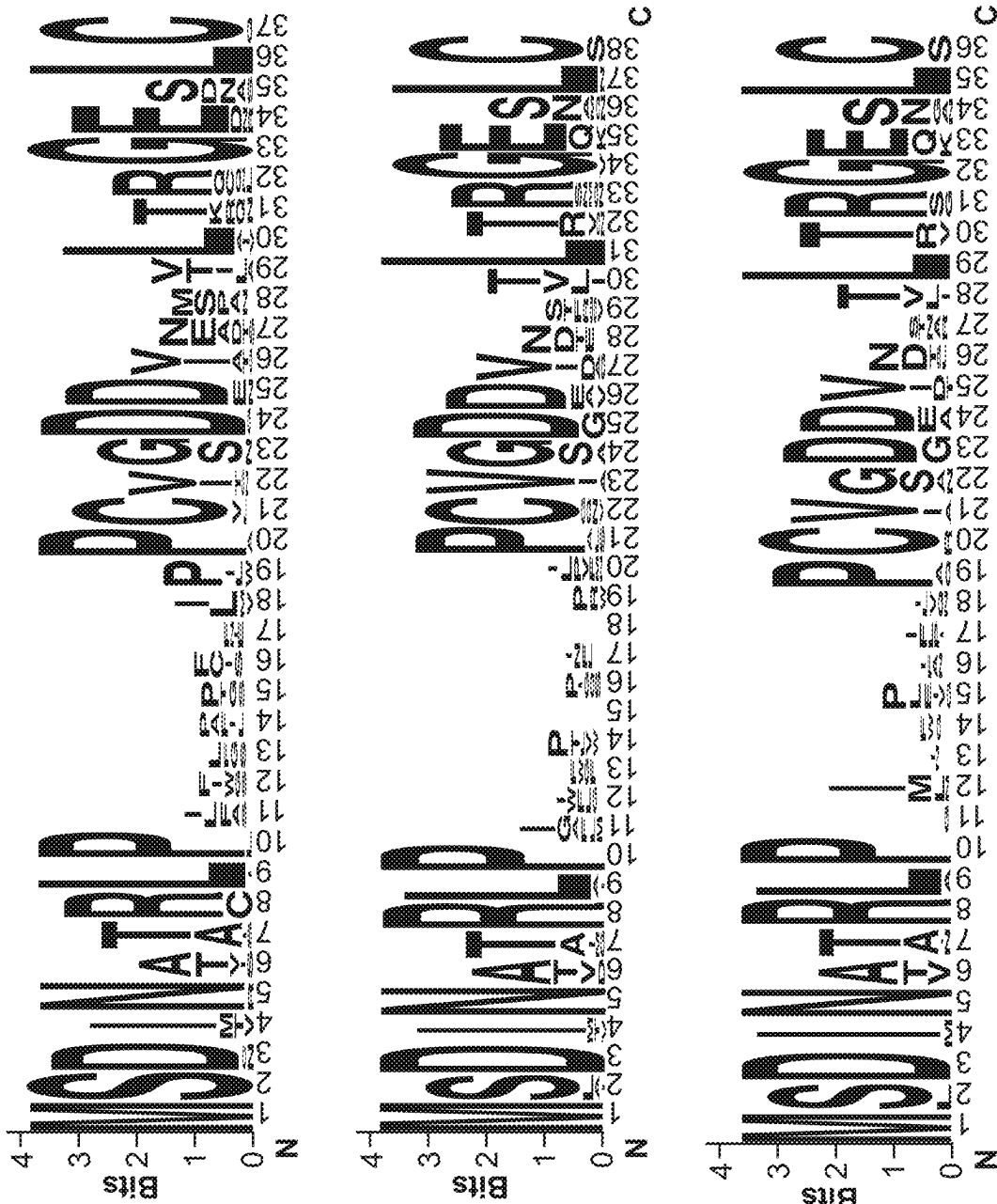


FIG. 13



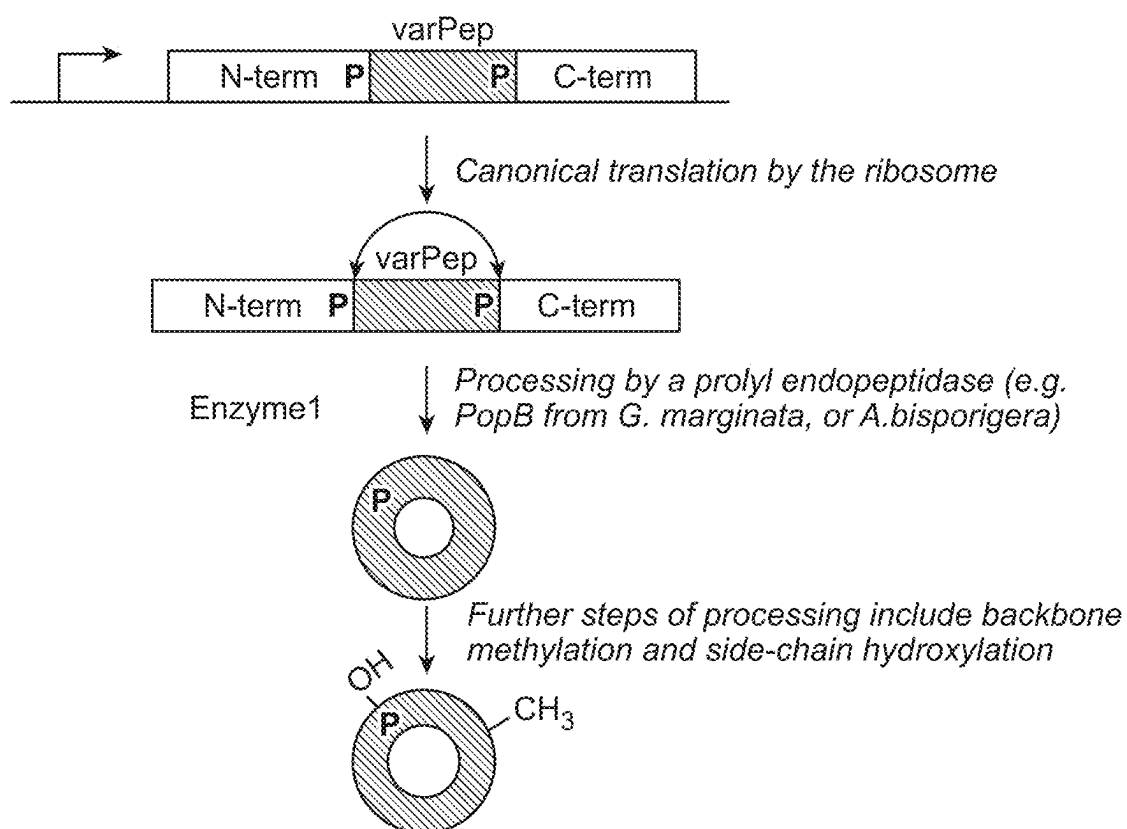


FIG. 15

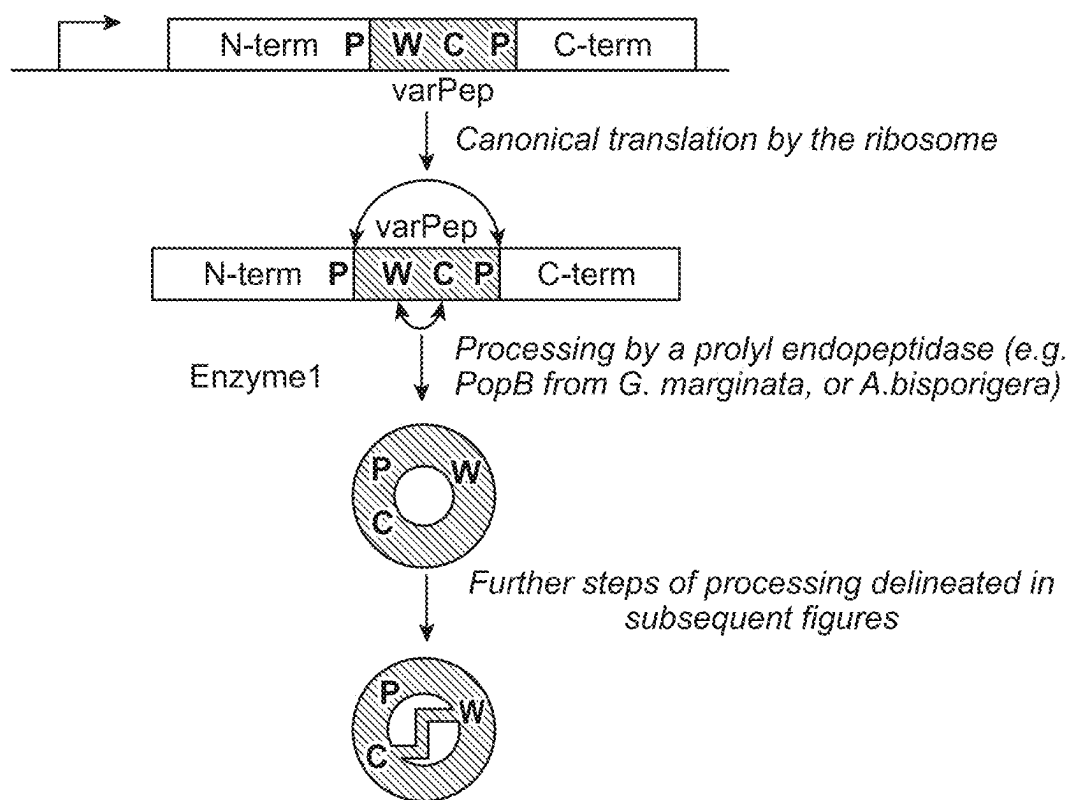
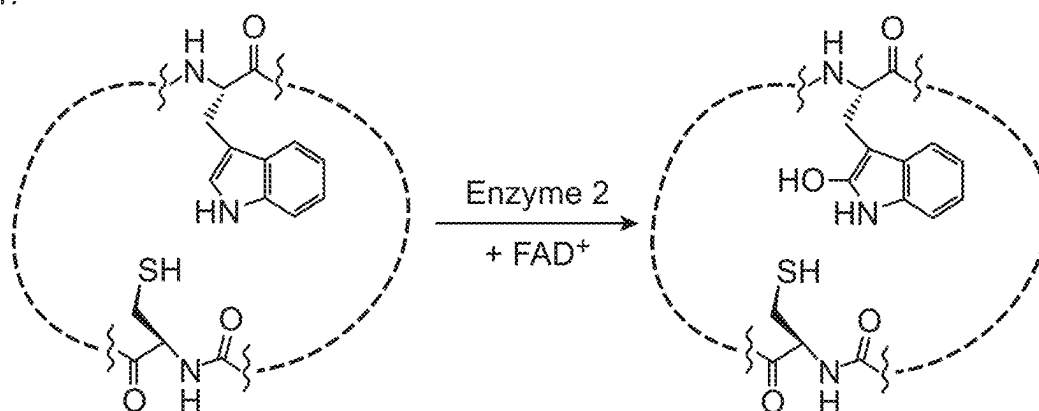
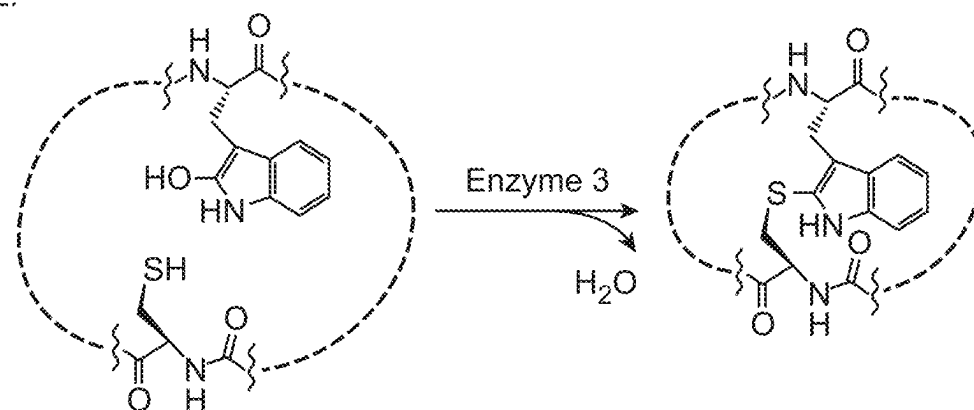


FIG. 16

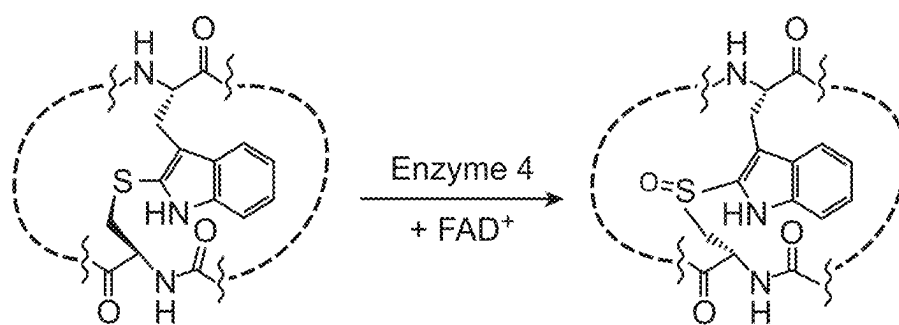
Step 1.



Step 2.



Step 3.



Step 4.

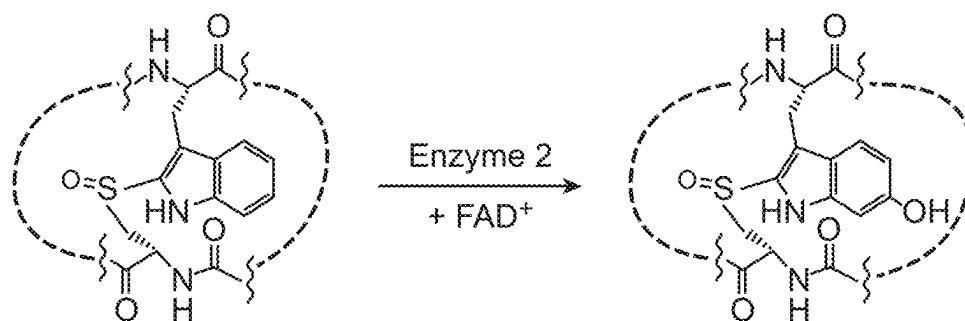


FIG. 17

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SELECTIVE MODULATION OF PROTEIN-PROTEIN INTERACTIONS

CROSS-REFERENCE

This application is a continuation application of U.S. Non-Provisional application Ser. No. 15/931,295, filed May 13, 2020, which is a continuation application of International Application No. PCT/US2018/061292, filed Nov. 15, 2018, which claims the benefit of U.S. Provisional Application No. 62/587,269 titled SELECTIVE DISRUPTION OF PROTEIN-PROTEIN INTERACTIONS, filed on Nov. 16, 2017 and U.S. Provisional Application No. 62/590,147 titled CYCLIC AND BICYCLIC PEPTIDES AND METHODS OF MAKING AND USING THEREOF, filed on Nov. 22, 2017, which are incorporated herein by reference.

SEQUENCE LISTING

The instant application contains a Sequence Listing which has been submitted electronically in XML format and is hereby incorporated by reference in its entirety. Said XML file, created on Oct. 31, 2023, is named 50607-703_302_SL.xml and is 96,288 bytes in size.

BACKGROUND

Disruption of protein-protein interactions in a precise manner can be a key for controlling cellular functions. Many pathological conditions are characterized by aberrant functions of cellular pathways, either because of precocious protein complex formation or the incorporation of malfunctional variants. Thus, compounds that can specifically and precisely prevent the formation of such protein complexes or the malfunction of faulty variants could be beneficial to treating various ailments. The selective disruption of precise protein-protein interactions is difficult to achieve using the traditional enzyme active-site/inhibitor-based drug development scheme. Accordingly, there is a need for development of methods and compositions that target protein-protein interactions in precise and selective ways.

SUMMARY

Disclosed herein, in certain embodiments, is a method for identifying a molecule that selectively disrupts an interaction between a first test protein and a second test protein in a host cell, comprising: expressing in the host cell a first fusion protein comprising the first test protein and a DNA-binding moiety; expressing in the host cell a second fusion protein comprising the second test protein and a gene activating moiety; expressing in the host cell a third fusion protein comprising the third test protein and a different DNA-binding moiety; and delivering a molecule from a library to the host cell, wherein a sequence of gene for expressing a death agent is disposed within the host cell and operably linked a promoter DNA sequence specific for the DNA binding moiety of the first fusion protein, wherein a positive selection reporter is disposed within the host and operably linked to a promoter DNA sequence specific for the DNA binding moiety of the third fusion protein, and wherein, in the absence of the molecule, the interaction between the first test protein and the second test protein causes the gene activating moiety to activate expression of the death agent, while the interaction between the second test protein and the third test protein causes the gene activating moiety to activate the expression of the positive

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selection reporter. In some embodiments, the molecule from the library is delivered exogenously. In some embodiments, the host cell comprises more than one sequence for expressing a positive control reporter that is activated by a promoter DNA sequence specific for a DNA binding moiety. In some embodiments, the host cell comprises more than one sequence for expressing a death agent that is activated by a promoter DNA sequence specific for a DNA binding moiety. In some embodiments, the host cell comprises an integrated DNA encoding the first fusion protein, an integrated DNA encoding the second fusion protein, an integrated DNA encoding the third fusion protein; a plasmid DNA encoding the death agent; and a plasmid DNA encoding a positive selection reporter. In some embodiments, the first test protein is a variation of KRAS. In some embodiments, the third test protein is KRAS. In some embodiments, the second test protein is c-Raf. In some embodiments, the first test protein is YAP or TAZ. In some embodiments, the third test protein is VGLL4. In some embodiments, the second test protein is TEAD. In some embodiments, the DNA binding moiety is derived from LexA, cI, Gli-1, YY1, Glucocorticoid receptor, TetR, or Ume6. In some embodiments, the gene activating moiety is derived from VP16, GAL4, NF- κ B, B42, BP64, VP64, or p65. In some embodiments, the death agent is an overexpressed product of genetic element selected from DNA or RNA. In some embodiments, the genetic element is a Growth Inhibitory (GIN) sequence such as GIN11. In some embodiments, the death agent is a ribosomally encoded xenobiotic agent, a ribosomally encoded poison, a ribosomally encoded endogenous or exogenous gene that results in severe growth defects upon mild overexpression, a ribosomally encoded recombinase that excises an essential gene for viability, a limiting factor involved in the synthesis of a toxic secondary metabolite, or any combination thereof. In some embodiments, the death agent is Cholera toxin, SpvB toxin, CARDS toxin, SpyA Toxin, HopU1, Chelt toxin, Certhrax toxin, EFV toxin, ExoT, CdtB, Diphtheria toxin, ExoU/VipB, HopPtoE, HopPtoF, HopPtoG, VopF, YopJ, AvrPtoB, SdbA, SidG, VpdA, Lpg0969, Lpg1978, YopE, SptP, SopE2, SopB/SigD, SipA, YpkA, YopM, Amatoxin, Phallacidin, Killer toxin KP1, Killer toxin KP6, Killer Toxin K1, Killer Toxin K28 (KHR), Killer Toxin K28 (KHS), Anthrax lethal factor endopeptidase, Shiga Toxin, Saporin Toxin, Ricin Toxin, or any combination thereof. In some embodiments, the host cell is a fungus or bacteria. In some embodiments, the fungus is *Aspergillus*. In some embodiments, the fungus is *Pichia pastoris*. In some embodiments, the fungus is *S. cerevisiae*. In some embodiments, the molecule is small molecule. In some embodiments, the small molecule is peptidomimetic. In some embodiments, the molecule is peptide or protein. In some embodiments, the peptide or protein is derived from naturally occurring protein product. In certain embodiments, the peptide or protein is synthesized protein product. In some embodiments, the peptide or protein is product of recombinant genes. In some embodiments, the molecule is a peptide or protein expressed from test DNA molecule inserted into the host cell, wherein the test DNA molecule comprises DNA sequences that encodes polypeptides, forming the library. In some embodiments, the library comprises polypeptides 60 or fewer amino acids in length. In some embodiments, the DNA sequence encodes a 3'UTR of mRNA. In some embodiments, the 3'UTR is the 3'UTR of sORF1. In some embodiments, the polypeptides comprise a common N-terminal sequence of Methionine-Valine-Asparagine. In some embodiments, the polypeptides in the library are processed into cyclic or bicyclic peptides in the host cell.

Disclosed herein, in certain embodiments, is a plasmid vector, comprising the components of PLASMID 1, or any combination of the components of PLASMID 1. In some embodiments, the plasmid vector comprises a DNA sequence encoding a first polypeptide inserted in frame with Gal4-DNA binding domain ("DBD"), a DNA sequence encoding a second polypeptide inserted in frame with LexA-DBD, and a DNA sequence encoding a third polypeptide inserted in frame with Dof1-AD. In certain embodiments, a host cell comprises the plasmid vectors.

Disclosed herein, in certain embodiments, is a library of plasmid vectors, each plasmid vector comprising: a DNA sequence encoding a different peptide sequence operably linked to a first switchable promoter; a DNA sequence encoding a death agent under control of a second switchable promoter; and a DNA sequence encoding a positive selection reporter under control of a third switchable promoter. In some embodiments, the different peptide sequence encodes a common N-terminal stabilization sequence. In some embodiments, the DNA sequence encodes a mRNA sequence comprising a 3'UTR. In some embodiments, the different peptide sequence is 60 amino acids or fewer in length. In some embodiments, the different peptide sequences are random. In some embodiments, the different peptide sequences are pre-enriched for binding to a target. In some embodiments is a library of host cells, each comprises a library of the plasmid vectors.

Disclosed herein, in certain embodiments, is a host cell configured to express: a first fusion protein comprising a DNA-binding moiety; a second fusion protein comprising a gene activating moiety; a third fusion protein comprising a different DNA-binding moiety; a death agent, wherein the expression of the death agent is under control of a promoter DNA sequence specific for one of the DNA-binding moiety; a positive selection reporter, wherein the expression of the positive reporter is under control of a promoter DNA sequence specific for the other DNA-binding moiety; and a polypeptide of 60 or fewer amino acids, wherein the polypeptide modulates an interaction between the first test protein and the second test protein; wherein the host cell optionally has a mutant background enabling uptake of small molecules; and wherein the host cell optionally has a mutant background enabling increased transformation efficiency. In some embodiments, the polypeptide encodes an N-terminal sequence for peptide stabilization. In some embodiments, the polypeptide is an encoded product of an mRNA, wherein the mRNA comprises a 3'UTR. In some embodiments, the mRNA is an encoded product of a DNA molecule, wherein the DNA molecule is delivered into the host cell exogenously. In some embodiments, synthetic compound libraries can be tested. In some embodiments, the host cell is a eukaryote or a prokaryote. In some embodiments, the host cell is animal, plant, a fungus, or bacteria. In some embodiments, the host cell is a haploid yeast cell. In some embodiments, the host cell is a diploid yeast cell. In some embodiments, the diploid yeast cell is produced by mating a first host cell comprising DNA sequences encoding the first chimeric gene, the second chimeric gene, and the third chimeric gene, to a second host cell comprising DNA sequences encoding the death agent, positive selection reporter, and the mRNA comprising a nucleotide sequence encoding a polypeptide. In some embodiments, the fungus is *Aspergillus*. In some embodiments, the fungus is *Pichia pastoris*. In some embodiments, the fungus is *S. cerevisiae*. In some embodiments is a kit, comprising: the plasmid vector and the library of plasmid vectors.

Disclosed herein, in certain embodiments, is a method for identifying a molecule that selectively facilitates an interaction between a first test protein and a second test protein, comprising: expressing in the host cell a first fusion protein comprising the first test protein and a DNA-binding moiety; expressing in the host cell a second fusion protein comprising the second test protein and a gene activating moiety; expressing in the host cell a third fusion protein comprising the third test protein and a different DNA-binding moiety; and delivering a molecule from a library to the host cell such that the molecule forms a bridging interaction between the first test protein and the second test protein; wherein a sequence of a gene for expressing a death agent is disposed within the host cell and operably linked a promoter DNA sequence specific for the DNA binding moiety of the third fusion protein; wherein a positive selection reporter is disposed within the host cell and operably linked to a promoter DNA sequence specific for the DNA binding moiety of the first fusion protein; and wherein the first test protein and second test protein to form a functional transcription factor that activates expression of the death agent when the molecule from the library forms the bridging interaction. In some embodiments, the molecule from the library is delivered exogenously. In some embodiments, the host cell comprises more than one sequence for expressing a death agent that is activated by the promoter DNA sequence specific for a DNA binding moiety. In some embodiments, the host cell comprises more than one sequence for expressing a positive control reporter that is activated by a promoter DNA sequence specific for a DNA binding moiety. In some embodiments, the host cell comprises an integrated DNA encoding the first fusion protein, an integrated DNA encoding the second fusion protein, an integrated DNA encoding the third fusion protein; a plasmid DNA encoding the death agent; and a plasmid DNA encoding a positive selection reporter. In some embodiments, the DNA binding moiety is derived from LexA, cI, Gli-1, YY1, Glucocorticoid receptor, TetR, or Ume6. In some embodiments, the gene activating moiety is derived from VP16, Gal4, NF-κB, B42, BP64, VP64, or p65. In some embodiments, the death agent is a genetic element wherein overexpression of genetic material results in growth inhibition of the host cell. In some embodiments, the death agent is an overexpressed product of DNA. In some embodiments, the death agent is an overexpressed product of RNA. In some embodiments, the sequence of the gene for expressing the death agent is a Growth Inhibitory (GIN) sequence such as GIN11. In some embodiments, the death agent is a ribosomally encoded xenobiotic agent, a ribosomally-encoded poison, a ribosomally-encoded endogenous or exogenous gene that results in severe growth defects upon mild overexpression, a ribosomally-encoded recombinase that excises an essential gene for viability, a limiting factor involved in the synthesis of a toxic secondary metabolite, or any combination thereof. In some embodiments, the death agent is Cholera toxin, SpvB toxin, CARDS toxin, SpyA Toxin, HopU1, Chelt toxin, Certhra toxin, EFV toxin, ExoT, CdtB, Diphtheria toxin, ExoU/VipB, HopPtoE, HopPtoF, HopPtoG, VopF, YopJ, AvrPtoB, SdbA, SidG, VpdA, Lpg0969, Lpg1978, YopE, SptP, SopE2, SopB/SigD, SipA, YpkA, YopM, Amatoxin, Phalloidin, Killer toxin KP1, Killer toxin KP6, Killer Toxin K1, Killer Toxin K28 (KHR), Killer Toxin K28 (KHS), Anthrax lethal factor endopeptidase, Shiga Toxin, Saporin Toxin, Ricin Toxin, or any combination thereof. In some embodiments, the first test protein is a variation of KRAS. In some embodiments, the third test protein is KRAS. In some embodiments, the second test

protein is c-Raf. In some embodiments, the first test protein is YAP or TAZ. In some embodiments, the third test protein is VGLL4. In some embodiments, the second test protein is TEAD. In some embodiments, the molecule is small molecule. In some embodiments, the small molecule is peptidomimetic. In some embodiments, the molecule is peptide or protein. In some embodiments, the peptide or protein is derived from naturally occurring protein product. In some embodiments, the peptide or protein is synthesized protein product. In some embodiments, the peptide or protein is product of recombinant genes. In some embodiments, the peptide or protein is expressed product of test DNA molecule inserted into the host cell, wherein the test DNA molecule comprises of DNA sequences that encodes polypeptides, forming the library. In some embodiments, the library comprises of sixty or fewer amino acids. In some embodiments, the peptide or protein is a product of post-translational modification. In some embodiments, the post-translational modification includes cleavage. In some embodiments, the post-translational modification includes cyclization. In some embodiments, the post-translational modification includes bi-cyclization. In some embodiments, the cyclization comprises reacting with prolyl endopeptidase. In some embodiments, the cyclization comprises reacting with beta-lactamase. In some embodiments, the bicyclization comprises reacting with hydroxylase and dehydratase. In some embodiments, the bicyclization is formed by a tryptathionine bridge. In some embodiments, the post-translational modification includes methylation. In some embodiments, the methylation comprises reacting with N-methyltransferase. In some embodiments, the post-translational modification includes halogenation. In some embodiments, the post-translational modification includes glycosylation. In some embodiments, the post-translational modification includes acylation. In some embodiments, the post-translational modification includes phosphorylation. In some embodiments, the post-translational modification includes acetylation. In some embodiments, the test DNA molecule comprises of gene sequence expressing modifying enzyme. In some embodiments, the test DNA molecule comprises of a gene sequence expressing N-terminal sequence of methionine-valine-asparagine. In some embodiments, the test DNA molecule comprises of a gene sequence encoding a 3'UTR. In some embodiments, the 3'UTR is 3'UTR of sORF1. In some embodiments, the host cell is a eukaryote or a prokaryote. In some embodiments, the host cell is animal, plant, a fungus, or bacteria. In some embodiments, the fungus is *Aspergillus*. In some embodiments, the fungus is *Pichia pastoris*. In some embodiments, the fungus is *S. cerevisiae*.

Disclosed herein, in certain embodiments, is a method of identifying a molecule that selectively modulates a first test protein and a second test protein in a host cell, comprising: expressing in the host cell a first fusion protein comprising the first test protein and a second fusion protein comprising the second test protein; delivering a first molecule to the host cell; modifying the first molecule while in the host cell via a modifying enzyme; and allowing the first molecule to modulate the interaction between the first test protein and the second test protein, wherein the first molecule is a product of an encoded DNA sequence, wherein the first molecule comprises a library and one or more modifying enzymes, and wherein the one or more modifying enzymes modify the library. In some embodiments, the first molecule is a small molecule. In some embodiments, the small molecule is peptidomimetic. In some embodiments, the first molecule is peptide or protein. In certain embodiments, the

peptide or protein is derived from naturally occurring protein product. In some embodiments, the peptide or protein is synthesized protein product. In some embodiments, the first molecule is encoded in the host cell. In some embodiments, the first molecule is delivered exogenously. In some embodiments, the one or more modifying enzymes cause cleavage of the library. In some embodiments, the one or more modifying enzymes cause cyclization of the library. In some embodiments, the one or more modifying enzymes cause bicyclization of the library. In some embodiments, the cyclization comprises reacting with prolyl endopeptidase. In some embodiments, the cyclization comprises reacting with beta-lactamase. In some embodiments, the bicyclization comprises reacting with hydroxylase and dehydratase. In some embodiments, the bicyclization comprises formation of a tryptathionine bridge. In some embodiments, the one or more modifying enzymes cause methylation. In some embodiments, the one or more modifying enzymes is a methyltransferase. In some embodiments, the one or more modifying enzyme is a halogenase. In some embodiments, the one or more modifying enzymes cause glycosylation. In some embodiments, the one or more modifying enzymes cause acylation. In some embodiments, the one or more modifying enzymes cause phosphorylation. In some embodiments, the one or more modifying enzymes cause acetylation. In some embodiments, the library comprises of sixty or fewer amino acids. In some embodiments, the first test protein is KRAS or a variation of KRAS. In some embodiments, the second test protein is c-Raf. In some embodiments, the first test protein is YAP, TAZ, or VGLL4. In some embodiments, the second test protein is TEAD. In some embodiments, the host cell is a eukaryote or a prokaryote. In some embodiments, the host cell is animal, plant, a fungus, or bacteria. In some embodiments, the fungus is *Aspergillus*. In some embodiments, the fungus is *Pichia pastoris*. In some embodiments, the fungus is *S. cerevisiae*.

INCORPORATION BY REFERENCE

All publications, patents, and patent applications mentioned in this specification are herein incorporated by reference to the same extent as if each individual publication, patent, or patent application was specifically and individually indicated to be incorporated by reference.

BRIEF DESCRIPTION OF THE DRAWINGS

Features of the disclosure are set forth with particularity in the appended claims. A better understanding of the features and advantages of the present disclosure will be obtained by reference to the following detailed description that sets forth illustrative embodiments, in which the principles of the disclosure are utilized, and the accompanying drawings of which:

FIGS. 1A and 1B illustrate a platform to identify a compound that specifically disrupts a protein-protein interaction. FIG. 1A illustrates a case where a peptide is unable to disrupt the pairwise interaction of interest (A-C). FIG. 1B illustrates a case where a peptide is able to disrupt the pairwise interaction of interest (A-C) without disrupting another interaction (A-B).

FIGS. 2A and 2B illustrate a platform to identify a compound that disrupts a protein-protein interaction (A-B') in a variant-specific manner. FIG. 2A illustrates a case where a peptide is unable to disrupt the pairwise interaction of interest (A-B'). FIG. 2B illustrates a case where a peptide is able to disrupt the pairwise interaction of interest between

one variant (B') and a protein (A) without disrupting the interaction between another variant (B) and the protein (A).

FIGS. 3A and 3B exemplify an embodiment of a platform to identify a compound that bridges two proteins in a variant- or protein-specific manner. FIG. 3A shows a case where a peptide is able to bridge one variant (B) and a protein (A). FIG. 3B illustrates a case where a peptide is able to bridge the secondary variant (B') and a protein (A).

FIGS. 4A and 4B show an embodiment of a platform to identify a compound that specifically disrupts a protein-protein interaction, where the platform includes two positive selection markers and a negative selection markers. In these figures, the compound may interact with the prey to disrupt the prey-bait interaction. FIG. 4A illustrates a case where a peptide is unable to disrupt the pairwise interaction of interest. FIG. 4B illustrates a case where a peptide is able to disrupt the Prey and BaitMut interaction by acting on the Prey without disrupting the interaction between BaitWT and Prey.

FIGS. 5A and 5B show an embodiment of a platform to identify a compound that specifically disrupts a protein-protein interaction in a system with two positive selections and a negative selection. In contrast with FIGS. 4A and 4B, in these figures, the compound may act on the bait (rather than the prey) to disrupt the prey-bait interaction. FIG. 5A illustrates a case where a peptide is unable to disrupt the pairwise interaction of interest. FIG. 5B illustrates a case where a peptide disrupts the Prey-BaitMut interaction by acting on the BaitMut without disrupting the interaction between BaitWT and Prey.

FIG. 6 illustrates two platforms to identify a compound that specifically disrupts a protein-protein interaction in cell culture by selecting for nutritional reporter.

FIG. 7 illustrates two platforms to identify a compound that specifically disrupts a protein-protein interaction in cell culture by selecting for cytotoxic reporter.

FIG. 8 shows an embodiment of an integration plasmid that encodes two bait proteins, each with its own DNA binding domain ("DBD"), and a prey protein with an activation domain ("AD").

FIG. 9 shows an embodiment of a selection and library plasmid that encodes two death agents, both with the same DNA binding sequence, a positive selection agent driven by another DNA binding sequence and an inducible stabilized peptide library.

FIG. 10 shows an embodiment of a confirmation plasmid that encodes two bait-prey fusion proteins, each with its own DBD.

FIG. 11 illustrates a cyclization process of families of RiPP scaffolds that lead to the generation of N-to-C cyclized, backbone N-methylated macrocycles by the action of (1) prolyl oligopeptidases belonging to the PopB family and (2) N-methyltransferases belonging to the omphalotin methyltransferase family. The variable peptide library region is embedded within the primary sequence of the modifying enzyme. N- and C-term sequences refer to consensus binding and processing recognition elements. The N-terminus comprises the enzymatic N-methyltransferase domain, a linker region, and the processing enzymes binding sites.

FIG. 12 illustrates a cyclization process of different families of RiPP scaffolds that lead to the generation of N-to-C cyclized, backbone N-methylated macrocycles by the action of a beta-lactamase (alanine-alanine transpeptidase) and an N-methyltransferase. The variable peptide library region is embedded within the primary sequence of the modifying enzyme. N- and C-term sequences refer to consensus binding and processing recognition elements. The N-terminus

comprises the enzymatic N-methyltransferase domain, a linker region, and the processing enzymes binding sites.

FIG. 13 shows a gene organization within two sets exemplary loci from *Rhizopogon vinicolor* that encode for modifying enzymes that could be used in the schematic depicted in FIG. 12.

FIG. 14 shows WebLogo alignments of a large variety of MSDIN family genes (toxin preproprotein sequences) identified in the genomes of *Amanita bisporigera* and *Amanita phalloides*. See Pulman, Jane A., et al. "Expansion and Diversification of the MSDIN Family of Cyclic Peptide Genes in the Poisonous Agarics *Amanita Phalloides* and *A. Bisporigera*." *BMC Genomics*, vol. 17, no. 1, 15 Dec. 2016, p. 1038., doi: 10.1186/s12864-016-3378-7.

FIG. 15 illustrates a cyclization process for members of the MSDIN family of RiPPs that leads to the generation of N-to-C cyclized macrocycles by the action of prolyl oligopeptidases belonging to the PopB family.

FIG. 16 illustrates a bicyclization process for members of the MSDIN family of RiPPs that leads to the generation of N-to-C cyclized macrocycles and bicyclic macrocycles internally bridged via a tryptathionine bridge.

FIG. 17 illustrates biochemical steps to create a tryptathionine bridge of a bicyclic ring.

DETAILED DESCRIPTION

Two-hybrid screening can be used to identify and characterize protein-protein interactions. The two-hybrid system was initially developed using yeast as a host organism. However, bacterial or animal cell two-hybrid systems can also be used to characterize protein-protein interactions. The present disclosure provides a system that can use a unified eukaryotic or prokaryotic two-hybrid system in which bait and prey expression plasmid is used in both organismal contexts. Additionally, an extensive series of leucine zipper fusion proteins of known affinities can be generated to compare the efficiency of interaction detection using both systems. The yeast system can produce a quantitative readout over a dynamic range. "Auto-activation" by baits can be less prevalent in the bacterial system. In addition, modified expression vectors disclosed herein can be used for expression of a protein of interest in both eukaryotes and prokaryotes.

Three-hybrid systems rely on similar principles as two-hybrid systems, but involve an additional factor to bridge a protein-protein interaction to result in a gene expression readout.

The present disclosure also provides a system for delivering molecules across the cell membrane. The cell membrane presents a major challenge in drug discovery, especially for biologics such as peptides, proteins, and nucleic acids. One potential strategy to subvert the membrane barrier and deliver biologics into cells is to attach them to "cell penetrating peptides" (CPPs). Despite three decades of investigation, the fundamental basis for CPP activity remains elusive. CPPs that enter cells via endocytosis generally exit from endocytic vesicles in order to reach the cytosol. Unfortunately, the endosomal membrane has proven to be a significant barrier towards cytoplasmic delivery of these CPPs such that often a negligible fraction of the peptides escapes into the cell interior. What are thus needed are new scaffolds and structures that impart peptides with highly proficient intrinsic cell penetrating ability to various cell types. Several naturally occurring polyketides and peptides exhibit remarkable cell permeability (e.g. cyclosporine and amanitins). These peptides are characterized by specific

modifications (e.g., N-methylation of the backbone and cyclization) that can play a crucial role in their cell membrane permeability. The compositions and methods disclosed herein describe methods and approaches that enable the general utilization of similar modifications to generate compositions that may be of high therapeutic value and that may be capable of disrupting select protein-protein interactions with high selectivity.

Definitions

As used herein, “reporter gene” refers to a gene whose expression can be assayed. Such genes include, for example, LacZ, β -glucuronidase (GUS), amino acid biosynthetic genes, the yeast LEU2, HIS3, LYS2, or URA3 genes, nucleic acid biosynthetic genes, the mammalian chloramphenicol transacetylase (CAT) gene, the green fluorescent protein (GFP) or any surface antigen gene for which specific antibodies are available. Reporter genes can result in both positive and negative selection.

An “allele” refers to a DNA sequence of a gene which includes a naturally occurring, or pathogenic variant of a gene. Expression of differing alleles may lead to different protein variants.

A “promoter” is a DNA sequence located proximal to the start of transcription at the 5' end of an operably linked transcribed sequence. The promoter can contain one or more regulatory elements or modules, which interact in modulating transcription of the operably linked gene. Promoters can be switchable or constitutive. Switchable promoters allow for reversible induction or repression of operably linked target genes upon administration of an agent. Examples of switchable promoters include but are not limited to the LexA operator and the alcohol dehydrogenase I (alcA) gene promoter. Examples of constitutive promoters include the human beta-actin gene promoter.

“Operably linked” describes two macromolecular elements arranged such that modulating the activity of the first element induces an effect on the second element. In this manner, modulation of the activity of a promoter element can be used to alter or regulate the expression of an operably-linked coding sequence. For example, the transcription of a coding sequence that is operably-linked to a promoter element can be induced by factors that activate the promoter’s activity; transcription of a coding sequence that is operably-linked to a promoter element can be inhibited by factors that repress the promoter’s activity. Thus, a promoter region is operably-linked to the coding sequence of a protein if transcription of such coding sequence activity is influenced by the activity of the promoter.

“In frame” as used herein throughout, refers to the proper positioning of a desired sequence of nucleotides within a DNA fragment or coding sequence operably linked to a promoter sequence, thereby permitting transcription and/or translation.

“Fusion construct” refers to recombinant genes that encode fusion proteins.

A “fusion protein” is a hybrid protein, i.e., a protein that has been constructed to contain domains from at least two different proteins. Fusion proteins described herein can be a hybrid proteins that possess both (1) a transcriptional regulatory domain from a transcriptional regulatory protein or a DNA binding domain from a DNA binding protein and (2) a heterologous protein to be assayed for interaction status. The protein that is the source of the transcriptional regulatory domain may differ from the protein that is the source

of the DNA binding domain. In other words, the two domains may be heterologous to each other.

A transcriptional regulatory domain of a prey fusion protein can either activate or repress transcription of target genes, depending on the biological activity of the domain. Bait proteins of the disclosure may also be fusion proteins, where the fusion protein is encoded by a fusion gene that can encode for a protein of interest that is operably linked to a DNA binding moiety.

“Bridging interaction” refers to an interaction between a first protein and a second that occurs only when one or both of the first protein and the second protein interact with a molecule, such as a peptide or small molecule from a library. In some cases, the bridging interaction between the first protein and the second protein is direct, while in other cases the bridging interaction between the first protein and the second protein is indirect.

“Expression” is the process by which the information encoded within a gene is revealed. If the gene encodes a protein, then expression involves both transcription of the DNA into mRNA, the processing of mRNA (if necessary) into a mature mRNA product, and translation of the mature mRNA into protein.

As used herein, a “cloning vehicle” is any entity that is capable of delivering a nucleic acid sequence into a host cell for cloning purposes. Examples of cloning vehicles include plasmids or phage genomes. A plasmid that can replicate autonomously in the host cell is especially desired. Alternatively, a nucleic acid molecule that can insert (integrate) into the host cell’s chromosomal DNA is useful, especially a molecule that inserts into the host cell’s chromosomal DNA in a stable manner, that is, a manner that allows such molecule to be inherited by daughter cells.

Cloning vehicles are often characterized by one or a small number of endonuclease recognition sites at which such DNA sequences may be cut in a determinable fashion without loss of an essential biological function of the vehicle, and into which DNA may be spliced in order to bring about its replication and cloning.

The cloning vehicle can further contain a marker suitable for use in the identification of cells transformed with the cloning vehicle. For example, a marker gene can be a gene that confers resistance to a specific antibiotic on a host cell.

The word “vector” can be used interchangeably with “cloning vehicle.”

As used herein, an “expression vehicle” is a vehicle or vector similar to the cloning vehicle that is especially designed to provide an environment that allows the expression of the cloned gene after transformation into the host. One manner of providing such an environment is to include transcriptional and translational regulatory sequences on such expression vehicles, such transcriptional and translational regulatory sequences being capable of being operably linked to the cloned gene. Another manner of providing such an environment is to provide a cloning site or sites on such vehicle, wherein a desired cloned gene and a desired expression regulatory element can be cloned.

In an expression vehicle, the gene to be cloned is usually operably-linked to certain control sequences such as promoter sequences. Expression control sequences will vary depending on whether the vector is designed to express the operably-linked gene in a prokaryotic or eukaryotic host and can additionally contain transcriptional elements such as enhancer elements, termination sequences, tissue-specificity elements, or translational initiation and termination sites.

A “host” refers to any organism that is the recipient of a cloning or expression vehicle. The host may be a yeast cell

or a cultured animal cell such as a mammalian or insect cell. The yeast host may be *Saccharomyces cerevisiae*.

A “host cell” as described herein can be a bacterial, fungal, or mammalian cell or from an insect or plant. Examples of bacterial host cells are *E. coli* and *B. subtilis*. Examples of fungal cells are *S. cerevisiae* and *S. pombe*. Non-limiting examples of mammalian cells are immortalized mammalian cell lines, such as HEK293, A549, HeLa, or CHO cells, or isolated patient primary tissue cells that have been genetically immortalized (such as by transfection with hTERT). Non-limiting example of the plant is *Nicotiana tabacum* or *Physcomitrella patens*. A non-limiting example of insect cell is a sf9 (*Spodoptera frugiperda*) cell.

A “DNA-binding domain (DBD),” or a “DNA-binding moiety” is a moiety that is capable of directing specific polypeptide binding to a particular DNA sequence (i.e., a “protein binding site”). These proteins can be homodimers or monomers that bind DNA in a sequence specific manner. Exemplary DNA-binding domains of the disclosure include LexA, cI, glucocorticoid receptor binding domains, and the Ume6 domain.

A “gene activating moiety” or “activation domain” (“AD”) is a moiety that is capable of inducing (albeit in many instances weakly inducing) the expression of a gene to whose control region it is bound (one example is an activation domain from a transcription factor). As used herein, “weakly” is meant below the level of activation effected by GAL4 activation region II and is preferably at or below the level of activation effected by the B42 activation domain. Levels of activation can be measured using any downstream reporter gene system and comparing, in parallel assays, the level of expression stimulated by the GAL4 region II-polypeptide with the level of expression stimulated by the polypeptide to be tested.

Screening for Disruptors to a Protein-Protein Interaction (PPI)

The often large and broad surfaces that can form the contact interface between two proteins can be potential targets of canonical small molecule inhibitors. However, the large and broad surfaces can have size limitations, and evolved resistance can occur readily. The specificity of antibodies can be combined with cell permeability in the form of short peptides, for example, peptides of less than 25 residues. Screening for short peptide disruptors of protein-protein interactions (PPIs) can be performed using technologies such as phage display or mRNA display. However, these screens are performed in vitro and require the purification of one of the interacting proteins of interest. Upon selection of a peptide sequence with affinity toward one of the proteins, secondary screens can be performed to validate that the peptide interferes with the binding interface of the second protein. This secondary screening can further rely upon the proper folding of the proteins and the replication of intracellular biophysical conditions in the assays.

Methods and systems of the disclosure can involve the intracellular selection of peptide disruptors of PPIs. Stated differently, various systems described herein can be used to screen for molecules that selectively disrupt an interaction between two proteins. A model organism, for example *Saccharomyces cerevisiae*, can be employed, and the coexpression of a PPI of interest with a test DNA molecule comprising a DNA sequence that encodes a randomized peptide library can allow for the selection of unbiased peptides that interfere with a specific PPI using selection mechanisms (e.g., a stringent viability readout selection mechanisms). The method can involve a permutation of a yeast two-hybrid system that can rely on the reconstitution

of a transcription factor that requires an interaction between one or two test proteins fused to one or two DNA binding domain(s) (DBDs) and a second test protein fused to a transcription activation domain (AD) or gene activating moiety.

Methods and systems of the disclosure can use the reconstitution of a transcription factor mediated by the interaction between a protein fused to an AD, for example, VP16, NF- κ B AD, VP64AD, BP64 AD, B42 acidic activation domain (B42AD), or p65 transactivation domain (p65AD) and another protein fused to a DBD, for example, LexA, cI, Gli-1, YY1, glucocorticoid receptor binding domain, or Ume6 domain.

Methods and system of the disclosure can also use two different proteins, or two variants of one protein, fused to different DBDs. These proteins may interact with the same protein fused to an AD to drive two different or identical reporters. The system can identify inhibitors against a specific PPI in a complex without affecting the rest of the complex integrity (see FIGS. 1A and 1B). This system can also be used to identify selective inhibitors that disrupt a PPI between a specific isoform and its binding partner without affecting another variant (see FIGS. 2A and 2B).

An efficient interaction between the two proteins of interest can direct RNA polymerase to a specific genomic site, and allow for the expression of a genetic element. The genetic element can be, for example, a gene that encodes a protein that enables an organism to grow on selection media. The selection media can be specific to, for example, ADE2, URA3, TRP1, KANR, or NATR, and will lack the essential component (Ade, Ura, Trp) or include a drug (G418, NAT). Markers that can detect when an interaction is no longer present (for example when the interaction is disrupted by an external composition) can be referred to as counter-selection markers, such as the URA3 gene, and can be poor or leaky (easily masked by the selection of mutants that escape the selection). This leakiness of the selection marker can lead to a high false positive rate.

Methods and systems of the disclosure can combine a strong negative selection marker with the intracellular stabilization of the production of short peptides to screen for blockers of PPIs. An inducible two-hybrid approach can be employed, which can drive the expression of any one or combination of several cytotoxic reporters (death agents) as well as positive selection markers. A method of the disclosure involving induced expression of a combination of cytotoxic reporters in a two-hybrid system can allow for a multiplicative effect in lowering the false-positive rate of the two-hybrid assay, as all of the cytotoxic reporters must simultaneously be “leaky” to allow for an induced cell to survive.

Disclosed herein, in certain embodiments, is a method for identifying a molecule that selectively disrupts an interaction between a first test protein and a second test protein in a host cell, comprising: expressing in the host cell a first fusion protein comprising the first test protein and a DNA-binding moiety; expressing in the host cell a second fusion protein comprising the second test protein and a gene activating moiety; expressing in the host cell a third fusion protein comprising the third test protein and a different DNA-binding moiety; and delivering a molecule from a library to the host cell; wherein a sequence of gene for expressing a death agent is disposed within the host cell and operably linked a promoter DNA sequence specific for the DNA binding moiety of the first fusion protein, wherein a positive selection reporter is disposed within the host and operably linked to a promoter DNA sequence specific for the

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DNA binding moiety of the third fusion protein, wherein, in the absence of the molecule, the interaction between the first test protein and the second test protein causes the gene activating moiety to activate expression of the death agent, while the interaction between the second test protein and the third test protein causes the gene activating moiety to activate the expression of the positive selection reporter.

FIGS. 1A and 1B show a platform to identify a compound that disrupts a protein-protein interaction within a complex in a specific manner. DBD1 and DBD2 are promoter specific DNA-binding domains. AD refers to an activation domain. A, B, and C refer to three proteins, wherein B and C each interact with A. Broken arrows indicate active expression of the reporter. +ve refers to positive selection markers, -ve refers to death agents (negative selection markers). Pep refers to a peptide, such as a peptide from a library. In some embodiments, a peptide library may be replaced with a library that includes compounds other than peptides like small molecules. In some embodiments, the small molecules are peptidomimetics. Two scenarios are shown; FIG. 1A illustrates a case where a peptide (e.g., a peptide from a library) is unable to disrupt the pairwise interaction of interest (A-C), and a death agent is expressed, leading to cell death. FIG. 1B illustrates a case where a peptide (e.g., a peptide from a library) is able to disrupt the pairwise interaction of interest (A-C) without disrupting the A-B interaction. Selective peptide disruption activity is assayed by survival due to (1) the absence of expression of the death agent and (2) expression of the positive selection reporter (which provides evidence of selectivity).

FIGS. 2A and 2B show a platform to identify a compound that disrupts a protein-protein interaction in a variant-specific manner. In other words, in contrast to FIGS. 1A and 1B in which B and C were unrelated protein, FIGS. 2A and 2B describe an analogous assay in which B and B' are related (but different) proteins (e.g., protein variants). As in FIGS. 1A and 1B, DBD1 and DBD2 are promoter-specific DNA-binding domains. AD refers to an activation domain. A, B, and B' refer to three proteins, wherein B and B' are two variants that are configured to interact with A. Broken arrows indicate active expression of the reporter. +ve refers to a positive selection marker, while -ve refers to a death agent. Pep refers to a peptide. Two scenarios are shown; FIG. 2A illustrates a case where a peptide is unable to disrupt the pairwise interaction of interest (A-B'), and a death is expressed, leading to cell death. FIG. 2B illustrates a case where a peptide is able to disrupt the pairwise interaction between A and B' without disrupting the interaction between the A and B. Selective peptides disruption is assayed by survival due to (1) the absence of expression of the death agent and (2) expression of the positive selection reporter (which provides evidence of selectivity).

Screening for Facilitators to Protein-Protein Interaction

The system can additionally be used to screen for molecules that "bridge" an interaction between two proteins in a selective manner. In some embodiments, the system can be used to identify molecules which can bind to one isoform, or one protein, and bridge its interaction with another macromolecule, such as a protein, RNA, or DNA. For example, the bridging could occur to link the protein to an E3 ligase to mediate its degradation. For example, bridging can occur between an oncogenic protein such as K-Ras oncogenic alleles, Cyclin D family, Cyclin E family, c-MYC, EGFR, HER2, PDGFR, Raf kinase, VEGF and beta-catenin, or oncogenic variants such as IDH1(R132H, R132S, R132C, R132G, and R132L) or IDH2(R140Q, R172K), and an E3 ligase. An E3 ligase can be chosen from a list including, but

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not limited to, Cereblon, Skp2, MDM2, FBXW7, DCAF15, VHL, AMFR, ANAPC11, ANKIB1, ARL1, ARIH1, ARIH2, BARD1, BFAR, BIRC2, BIRC3, BIRC7, BIRC8, BMI1, BRAP, BRCA1, CBL, CBLB, CBLC, CBLL1, CCDC36, CCNB1IP1, CGRRF1, CHFR, CNOT4, CUL9, CYHR1, DCST1, DTX1, DTX2, DTX3, DTX3L, DTX4, DZIP3, E4F1, FANCL, G2E3, HACE1, HECTD1, HECTD2, HECTD3, HECTD4, HECW1, HECW2, HERC1, HERC2, HERC3, HERC4, HERC5, HERC6, HLF, HUWE1, IRF2BP1, IRF2BP2, IRF2BPL, Itch, KCMF1, KMT2C, KMT2D, LNX1, LNX2, LONRF1, LONRF2, LONRF3, LRSAM1, LTN1, MAEA, MAP3K1, MARCH1, MARCH10, MARCH11, MARCH2, MARCH3, MARCH4, MARCH5, MARCH6, MARCH7, MARCH8, MARCH9, MDM4, MECOM, MEX3A, MEX3B, MEX3C, MEX3D, MGRN1, MIB1, MIB2, MID1, MID2, MKRN1, MKRN2, MKRN3, MKRN4P, MNAT1, MSL2, MUL1, MYCBP2, MYLIP, NEDD4, NEDD4L, NEURL1, NEURL1B, NEURL3, NFX1, NFXL1, NHLRC1, NOSIP, NSMCE1, PARK2, PCGF1, PCGF2, PCGF3, PCGF5, PCGF6, PDZRN3, PDZRN4, PELI1, PELI2, PELI3, PEX10, PEX12, PEX2, PHF7, PHRF1, PJA1, PJA2, PLAG1, PLAGL1, PML, PPIL2, PRPF19, RAD18, RAG1, RAPSN, RBBP6, RBCK1, RBX1, RC3H1, RC3H2, RCHY1, RFFL, RFPL1, RFPL2, RFPL3, RFPL4A, RFPL4AL1, RFPL4B, RFWD2, RFWD3, RING1, RLF, RLIM, RMND5A, RMND5B, RNF10, RNF103, RNF11, RNF111, RNF112, RNF113A, RNF113B, RNF114, RNF115, RNF121, RNF122, RNF123, RNF125, RNF126, RNF128, RNF13, RNF130, RNF133, RNF135, RNF138, RNF139, RNF14, RNF141, RNF144A, RNF144B, RNF145, RNF146, RNF148, RNF149, RNF150, RNF151, RNF152, RNF157, RNF165, RNF166, RNF167, RNF168, RNF169, RNF17, RNF170, RNF175, RNF180, RNF181, RNF182, RNF183, RNF185, RNF186, RNF187, RNF19A, RNF19B, RNF2, RNF20, RNF207, RNF208, RNF212, RNF212B, RNF213, RNF214, RNF215, RNF216, RNF217, RNF219, RNF220, RNF222, RNF223, RNF224, RNF225, RNF24, RNF25, RNF26, RNF31, RNF32, RNF34, RNF38, RNF39, RNF4, RNF40, RNF41, RNF43, RNF44, RNF5, RNF6, RNF7, RNF8, RNFT1, RNFT2, RSPRY1, SCAF11, SH3RF1, SH3RF2, SH3RF3, SHPRH, SLAH1, SLAH2, SLAH3, SMURF1, SMURF2, STUB1, SYVN1, TMEM129, TOPORS, TRAF2, TRAF3, TRAF4, TRAF5, TRAF6, TRAF7, TRAF, TRIM10, TRIM11, TRIM13, TRIM15, TRIM17, TRIM2, TRIM21, TRIM22, TRIM23, TRIM24, TRIM25, TRIM26, TRIM27, TRIM28, TRIM3, TRIM31, TRIM32, TRIM33, TRIM34, TRIM35, TRIM36, TRIM37, TRIM38, TRIM39, TRIM4, TRIM40, TRIM41, TRIM42, TRIM43, TRIM43B, TRIM45, TRIM46, TRIM47, TRIM48, TRIM49, TRIM49B, TRIM49C, TRIM49D1, TRIM5, TRIM50, TRIM51, TRIM52, TRIM54, TRIM55, TRIM56, TRIM58, TRIM59, TRIM6, TRIM60, TRIM61, TRIM62, TRIM63, TRIM64, TRIM64B, TRIM64C, TRIM65, TRIM67, TRIM68, TRIM69, TRIM7, TRIM71, TRIM72, TRIM73, TRIM74, TRIM75P, TRIM77, TRIM8, TRIM9, TRIML1, TRIML2, TRIP12, TTC3, UBE3A, UBE3B, UBE3C, UBE3D, UBE4A, UBE4B, UBOX5, UBR1, UBR2, UBR3, UBR4, UBR5, UBR7, UHRF1, UHRF2, UNK, UNKL, VPS11, VPS18, VPS41, VPS8, WDR59, WDSUB1, WWP1, WWP2, XIAP, ZBTB12, ZFP91, ZFPL1, ZNF280A, ZNF341, ZNF511, ZNF521, ZNF598, ZNF645, ZNRF1, ZNRF2, ZNRF3, ZNRF4, Zswim2, and ZXDC. The peptide-mediated bridging event can be specific to a mutant variant, or to one member of a complex, without disrupting the integrity of the WT variant or the rest of the complex.

Disclosed herein, in certain embodiments, is a method for identifying a molecule that selectively facilitates an interaction between a first test protein and a second test protein comprising: expressing in the host cell a first fusion protein comprising the first test protein and a DNA-binding moiety; expressing in the host cell a second fusion protein comprising the second test protein and a gene activating moiety; expressing in the host cell a third fusion protein comprising the third test protein and a different DNA-binding moiety; and delivering a molecule from a library to the host cell such that the molecule forms a bridging interaction between the first test protein and the second test protein; wherein a sequence of a gene for expressing a death agent is disposed within the host cell and operably linked a promoter DNA sequence specific for the DNA binding moiety of the third fusion protein; wherein a positive selection reporter is disposed within the host cell and operably linked to a promoter DNA sequence specific for the DNA binding moiety of the first fusion protein; and wherein the first test protein and second test protein to form a functional transcription factor that activates expression of the death agent when the molecule from the library forms the bridging interaction.

FIGS. 3A and 3B show a platform to identify a compound that bridges a protein-protein interaction in a variant- or protein-specific manner. DBD1 and DBD2 are promoter specific DNA-binding domains. AD refers to an activation domain. A, B, and B' refer to three proteins, wherein B and B' are two variants to be assayed for interaction with A. Broken arrows indicate active expression of the reporter. +ve refers to positive selection markers, -ve refers to death agents. Pep refers to a peptide (e.g., a peptide from a library). Two scenarios are shown; FIG. 3A illustrates a case where a peptide is able to bridge an interaction between one variant (B) and a protein (A). In this case, this is the control variant (B) and it leads to expression of the death and cell death. FIG. 3B illustrates a case where a peptide is able to bridge the secondary variant (B') and a protein (A). In this case, the peptide is bridging the variant of interest (B') and enables its binding to protein A. This combination of results leads to (1) the activation of a positive selection marker and (2) the lack of activation of the death agent (due to the lack of bridging to the unintended variant).

In some embodiments, the host cell disclosed herein further comprises an integrated DNA encoding the first fusion protein, an integrated DNA encoding the second fusion protein, an integrated DNA encoding the third fusion protein; a plasmid DNA encoding the death agent; and a plasmid DNA encoding a positive selection reporter.

To identify peptides that can disrupt or facilitate a PPI, a PPI integration plasmid (PLASMID 1; FIG. 8), a selection and library plasmid (PLASMID 2; FIG. 9), and a confirmation plasmid (PLASMID 3; FIG. 10) can be used. The integration of PLASMID 1 into the genome of the host cell (as confirmed using PLASMID 3) can be followed by transformation of a library of PLASMID 2 encoding random peptides with, for example, NNK or NNN codons. Expression of Fusion Proteins for PPI

In some embodiments, the host cell disclosed herein comprises a plasmid vector, which comprises the components of PLASMID 1 (FIG. 8), or any combination of the components of PLASMID 1. PLASMID 1 can contain, for example, two restriction sites that enable the integration of two proteins that constitute the PPI of interest. The PPI of interest can involve a pair of domains having known importance for carcinogenesis, such as p53-MDM2, RAS-RASBDPs, and MYC-MAX. The PPI of interest can also

involve the interaction of an oncogene (such as Cyclin E family, Cyclin D family, c-MYC, EGFR, HER2, K-Ras, PDGFR, Raf kinase, and VEGF) or a tumor suppressor (such as BRCA1, BRCA2, cyclin-dependent kinase inhibitor 1C, PTEN, p16, p27, p53, p73, and Retinoblastoma protein (pRb)) with a known cellular interaction partner. The PPI of interest can involve the interaction of a protein involved in the DNA repair pathway (such as ATM, ATRX, BRCA1, BRCA2, ERCC1, FANCB, FANCF, FEN1, HMGAI, HMGAI, MDC1, MGMT, MLH1, MSH2, MSH4, Mre11A, NBS1, NEIL1, PARP1, PARP2, PMS2, RAD51, RAD52, RAD54, RAD51AP1, WRN, and XPF) with another cellular factor.

PLASMID 1 can be configured to express two proteins that constitute a PPI and an additional factor, for example, a variant of one of the proteins, such as KRAS (G12D, G12V, G12C, G12S, G13D, Q61K, or Q61L, etc.) and WT KRAS along with BRAF. The additional factor can also be another protein bound to one of the components of the PPI, or as member of a larger complex (such as YAP or TAZ disruption from TEAD without compromising VGLL4 binding to TEAD, or maintaining binding of BAX to BAK but preventing binding of BAX to BCL-2).

In some embodiments, the host cell disclosed herein comprises PLASMID 1, wherein a DNA sequence encoding a first polypeptide is inserted in frame with Gal4-DBD, a second polypeptide is inserted in frame with LexA-DBD, and wherein a DNA sequence encoding a third polypeptide is inserted in frame with VP64-AD.

In some embodiments, the first test protein is a variant of KRAS, the second test protein is c-Raf, and the third test protein is KRAS.

In some embodiments, the first test protein is YAP or TAZ, the second test protein is TEAD, and the third test protein is VGLL4.

PLASMID 1 can encode for the fusion of an activation domain or another gene activating moiety and a DBD to each protein driven by either a strong promoter and terminator (such as ADH1), or by an inducible promoter (such as GAL1). Other exemplary activation domains include those of VP16 and B42AD. In some embodiments, the DNA binding moiety is derived from LexA, TetR, LacI, Gli-1, YY1, glucocorticoid receptor, or Ume6 domain and the gene activating moiety is derived from Gal4, B42, or VP64, Gal4, NF- κ B AD, Dof1, BP64, B42, or p65. Each protein fusion can be tagged for subsequent biochemical experiments with, for example, a FLAG, HA, MYC, or His tag. PLASMID 1 can also include bacterial selection and propagation markers (i.e. ori and Amp^R), and yeast replication and selection markers (i.e. TRP1 and CEN or 2 μ m). The plasmid may contain multiple bait proteins fused to different DBDs. The plasmid can also be integrated into the genome at a specified locus.

Disclosed herein, in certain embodiments, is a library of plasmid vectors, each plasmid vector comprising: a DNA sequence encoding a different peptide sequence operably linked to a first switchable promoter; a DNA sequence encoding a death agent under control of a second switchable promoter; and a DNA sequence encoding a positive selection reporter under control of a third switchable promoter. Expression of Selection Markers

Positive Selection Markers

An efficient interaction between the two test proteins can direct RNA polymerase to a specific genomic site, and allow expression of a protein that enables an organism to grow on selection media. The selection media can be specific to, for example, ADE2, URA3, TRP1, KAN^R, or NAT^R, and can

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lack the essential component (Ade, Ura, Trp) or can include a drug (G418, NAT). PLASMID 2 (FIG. 9) can encode for one or more positive selection markers that enable an organism to grow on selection media.

FIG. 6 shows the results of two systems for identifying a compound that specifically disrupts a protein-protein interaction in cell culture using a positive selection marker. In the first platform, KRas and KRas(G12D) fused to DBDs and c-Raf fused to AD were expressed in yeast cells. In the absence of any inhibitors, the DBD fusion protein and AD fusion protein pairs maintain interaction to drive expression of nutritional reporters 1 and 2. A 5-fold dilution series starting at 10^4 cells were spotted onto selective media with or without inhibitor and visualized after 2 days of growth at 30°C . The results showed that the cells grown on media that selected for nutritional reporter 2 had particularly poor survival rate when the inhibitor was added, illustrating both (1) the specificity of the inhibitor for KRas(G12D) and c-Raf and (2) the validity of the screening assay.

In the second platform, VGLL4 and YAP fused to DBDs and TEAD fused to AD were expressed in cells. In the

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absence of any inhibitors, the DBD fusion protein and AD fusion protein pairs maintain interaction to drive expression of nutritional reporters 1 and 2. A 5-fold dilution series starting at 10^4 cells were spotted onto selective media with or without inhibitor and visualized after 2 days of growth at 30°C . The results showed that the cells grown on media that selected for nutritional reporter 2 had particularly poor survival rate when the inhibitor was added, illustrating both (1) the specificity of the inhibitor for the YAP and TEAD interaction and (2) the validity of the screening assay.

Negative Selection Markers

An inducible two-hybrid approach can be employed, which can drive the expression of any one or combination of several cytotoxic reporters (death agents) as well as positive selection markers. A method of the disclosure involving induced expression of a combination of cytotoxic reporters in a two-hybrid system can allow for a multiplicative effect in lowering the false-positive rate of the two-hybrid assay, as all of the cytotoxic reporters must simultaneously be "leaky" to allow for an induced cell to survive. The cytotoxic reporters can be, for example:

TABLE 1

Cholera toxin (CtxA)	SEQ ID NO.: 1	MVKIIFVFFIFLSSFSYANDDKLYRADSRPPDEIKQSGGLMPRGQSEYFDRGTQMN INLYDHARGTQTGFVRHDDGYVSTISLSRAHLVGQTLISGHSTYYIYVIATAPNM FNVNDVLGAYSHPHDEQEVSAALGGIPYSQIYGWYRVHFGVLDQELHNRNGYDRYY SNLDIAPAADGYGLAGFPPEHRAWREEPWIHHAPPGCCGNAPSSMSNTCEKTSQSL GVKFLDEYQSKVKRQIFSGYQSDIDTHNRKIDEL
SpvB toxin (<i>Salmonella enterica</i>)	SEQ ID NO.: 2	MLILNGFSSATLALITPPFLPKGGKALSQSGPDGLASITLPLPISAERGFPAPALAL HYSSGGNGPFGVGWSCATMSIARRTSHGVQYNDSDFLGPDGEVLVQTLSTGDA PNPVTCTFAYGDVSPQSYTVTRYQPRTESSFYRLEYWVGNSNGDDFWLLHDSNGIL HLLGKTAARLSDPQAASHTAQWLVEESVTPAGEHIYYSYLAENGDNVDLNGNEAG RDRSAMRYLSKVQYGNATPAADLYLWTSATPAVQWLFTLVFDYGERGVDPQVPPAF TAQNSWLARQDPFSLYNYGFEIRLHRLCRQVLMFHHFDELGEADTLVSRLLLEYD ENPILTLCAARTLAYEGDGYRRAPVNNMPPPPPPPPMGGNSRRPKSKWAIYE ESKQIQALRYSAQGYSVINKYLRGDDYPETQAKETLLSRDYLSTNEPSDEEFKNA MSVYINDIAEGLSSLPETDHRVYRGLKLDKPAKSDVLKEYTTIGNIIIDKAFMST SPDKAWINDTILNIYLEKGHGKRIIGDVAHFKEAEMLFPPNTKLIKIESIVNCGSQ DFASQSLKRLSDDATADTNRIKRIINMRVLNS
CARDS toxin (<i>Mycoplasma pneumoniae</i>)	SEQ ID NO.: 3	MSENLYFQGHMNPVRVYRVDLRSPPEIEFHGFSTLGDVRNFFEHILSTNFGRSY FISTSETPTAARFFGWSLREYVPEHPRAYLYEIRADQHFYNARATGENLLDLNR QRQVVDSDGREMAQMGIRALRTSFAYQREWTFDGPAAANVRSAWLVDAVPEPG HAHHPAGRVVETTRINEPEMHNPHYQELQTQANDQPWLPTPGIATPVHLSIPQAAS VADVSEGTASLSFACPDWSPPSSNGENPLDKCIAEKIDNYNLQSLPQYASSVKEL EDTPVYLRGIKTQKTFMLQADPQNNVFLVEVNPQKQSSFPQTIFFWDVYQRI CLK DLTGAQISLSLTAFQTQYAGQLKVHLSVAVNAVQKWKMTQDIAITQFRVYSEL LGQTENGLFWNTKSGSQHDLYVCLKNPPSDLEELQIIVDECTTHAQFVMTRAAS TFPVDVQLGWYRGYYTPQLSGWSYQMKTPDGQIPYDLKTSKIPFVQDNQNVFPL HNKLNKQTYGSWDWVWLKHDMDNEDKENFKWYFSRDDLTIPSVGEGLNFRHRCYA DNQQLKVIIISGSRWGGWYSTYDKVESNVEDKILVKDGFDRF
SpyA Toxin (<i>Streptococcus pyogenes</i>)	SEQ ID NO.: 4	MLKKRYQLAMILLSCFSLVWQTEGLVELFVCEHYERAVCEGTPAYFTFSDQKGAE TLIKRWGKGLVYPRAEQEAMAAATCQAGPINTSLDKAKGKLSQLTPELRDQVAQ LDAATHRLVIPWNIIVYRYVYETFLRDIGVSHADLTSSYRNHQFNPHLCKIKLGT RYTKHFSMTTALKNGAMTHRPVEVRI CVKKGAKAAPVEPYSAVPSVEVELLPKRC QLEVVGVYVSQDHKKLHIEAYFKGSL
HopU1 (<i>Pseudomonas syringae</i>)	SEQ ID NO.: 5	MNINRQLPVSGERLLTPDVGVSRQACSERHYSTGQDRHDFYRFAARLHVDAQCFG LSIDDLMDKFSKDHFAEHPEYRDVYPEECSAIYMHTAQDYSSHLVRGEIGTPLYR EVNNYLRLQHENS GREAEIDNHDEKLSPHIKMLSSALNRLMDVAARFGTVYRGIRG DLDTIARLYHLDFTGGRYVEPAFMSTTRIKDSAQVFEPGTPNNIAFQISLKGAD I SGSSQAPSEEEIMLPMMSEFVIEHASALSEGKHLFVLSQI
Chelt toxin	SEQ ID NO.: 6	MKTIISLIFIMFPLFVSAHNGNFYRADSRSPNEIKDLGGLYPRGYDDFERGTGPM ISLYDHARGAPSGNTRYDDGFVSTTTDIDSAHEIGQNLISGYTEYIYLIAPAPNL LDVNAVLRYSHPQENEYSALGGIPWTQVIGWYVNVNGVLDNRHNRQFRADLF NNLSPALPSESQYFAGFEPEHPAWRQEPWINFAPPGCCGRNRLTKHINQDQCSNSQ EELVYKQLDLRTQPKVDKLLVNTSSNNIIFPNHDFIREWVDLDGNGDLSYCG FTVDSGSRKIVCAHNNNGFTYSSINISLSYDYGWPKGQRFIDANGDGLVYRCVQ YVWTHLYCSLSLPQYFSLDKDAGYLDAGYNNNSRAWAKVIGTNKYSFCRLTSNGYI CTDIDSYSTAFKDDQGWADSRVWMDIDGNGGDDYCRVYNWTHLRCNLQKGDGLW KRVESKYLDGGYPSLRFKIKMTSNKDNCRIVRNHRVMECAVYVNDGEPHNSLNM PFSLYNKNDIQFIDIDGDNRDDICRYNSAPNTMECYLNQDKSFSQNKLVLYLSAKP ISSLSGSSSKIIRTFNSEKNSSAYCYNAGYGLRCDFFVIY

TABLE 1-continued

Certhrax toxin	SEQ ID NO.: 7	MKEIIRNLVRLDVRSDVDENSKKTQELVEKLPHLEVLELYKNVGGEIYITDKRLTQH EELSDSSSHKDMFIVSSEKGSFPLREHFVFAKGGKEPSLIHAEDYASHLSSEVEVY ELGKAIIRDTPPLNQKELGNPKFINAINEVNNQKEKGVNAKADEEDGRDLLFGKEL KKNLEHGQVLVDLISGNLSEFQHVFAKSFALYYEPHYKEALKSYPALFNYMLEL DQMRFKESIDDDVEKNKNVLDKFWYTRKAESWGVQTFKNWKENLTISEKDIITGYT GSKYDPINEYLKYDGEIIPNIGDDLKKSCKALEKIENQIKNLDAALQKSKITEN LIVYRRVSELQFGKKYEDYNLRQNGIINEEKVMELESNFKGQTFIQHNYMSTSLVQ DPHQSYSDNDRYPILLEITIPGVHGAZIADMSEYPGQYEMLINRGYTFKYDKFSIV KPTREEDKGKEYLKVNLSTYLGNLNREK
EFV toxin	SEQ ID NO.: 8	MSQLNKWQKELQALQKANYQETDNQLFNVYRQSLIDIKRLKVYTENAESLSFSTR LEVERLFSVADEINAILQLNSPKVEKTIKGYSAKQAEQGYGLWYTLQESQNIALS MPLINHDIYIMNLVNAPVAGKRLSKRLYKYRDELAQNVTNNIITGLFEGKSYAEIAR WINEETEASQKALRIARTEAGRTQSVTTQKGYEEAKELGINIKKKWLATIDKHTR RTHQELDGKEVDVDEEFTIRGHSAGKPRMFGVASEDVNCRCTTIEVVDGISPELRK DNESKEMSEFKSYDEWYADIRQNESKPKPNFTELDFGQSDLDQDDSDKWVAGLKP EQVNAMKDYTSDAFAMNKILRNEKYNPREKPYLVNIQNLDDAISKFKLKHDIIT YRGVSANEYDAILNGNVFKFKSTSINKKVAEDFLNFTSANKDGRVVKFLIPKGTQ GAYIGTNSMMKKESEFLLNRNLKYTVEIVDNILEVTILG
ExoT	SEQ ID NO.: 9	MHIQSSQQNPSFVAELSQAAGRLGQVEARQVATPREAQQLAQRQEAPKGEGLLSR LGAALARPFAVIEIWLGLKLGSRAHAATQAPLSRQDAPPAASLSAAEIKQMMMLQKA LPLTLGGLGKASELATLTAERLAKDHTRLASGDGALRSLATALVGIIRDGSLIEASR LQAARLLEQSVGGIALQQWGTAGGAASQHVLSASPEQLREIAVQLHAVMDKVALLR HAYSEVKGPEVDKALADGLVEHFGLEAEQYLGEHPDGPYSDAEVMALGLYTNGEY QHLNRLSLRQGRELDAGQALIDRGMSAAFEKSGPAEQVVKTFRGQTQRDAFEAVKEG QVGHADGYLSTSRDPSVARFAGLGTITTLFGRSGIDVSEISIEGDEQELLYDKGT DMRVLLSAKDGQGVTRRVLEATLGERSGHSEGLLDALDLATGDRSGKPQEQDLR LRMRGLDLA
CdtB	SEQ ID NO.: 10	MKKIICLFLSFNLAFANLENFNVTWNLQGSAAATESKWSVSVRQLVSGANPLDIL MIQEAGTLPRATPTGRHVQGGTPIDEYEWNLGTLSPDRVFIYYSRVDVGANRV NLAIIVSRMQAAEVIVLPPPTTVSRPIIGIRNGNDAFFNIHALANGGTDVGAIITAV DAHFAIMPQVNMWMIAGDFNRDPSTITSTVDRELANRIRVVFPTSATQASGGTLDYA ITGNSNRQQTYTPPLAAILMLASLRSHIVSDHFPVNFRRK
Diphtheria toxin	SEQ ID NO.: 11	MSRKLFAFILIGALLGIGAPPSAHAGADDVDSKSFVMENFSSYHGTPGYVDSI QKGIQKPKSGTQGNYYDDWKGFYSTDNKYDAAGYSVDNENPLSGKAGGVVKVYTPG LTKVLALKVDNAETIKKELGLSLTEPLMEQVGTEEFIKRFGDGASRVVLSLPPFAEG SSSVEYINNWEQAKALSVELEINFETRGRKRGQDAMYEMAQACAGNRVRRSVGSSL SCINLDWDVIRDKTKTKIESLKEHGPIKNKMSESPNKTVSEKAKQYLEEFHQAL EHPSELKTVTGTNPVFAGANYAAVAVNVAQVIDSETADNLEKTTAALSILPGIG SVMGIADGAVHHNTEEIVAQSIASSLMVAQAIPLVGLVLDIGFAAYNFVESIINL FQVVHNSYNRPAYSPGHKTQPFLLHDGYAVSWNTVEDSIIRTGFQGESGHDIKITAE NTPLPAGVLLPTIPGKLDVNKSKTHISVNGRKIRMRCAIDGDVTFCRPKSPVYV NGGVHANLHVAFHRSSSEKIHSEISSDSIGVLGYQKTVDHTKVNSKLSLFFFEIKS
ExoU/VipB	SEQ ID NO.: 12	MKLAIEIMTKSRKLKRNLLLEISKTEAGQYSVSAPEHKGLVLSGGGAKGISYLGMIQA LQBRGKIKNLTHVSGASAGAMTASILAVGMDIKDIKKLIEGLDITKLLDNSGVGF ARGDRFRNILDVIMMQMKKHLESVQQPIPEBQQMNYGILKQKIALYEDKLSRAGI VINNDVDIINLTKSVKDLKLDKALNSIPTELKGAKEQLENPRLTGLDLGRRLREL LPEENKHLIKNLSVVVTNQTKHELERYSEDTPQQSIAQVQVQWSGAHPVLFVPGRN AKGEYIADGGILDNMPEIEGLDREEVLCVKAEGATFEDRVNKAQKSAMEAISWFK ARMDSLVEATIGGKWLHATSVLNREKVYVINDMIYINTGEVTTNTSPTPEQRA RAVKNGYDQTMQLLDSHKQTFDHPMLAILYIGHDKLKDALIDEKSEKEIFEASAH QAILHLQEQIVKEMNDGDYSVQNYLDQIEDILTVDAMDDIQKEKAFALCIKQVN FLSEGKLETYLNKVEAEAKAAAEPSWATKILNLLWAPIEWVVSFLKGPQDFKVEV QPEPVKYSTSENQETVSNQKDINPAVEYRKIAEVRREHTDPSPSLQEKERVGLST TFGGH
HopPtoE	SEQ ID NO.: 13	MNRVSGSSSATWQAVNDLVEQVSERTTLSTTGQYQTAMGRNLNKEKSDADALMTMR AQQYTDSAKRTYISETLMNLADLQQRKIYRTNSGNLRGAIEMTPTQLTDCVQKCRE EGFSNCDIQALEIGLHLRHKLGISDFTIYSNRKLSHNYVVIHPSNAPFGAIVDSW TGQGVVELDFKTRLKFKHREENYAVNANMHEWIERYGQAHVID
HopPtoF	SEQ ID NO.: 14	MGNICGTSGRHVYSPSHTQRITSAPSTSTHVGGDTLTSIHQLSHSQREQFLNMHD PMRVMLGDHDTLFRITDSRYIKNDKLAGNPQSMASILMHEELRPNRFASHTGAQP HEARAYVPKRIKATDLGVPSLNVMTGSLARDGIRAYDHMSDNQVSVKMRLGDFLER GGKVYADASSVADDGETSQALIVTLPGKQKVPVERV
HopPtoG	SEQ ID NO.: 15	MQIKNSHLYSASRMVQNTFNASPKMEVTNAIAKNNEPAALSATQTAKTHEGDSKQG SSNNSKLPPFRAMYAAYLAGSAYLYDKTANNPFLSTTSLHDGKGGFTSDARLNDQ DKARKRYQNNHSTLENKNSLLSPLRLCGENQFLTMDYRAATKIYLSLDLVDTEQA HTSILKNIMCLKGELTNEEAIKKLNPEKTPKDYDLTNSAYISKNKYSITGVKNEE TGSTGYTSRSITKPFVEKGLKHFIKATHGEKALTPKQCMETLDNLLRKSITLNSDS QFAAGQALLVFRQVYAGEDAWGDAERVILKSHYNRGTVLQDEADKIELSRPFSEQD LAKNMFKRNTS IAGPVLHYAYIYIQEKIFKLPPDKIEDLKHKSMAIDLKNLPLTHVK LSNSGVGFEDASGLGDSFTALNATSCVNHARIMSGEPPLSKDDVVLIGCLNAVYD NSSGIRHSLREIARGCFVGAGFTVQDGDFFYKQICKNASKQFYNG

TABLE 1-continued

VopF	SEQ ID No.: 16	MFKISVSQQANVMSTSDTAQRSSLKISIKSICNKSLSKKLHTLAEKCRFSQELKE HTASKKQIVEQATTTVRESSLTKSDSELGSSRSLTSDVLSSSSSHEDLTAVNLED NDSVFVTIESSSELIVKQDGSIPPAPPLPGNIPPAPPLPSAGNIPTAPGLPKQKAT TESVAQTSNDRSKLMEEIRQGVKLRAATPKSSSTEKASADPHSKLMKELINHGAKLK KVSTSDIPVPPPLPAAFASKPTDGRSALLSEIAGESKDRLRKAGSSETLNVSQPTV AESSIPeAYDLLSDMFNLSPKLSETELNTLADSLADYLFKAADIDWMQVIAEQT KGSTQATSLKSQLEQAPEYVKAFCDEILKFPDCYKADVASPESPKAGPSSVIDVA LKRLQAGRNLRFSTIDAKGTNELKKGEALLESAINAARSVMTAEQKSALLSSNVKS ATFKVSELPCEMGFAEQNGKAFAFNALRLAFYSSISQSGDTAQQDIARFMKENLATG FSGYSYLGTSRVAQLEAQLAALTTK
YopJ	SEQ ID No.: 17	MIGPISQINISGGLSEKETSSSLISNEELKNIITQLETDISDGSWFHKNYSRMDVEV MPALVIQANNKYPENMLNLVTSPLDLSIETKNVIENGVRSSRFIINMGEGGIHFSV IDYKHINGKTSLILFEPANFNMSGPAMLAIRTKTAIERYQLPDCHFMSVEMDIQRS SSECGIFSPALAKKLYIERDSLKKIHEDNIKGILSDGENPLPHDKLDPYLPVTFYK HTQGKKRLNEYLNTNPQGVGTVVNKKNETIVNREDNNKSIVDGKELSVSVHKKRIA EYKTLKKV
AvrPtoB	SEQ ID No.: 18	MAGINGAGPSGAYFVGHTDPEPASGGAGHSSSGASSNSPRLPAPDPAPASQARDR REMLLRARPLSRQTREWVAQGMPTAEAGVPIRQESAEAAAPQARAEERHTPEAD AAASHVTEGGRTQPALAGTSPRHTGAVPHANRIVQQLVDAGADLAGINTMIDNAM RRHAIALPSRTVQSILIEHFPHELLAGELISGSELATAFRAALRREVRQOEASAPPR TAARSSVRTPERSTVPPTSTESSSGSNQRTLLGRFAGLMTPNQRRPSSASNASASQ RPVDRSPPRVNPVPTGANRVVMRNHGNNEADAALQGLAQQGVDMEDLRAALERHIL HRRPIPMIDIAALQGVGIAPSIDTGESLMENPLMNLVSVALHRLGPRPARAQAPRP AVPVAPATVSRPDSARATRLQVIPAREYENNVAYGVRLLSLNPAGAVRETVAAP VNRYERQAVVADIRAALNLSKQFNKLRTVSKADAASNKPGFKDLADHPDDATQCL FGEELSLTSSVQQVIGLAGKATDMSESYSREANKDLVFMMDKKLAQFLAGKPEHPM TRETNAENIAYAFRIVP
SdbA	SEQ ID No.: 19	MHKKNYYSLEKEKKTFWQHILDILKAPFRLPGWVVSFFLARNITHVALNPNNIPQ QRLIHLTKTSNRPEDDIVVINFKRPPHKWENDTLIKIANTIAALPFVTPRLRTRL HYDNENDINHVNKLLAEIDALVQGKSKQKYCKGRAFDWSKIHLLGLEFLDPKMRGY VYEQLHEKYGYVSYTTKRKPNIEFFTLKTPDGSSELDVQVTGEDEKKPMGERKFI ITCIARDQNFINWIKDLNLTAKNLGATAISFNRYRGVDYSRGLVWTENNLVDDILAQ VQRLISLGADPKNICLDGMCIGGAVATIAAAKLHEKGMKVKLNNERSFTLSLSLVF GFIVPELQTANWWSPLTYGRFLLAGVVYALLTPLIWLAWPVDVTKAWNRIPAQDK MYSVVVRDKDNGLYDGVHDFCSIASLVDSQINSILYKLSTDPQLTEEEKQILCDD QFHHFKPSQSVLKNPKYKGPFI SRQDLVAELGHREEYTNHDFLDRLREKQFLD RATRVALAEDGEKDIDGISSQLSNNKERPLIIASSGGTGHSATHGIINDLQSKT DNVVITQHHAEYKKNPFISITSVLIRIGVWFTSLPILEDILKGVMRFIGYVPLPSS SIFWDMQSKIQQSETKKENGIEGTGRTRPYVDMLLDIYPEGYEYTAFFNATHLTSSI EDIQTMISFKGHVEEDNRNIVYQNILQRLMHAAKQNTPYTRLISTQALSLGAICDA VKYYNTVFLPVYNAERGTSYQPIAIDQYMTDLPSLGCIFHMNNLEELTSEQRQLM IHAVNMSPEPFKEAHFGKEQGFKAVHNIDPRNNPMIRNAFKDPSLTLYLTKTQSF HFNIVYKKEKQNALPVLNGKEKITIKPHAKIASIMIGSLAANASADYAKVLLNQGYE HIFLFGGLNDSIAARIDQIINSYPAPTRDEIRKKIILLGNQSDVEMAPIMTRSNCV VIRGGGLSVMEQMAMPIMDDKIVLLHHEDNEBGLTSLGLSWEDGNSDKLIEYLSEK GAYAKKTSPLGCSGHLHEAEKSEFKKYHGLKSTETKKKVDLTIPQOETYSLKKEW DRKTGYTESGHILSHQHRFENTIPEVREPFCSEKDLHNNELSSQSLVSVSAG
SidG	SEQ ID No.: 20	MSRSKDEVLEANDSLFGITVQWTGNTDRPSNGMMNFADQQFFGGDVGHASINMKLP VTDKTKQWIEKYCSQTYDQFKKVGKNEKTYEYELKTAKRLIPVELKTQVTRKAQ YDSNGNLVTHEKAYEQIYFDIDWSWWPGRQLQNTEDDMVWEREGKHFEYDEKWKY LQPEQVRHVRGKLGSRKMDYAPTSIHQRDIPTSELEKTRDHKIHTIEEKLNVVKL LQSKIDEMPHTKMSPSMELMFKNLGINVEKLLDETNDNGVDPNTLEAMREYLTNRL KTRKLELETESAEKKEVDSTQVKNKVEDVYDFEYKLNQVRKKMEEVNSQLEKMD SLLHKLEGNSTGPIPYTAIDELMSVLPFLKEELELENGTSLSPKSIENLIDHIDEL KNELASKQEKKNERNLNLIKKYEELCEQYKDEEGLEEALWEEGIDVEEVNSAKKD ISKPAPEIQKLTDLQEQLRNHKESGVKLSSELEETLNSSVKMWTKIDSPCQVISE SSVKALVSKINSTRPVLVKEKEQLPEQEESSLKEAKKAQEELIKIQEFSSQFYSNS SAYMVIGLPPHHQVSLPLAVNGKRGHLHPEAMLKMHLELVAGPEKKEENLHTNNCSL TSEVLVSAGAQHDPLLSHIMGTRALGFFGTQQOVLENAKLTSKTINEGKKSNIPT LVFTASPLDRALGYAMSIYMDPEASKAKQNALGALGVLVGLAKTPGIIIGSLLNPKQ GFNDILNTLNLVYSRNSTGLKVLTLMALPAMIVLAPLAAIQKGVEVIAETIAKPF KLIANLFKQKPESTDEITVSVGSKVAEKEGYSNTALAGLVNSKIKSIDENTIT VEPQKSPQKMIIEFESQLKENPGKVVLSEKAHNAVLKPVSKSDDEALKQKPYDCC NQSVARSQKFAKTRDEIDELVEEVSTSDKTELTSRQEPSMSSTIDEENIDSE HQIETGETSMRI
VpdA	SEQ ID No.: 21	MKTQKEVSQQDKLKDSKSTPLQTKETWFI SDALNITFDPYDFSISVTEQAPMPYR IVESGGGSRILAHIGALDELTRHGLKFTFSGSSAGAMVAAPAYLGYNCSSEIKQII SWFNEDKLLDSPLIFNFNNIKQIFNKGGLSSAKLMRQAANYVILKKVMDIISDEKF KTRFAKQNFLEENIYRCPENITFQTLARIKEICPECELGEKLFITGTNLSTQKHE VFSIDTTPSMALADAIISANLPIAFERICYQGNVYSDGGISNNLPAHCFSEKGHK TTFLLKHKDDVDFSVLALQFDNGLEENALYSQNPPIKWSWLSNTFYSLITGHPNVE NWYEDLQILRRHAHQSIILIKTPTIALTNLTISQDTKKALVESGRATAKYLELHEF YTDYDNIIRHNECLHEKFKPEELDYCVLHSHFELLKKIKQAISSCSQYLEKGYKH YLCELCDNLLPPQLKCFNEGSGTEQPEIKLEKDTIICEKNNSGLTFSMTFFGVPS

TABLE 1-continued

		PLVKTNLNQDSPELKIKLFTGLYPILIQNWQNLCPVSGISGILNSIRMSFVEISSTD TCIKTLIDKLNIEIGHFLIFVFKAAALKNYDKHDFILLLKLNKHLHHSIELIRNKP FHSDDRFYQGWSEFEGHDPKRILEFIKSDDISGLMTILEDKKALPNNKPN
Lpg0969	SEQ ID NO.: 22	MVSLHEHIQKLISECRKLGKDGLDNGTNGLIPELEIDVVPSPAFLGVGNPAPFVNS KTYKLMRTTHEKWVENKTIIVEKSYLLSQPAIKIIGAIVHETGHAFNVAAKIPNTEA NACIFEIEVLMRLFPQVKSPLLLGCTELDMQSYFKSRLTDYNKCVKDCQCLAEMVEF ITHQFKLDEVSTISEKENQIPLLSISNKWPGLFAKKQIAPDMDKLLTSPVTTITPEVK ILFYQLVKEHFHSPETETKLDI
Lpg1978	SEQ ID NO.: 23	MYKIISYLGWRIDMKTENLPQAGQEAQIDKKIHFIWVGHIMPQKNIQVVSEWAEKN PGYETIIWVDKKIAPAKELDLFILDMSKSGITVKDINEEGVCRDSIRHELDQESPN YGMVSDMLRLNLAAEGGIYLDSDILCSAPFPDEIYAPFGFLLSPWSQGANNTLCN DIILCSKGNQIIQQLADAIEQSYIARDSFEFTHEYASMKETKGERIAKTLGVTGPG FLFHQLKKMGI LNDKSEMAIHWELQDQRYLIDGSKPEPDYFYVPPQNNNDASWVP SIRKPGIENMSFQERLENVQLIAPDIQKTGLFNLDHYANELKVQNSWCIAAETS PELKPDSYLLIRPRDKTGEWTLVYVDEDKKLNPNVTLFVIKGAIKLSEVSDPLRKFH TLLSQVSDPVNPATAHELKQIGRALIELKPRQDEWHCKNKWSGAEEIAQELWQRITS NETLRAQIKQCFTQFESLKPRVAELGLTRASGAGTEVEAHESTVKEQEIISQNTVG EEGTKEKNSVQLASENSSDEKIKTAHDLIDEIIQDVIQLDGKLGLLGGNTRQLEDG RVINIPNGAAMIFDDYKKYKQGELETAESALESMIKIAKLSNQLNRHTFFNQRPET GQFYKKAADLDQTTIAEYDNNHGLRI
YopE	SEQ ID NO.: 24	MKISSFISTSLPLPTSVSGSSSVGEMSGRSVSQQTSDQYANNLAGRTESPQGSLLA SRIIERLSSVAHSVIGFIQRMFSEGSHKPVVTPAPTPAQMPSPTSFSDSIKQLAAE TLPKYMQQLNSLDAEMLQKNHDQFATGSGPLRGSITQCQGLMQFCGGELQAEASAI LNTPVCGIPFSPQWGTIGGAASAYVASGVDLTQAANEIKGLAQMQKLLSLM
SptP	SEQ ID NO.: 25	MLKYEERKLNLTLSFSFKVGVSNDRALYIAKENTDKAYVAPEKFSKSVLTWLGKM PLPKNTEVVQKHTEINIRVDQKILQTFHLALTEKYGETAVNDALLMSRINMKNPLT QRLAVQITECVKADEGFINLIKSDNVGVRNAALVIKGGDTKVAEKNNDVGAEK QPLLDIALKGLKRTLPLEQMDGNSLRENFEQEMASGNGPLRLSLMTNLQNLNKIPEA KQLNDYVTTLTNIQVGVARFSPQWGTGCGEVERWVDKASTHETQAVKKIHVIAKEL KNVTAELEKIEAGAMPQMTMSGPTLGLARFAVSSIPINQQTQVKLSDGMPVPVNTL TFDGKPVALAGSYPKNTPDALEAHMKMLLEKECSCLVVLTSQDQMAKQLPPYFRG SYTFGEVHTNSQKVSASQGEAIDQYNMQLSCGEKRYTIPVLHVKNWPDHQPPLPST DQLEYLADRVKNSNQNGAPGRSSDKHLPMIHCLGGVGRGTGTMAALVLKDNPHSN LEQVRADPRDSRNNRMLDASQFVQLKAMQQLLMTTAS
SopE2	SEQ ID NO.: 26	MTNITLSTQHYRIHRSDVEPVKEKTTTEKDI FAKSITAVRNSFISLSTSLSDRFSLH QQTDIPTTHFRGNASEGRAVLTSKTVKDFMLQKLSLDIKGNASKDPAYARQTCE AILSAVYSNNKDQCCKLLISKGVSIPTPLKEIGEAQNAAGLPGEIKNGVFTPGGAG ANPFVPLIASASIKYPHMFINHNOQVSFKAYAEKIVMKEVTPLFNKGMTPTPQQF QLTIENIANKYLQNAS
SopB/SigD	SEQ ID NO.: 27	MQIQSFYHSASLKTQEAFLSKLQKTLYNGMQILSGQGKAPAKAPDARPEIIVLREPG ATWGNVYLQHQKASNHSLNHLNQLRDLTVAATVLGKQDPVLTSMANQMELAKVKA DRPATKQEEAAAKALKKNLIELIAARTQQQDGLPAKEAHRFAAVAFRDAQVKQLNN QPWQTIKNTLTHNGHHYTNQLPAAEMKIGAKDIFPSAYEGKGVCSWDTKNIHHAN NLWMTSVSVHEDGKDKTLFCGIRHGVLSPYHEKDPLLRHVGAENKAKEVLTAALFS KPELLNKALAGEAVSLKLVSVGLLTASNIFGKEGTMVEDQMRAWSLTQPGKMIHL KRNKDGDLQTVKIKPDVAAPNVGVNELALKLGFGLKASDSYNAEALHQLLGNLDR PEARPGGWGEWLAQYPDNYEVVNTLARQIKDIWKNNQHHKDGGEPYKLAQRLAML AHEIDAVPAWNCKSGKDRTGMMDSSEIKREIISLHQTHMLSAPGSLPDSGGQKIFQK VLLNSGNLEIQKQNTGGAGNKVMKNLSPEVLNLSYQKRVGDENIWOQSVKGISSLIT S
SipA	SEQ ID NO.: 28	MVTSVRTQPPVIMPGMQTEIKTQATNLAANLSAVRESATTTLSGEIKGPQLEDFFA LIKQASLDALFKCGKDAEALKEVFTNSNNVAGKKAIMEFAGLFRSALNATSDSPEA KTLMLKVGAEYTAQIIKDGLKEKSAFGPWLPETKKAELKENLEKQLLDI IKNTG GELSKLSNLVMQEVMPYIASCIEHNFPGCTLDPLTRSNLTHLVDKAAAKAVEALDM CHQKLTQEQGTSVGREARHLEMQTLIPLLLRNVFAQIIPADKLPDPKIPPEAAGPVP DGGKKAEPGTGINININIDSSNHSVDNSKHINNSRSHVDNSQRHIDNSNHDNSRKT IDNSRTFIDNSQRNGESHSTNSNVSHSHSRVDSHTHQETATAHSASTGAIDHGIAG KIDVTAHATAEAATNASSEKDGKVVTSSEKGTGTGETTSFDEVDGVTSSKSIIGKPVQ ATVHGVDNKKQSQTAIEIVNVKPLASQLAGVENVKDTLQSDTTVI TGNKAGTTDN DPSQDKTGTFPSGLFKPQNSPLSTVPSVTNMHSMHFDARETFLGVIKALEPDTST PPFVRRADFGLRAEILPNDTIKSAALKAQCSDDIDKHPCLKAKMETLKEVITHHPQK EKLAEIALQFAREAGLTRLKGETDYVLSNVLDGLIGDGSWRAGPAYESYLNKPGVD RVITTVVDGLHMQR
YpkA	SEQ ID NO.: 29	MKSVKIMGTMPSPISLAKAHERISQHWQNPVGELNIGGKRYRIIDNQVLRNLNPHSG FSLFRREGVGKIFSGKMFNFSIARNLTDTLHAAQKTSQELRSDIPNALSNLFGAKP QTEPLGWKGEPLSGAPDLEGMRVAETDKFAEGESHSI IETKDKQLRVAKIERSI AEGHLFAELEAYKHIIYKTAGKHPNLANVHGMAVVPYGNRKEEALLMDEVGWRCS DLRLTADSWKQKGINSEAYWGTIKFIAHRLLDVTHLAKAGVVHNDIKPGNVVDFR ASGEPVVIDLGLHSRSGEQPKGFTESFKAPELGVGNLGASEKSDVFLVSTLLHCI EGFEKNPEIKPNQGLRFITSEPAHVMDENGYPPIHRPGIAGVETAYTRFITDILGVS ADSRPDSNEARLHEFLSDGTIDEESAKQILKDTLTGEMSPLSDTVRRI TPKKLREL SDLLRTHLSAATKQLDMGGVLSDDL TMLVALDKAEREGVDKDQLKSFNSLILKT

TABLE 1-continued

		YRVIEDYVKGREGDTKNSSTEVSYPHRSNFMLSIVEPSLQRIQKHLDDQTHSFSDIG SLVRAHKHLETLLEVLVTLSSQQGPVSSETYGFLNRLAEAKITLSQQLNTLQQQQE SAKAQLSILINRSGSWADVARQSLQRFDSTRPVVKFGTEQYTAIHRQMAAAHAAT LQEVSEFTDDMRNFTVDSIPLLIQLGRSSLMDEHLVEQREKLRELTITIAERLNRLE REWM
YopM	SEQ ID NO.: 30	MFINPRNVSTFLQEPLRHSSNLTEMPVEAENVKSKTEYNAWSEWERNAPPGNGE QREMAVSRLRDCLDRAHELELNNLGLSSLPPELPHLESIVASCNSLTPELPELPQS LKSLQVENNNLKALPDLPSSLKKLHVRENDLTDLPELPQSLESRLVDNNNLKALS LPPSLEYLTASSNKLEELPELQNLPLAAIYADNNLLETLPDLPPSLKKLHVREND LTDLPPELPQSLESQVDNNNLKALSLLPPSLEYLTASSNKLEELPELQNLPLAAI YADNNLLETLPDLPPHLEILVASYNSTELPELPQSLKSLRVDNNNLKALSLLPPS LEYLTASSNKLEELPELQNLPLAAIYADNNLLETLPDLPPSLKKLHVRENDLTDL PELPQSLTFLDVSDDNNSGLSELPPNLYLDASSNEIRSLCDLPSSLVDNLVKSNO LSELPALPPHLERLIASFNYLAEVPELPQNLKQLHVEQNALREFPDIPESLEELEM DSERVVDPYEFAHETTDKLEDDVFE
Amatoxin	SEQ ID NO.: 31	MSDINATRLPIWGICNCPVGGDDVTLLTRGEALC
Phallacidin	SEQ ID NO.: 32	MSDINATRLPAWLVDPCVGGDDVNRLLTRGESLC
Killer toxin KP1	SEQ ID NO.: 33	MIKPERSILTILIGILCLLAYVLANGEPHDGDNEWSSYCSDDQGFRRSDDGLVTTDP VGQESIGKNSINGSELVDYLQCLKVLNGQKQVVSNDGWLILLVQEPSVNVTKAM SECNYNVSSGHKAGSYIQVTNTPADYKVISRRGSYEGDQLPEDVKPYFGVQKTSY RPI SKRINPNLTLRQLAYNFAALNMCSLWCNCSICSRSPYIAELTVHVNNIHGT VWLHHFCRNASPGGGLYSTLTISHKDTAYYVGTGWWKVRSTAATTNDVAGDWYPA SWNQYWCAPHY
Killer toxin KP6	SEQ ID NO.: 34	MLIFSVLMYLGLLLAGASALPNGLSPRNNAFCAGFLSCKWECWCTAHGTGNELRY ATAAGCGDHLSKSYDARAGHCLFSDDLRNQFYSHCSSLNNNMSCRSLSKRTIQDS ATD TVDLGAELHRDDPPPTASDIGKRGRPRPVMCQCVDTTNGGVRLDAVTRAACS IDSFIDGYYTEKDGFCRAKYSWDLFTSQGFYQACLRYSHAGTNCQPDQPYE
Killer Toxin K1	SEQ ID NO.: 35	MTKPTQVLVRSVSILFFITLLHLVVALNDVAGPAETAPVSLPREAPWYDKIWEVK DWLLQQRATDGNWKSITWGSFVASDAGVIFGINVCKNCVGERKDDISTDCGKQTL ALLVSI FVAVTSGHHLIWGNRPVVSQSDPNAGTVARRDISTVADGIDPLDFSALND ILNEHGISILPANASQYVKRSDTAEHTTSFVVNTNNTSLHTDLIHHGNGTYTFTT PHIPAVAKRYVYPMCEHGIKASYCMALNDAMVSANGNLYGLAEKLFSEDEGQWETN YYKLYWSTGQWIMSMKFI EESIDNANNDFEGCDTGH
Killer Toxin K28 (KHR)	SEQ ID NO.: 36	MGHLLAIFSLIIAVLNIATAVASSDSIYKGHVRGQDIDSILYRVYDNGTMYPTFNE WLNDLTGMNDLATNNATILKRDSSDVSVCVTETCQYVDYHVDDEGVI TIDISTYRIP VEWDSGAGNASYGVSKRDTKYETFCKKKICGINVSGFCNAYDAVHAFDEGGSVY NPVSGITDRIKEATKRDKTECLGYELDHVRIDPAVDWSISISTWKQGSANCDTQAS ADSLKCAAQKALESEHNHQTAFCHLDNGGSFNLDIRLISELSFKYNPWALPCP KYKGSNSWQVVSDFCQ
Killer Toxin K28 (KHS)	SEQ ID NO.: 37	MPRFAII FALLIAYSFLSTLFTGSI PDRANTVTSNAPCQVVIWDWIRTRRICNC SRLCYSLLGRSNLSRTAKRGVCTIAGAVLATAAIVAAVLVGKSSGSATKRLTKT ISVLNHTIPFTDHLNGQTLNNGTGSNFVTIGFSYAVHATIKRATTDI ISWVIP ESMEPTLARVASVSSSSINLAAPDPTGGNASALS PQNAVQEFATSWVSMTYDQSY GDLRNVANDEGGEEIILMRKRSYRISFQVIE TGSTALLLRTRRVVSQILITMTYLV TVQARVGIQIGDIFQHYGGIDNYVMTSISVLRTELDKAFHENKLLIVREPPNKSNO DANQSYRLRPFSANDLIQNLKSVDIGFLAFCSFFDKYAHYPEIIMMKITIFISKGN LWSIIYVIQARYVRKRVKVRGQMPGGLLTNMESLLNIVSTPNLNI SEFHIQTHSM SQSKPMYFQKQCYSSQNNIIYIYNSIHITCGAVYIVHDVTRTPSVFVLIELRNCKP LKNWCETTKTSRDTKI KKNENETVCRAGALLDGRVTRIRFLMMRTHWSRVKG VSCNTANRLSRFCNHVVSYPQNAIHLPTS LRAESLEQQYTRPLSSSNRFC CLKSIFINNCKKACESPSLVSNLQQTAELLMVYYLYICEACYVSRNHDLKQCM STVRAYVVARMLPKERSTFPCMPRLCWLNVGVVVV
Anthrax lethal factor endopeptidase	SEQ ID NO.: 38	MHVKEKEKNDENKRKDEERNKTQEEHLKEIMKHIVKIEVKGEEAVKKEAAEKLE KVPSDVLEMYKAIGGIYIVDGDITKHISLEALSEDKKIKIDYIGKDALLHEHYVY AKEGYEPLVVIQSSSEYVENTEKALNVYYEIGKILSRDILSKINQPYQKFLDLVNT IKNASDSGQDQLFTNLQKEHPTDFSVFLEQNSNEVEQVFAKAFAYYIEPQHRDV LQLYAPEAFNYMDKFNEQEINLSLEELKQDQMLSRYEKWEKIKQHYQHWSDSLSE GRLLKKLQIPIEPKKDDIIHLSQEEKELLKRIQIDSSDELSTEKEFLKKLQID IRDSLSEEEKELLNRIQVDSNPLSEKEKEFLKKLKLIDIPYDINQRLQDTGGLID SPSINLDVRKQYKRDIQNDALHQSIGSTLYNKIYLYENMNINNLATLGLADLVD STNTKINRGI FNEFKKNFKYISSNYMIVDINERPALDNERLKWRIQLSPDTRAG YLENGKILQRNIGLEIKDVQIIKQSEKEYIRIDAKVVPKSKIDTKIQEAQLNINQ EWNKALGLPKYTKLITFNVHNRYASNIVESAYLILNEWKNNIQSDLIKVKVTNYLVND GNGRFVFTDITLPNIAEQYTHQDEIYEQVHSGKGLYVPESRSILLHGSPKGVELRND SEGFIHEFGHAVDDYAGYLLDKNQSDLVNTNSKKFIDIFKEEGSNLTSYGRTNIEAF FAEAPRLMHSTDAERLKVQKNAPKTFQFINDQIKFIINS

TABLE 1-continued

Shiga Toxin	SEQ ID NO.: 39	MKCILLKWVLCLLGFSSVSYSREFTIDFSTQQSYVSSLNSIRTEISTPLEHISQGTTSVSVINHTPPGSGYFAVDIRGLDYYQARFDHLRLIEQNLYVAGFVNTATNTFYRFSDFAHISVPGVTTVSMTTDSSYTTLQ RVAALERSGMQISRHSVSSYLALMEFSGNTMTRDASRAVLRFFVIVTAEALRFQIQREFRQALSETAPVYTMTPGDVDLTLNWGRISNVLPEYRGEDGVRVGRISFNNSAILGTVAVILNCHHQGARSVRVAVNEESQPECQITGDRPVIKINNTLWESNTAAFLNRKSQSLYTTGE
Saporin Toxin	SEQ ID NO.: 40	MKSWIMLVVTLIIILQTTVTAVIIYELNLQGTTKAQYSTFLKQLRDDIKDPNLHYGTGNLPVIRKPVGPPKFLRVNLKASTGTVSLAVQSRNLYVAAYLAKNNKQFRAYYFKGFQITTNQLNNLFPEATGVSNQOELGYGESYPQIQNAAGVTRQQAGLGKKLAESMTKVNGVARVEKDEALFLLIIVVQMVGEAARFKYIENLVLNNFDTAKEVEPVDPDRVILENNWGLLSRAAKTANNGVFQTPVLVTSYAVPGVEWRVTTVAEVEIGIFLNVDDNGLPSIIYNNIISGAFGDTY
Ricin Toxin	SEQ ID NO.: 41	MYAVATWLCFGSTSGWSFTLEDNNIFPKQYPIINFTTAGATVQSYTNFIRAVRGRLTTGADVDRHDIPVLNVRVGLPINQRFILVELSNHAEISVTLALDVTNAYVVGVRAGNSAYFFHPDNQEDAEAI THLFTDVQNRVYTFAGGNYDRLEQLAGNLRENIELGNGLGTEAISALYYSTGGTQLPTLARSFII CIQMI SEARFQYIEGEMRTRIRYNRRSAPDPSVITLNSWGLRLSTAIQESNQGAFASPIQLQRNGSKFSVYDVSILPIIALMVYRCAPPPSSQFSLIRPVVNFNADVCMDEPIVIRVGRNGLCVDVDRDGRFHNGNAIQLWPCKSNTDANQLWTLKRDNTIRSNKCLTTYGSPGVYVMIYDCNTAATDATRWQLWDNGTI INPRSSLVLAATSGNSGTTLTQVQTNIIYAVSQGWLPNTNTQPFVTTIVGLYGLCLQANSQQVWIEDCSSEKAEQQWALYADGSI RPPQNRDNCCLTSDSNIRETVVKILSCGPASSGQRWFMKNDGTILNLYSGLVLDVRRSDPSLKQI ILYPLHGDPNQIWLPLF

In some embodiments, the death agent is an overexpressed product of genetic element selected from DNA or RNA. In some embodiments, the genetic element is a Growth Inhibitory (GIN) sequence such as GIN11.

In some embodiments, the death agent is a ribosomally encoded xenobiotic agent, a ribosomally encoded poison, a ribosomally encoded endogenous or exogenous gene that results in severe growth defects upon mild overexpression, a ribosomally encoded recombinase that excises an essential gene for viability, a limiting factor involved in the synthesis of a toxic secondary metabolite, or any combination thereof. In some embodiments, the ribosomally encoded death agent is Cholera toxin, SpvB toxin, CARDS toxin, SpyA Toxin, HopU1, Chelt toxin, Certhrax toxin, EFV toxin, ExoT, CdtB, Diphtheria toxin, ExoU/VipB, HopPtoE, HopPtoF, HopPtoG, VopF, YopJ, AvrPtoB, SdbA, SidG, VpdA, Lpg0969, Lpg1978, YopE, SptP, SopE2, SopB/SigD, SipA, YpkA, YopM, Amatoxin, Phallacidin, Killer toxin KP1, Killer toxin KP6, Killer Toxin K1, Killer Toxin K28 (KHR), Killer Toxin K28 (KHS), Anthrax lethal factor endopeptidase, Shiga Toxin, Saporin Toxin, Ricin Toxin, or any combination thereof.

Along with one or more positive selection markers, a plasmid (such as PLASMID 2) can also include one or more negative selection markers under control of a different DNA binding sequence to enable binary selection. The plasmid (e.g., PLASMID 2) can encode for one or more of negative selection markers in Table 1 driven by a promoter which depends on the DBD present in the PPI integration plasmid—DNA Binding Sequence (DBS), for example, the LexAop sequence (DBS) which can become bound by LexA (DBD). In some embodiments, to ensure repression of the ‘death agents,’ the plasmid (e.g., PLASMID 2) can include a silencing construct such as a TetR¹-Tup11 fusion driven by a strong promoter (such as ADH1) to bind the DBD and silence transcription in the presence of doxycycline. The plasmid (e.g., PLASMID 2) can comprise bacterial selection and propagation markers (i.e. ori and AmpR), and yeast replication and selection markers (i.e. LEU2 and CEN or 2 um) as well.

FIG. 7 shows two platforms to identify a disrupting compound in cell culture using both a negative and a

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positive selection marker. In the first platform, KRas and KRas(G12D) fused to DBDs and c-Raf fused to AD were expressed in yeast cells. In the absence of inhibitors, the KRas(G12D) and c-Raf maintain an interaction to drive expression of a cytotoxic reporter, resulting in a low amount of cell growth/survival. A nutritional reporter was controlled by KRas and c-Raf interaction. The cells were patched onto selective media for a nutritional marker with or without inhibitor and visualized after 4 days of growth at 30° C. In cell populations with KRas(G12D) and c-Raf interaction, only those with the inhibitor showed enhanced cell viability, illustrating the specificity of the inhibitor for the KRas (G12D) and c-Raf interaction.

In the second platform, VGLL4 and YAP fused to DBDs and TEAD fused to AD were expressed in cells. In the absence of inhibitors, the YAP and TEAD maintain an interaction to drive expression of cytotoxic reporter. A nutritional reporter was controlled by VGLL4 and TEAD interaction. The cells were patched onto selective media for a nutritional marker with or without inhibitor and visualized after 4 days of growth at 30° C. In cell populations with YAP and TEAD interaction, only those with the inhibitor showed enhanced cell viability, illustrating the specificity of the disruptor to YAP and TEAD interaction.

In some embodiments, the host cell can further comprises more than one sequence for expressing a positive control reporter that is activated by a promoter DNA sequence specific for a DNA binding moiety. In some embodiments, the host cell further comprises more than one sequence for expressing a death agent that is activated by a promoter DNA sequence specific for a DNA binding moiety.

FIGS. 4A and 4B illustrate a platform with more than one positive selection markers and a negative selection marker for identifying a compound that disrupts a protein-protein interaction within a complex in a specific manner. The multiple selection markers can reduce false positive rate due to mutations that leads to avoidance of selection. DBD1 and DBD2 are promoter specific DNA-binding domains. AD refers to an activation domain. Prey, BaitWT, and BaitMut refer to three proteins, wherein BaitWT and BaitMut each interacts with Prey. Broken arrows indicate active expression of the reporter. Markers 1 and 2 refer to positive

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selection markers, toxins refers to death agent. The ring comprising smaller circular items illustrates a potential inhibitor (e.g., a randomly produced peptide). Two scenarios are shown; FIG. 4A illustrates a case where a peptide is unable to disrupt the pairwise interaction of interest, and death agent is expressed, leading to cell death. FIG. 4B illustrates a case where a peptide is able to disrupt the Prey and BaitMut interaction by acting on the Prey without disrupting the BaitWT and Prey interaction. Peptide disruption activity is assayed by survival, and specificity is assayed by growth on the expressed positive selection reporter, marker 2.

FIGS. 5A and 5B show a platform that is analogous to the platform of FIGS. 4A and 4B, where the inhibitor binds to the bait instead of the prey. More specifically, the platform of FIGS. 5A and 5B has more than one positive selection marker and one negative selection marker for identifying a compound that disrupts a protein-protein interaction within a complex in a specific manner. The multiple selection markers can reduce false positive rate due to mutations that leads to avoidance of selection. DBD1 and DBD2 are promoter specific DNA-binding domains. AD refers to an activation domain. Prey, BaitWT, and BaitMut refer to three proteins, wherein BaitWT and BaitMut each interacts with Prey. Broken arrows indicate active expression of the reporter. Markers 1 and 2 refers to positive selection markers, toxins refers to death agent. The ring comprising smaller circular items illustrates a randomly produced peptide. Two scenarios are shown; FIG. 5A illustrates a case where a peptide is unable to disrupt the pairwise interaction of interest, and death agent is expressed, leading to cell death. FIG. 5B illustrates a case where a peptide is able to disrupt the Prey and BaitMut interaction by acting on the BaitMut without disrupting the BaitWT and Prey interaction, and peptide disruption activity is assayed by survival and specificity is assayed by growth on the expressed positive selection reporter, marker 2.

A plasmid (e.g., PLASMID 3) can be used to confirm expression of the reporters and the successful construction of the strains. PLASMID 3 can include a direct fusion between the AD and one or multiple DBDs. The plasmid (e.g., PLASMID 3) can further include bacterial selection and propagation markers (i.e. ori and AmpR), and yeast replication and selection markers (i.e. TRP1 and CEN or 2 um).

FIG. 10 illustrates a confirmation plasmid. The depicted plasmid show the integration of two bait-prey fusion proteins, each with its own DBD. Protein tags may be included to enable detection of the proteins. The plasmid may also include propagation and selection markers for growth in hosts.

Disclosed herein, in certain embodiments, is a library of plasmid vectors, each plasmid vector comprising a DNA sequence encoding a different peptide sequence operably linked to a first switchable promoter; a DNA sequence encoding a death agent under control of a second switchable promoter; and a DNA sequence encoding a positive selection reporter under control of a third switchable promoter. Addition or Expression of Modulators

A molecule from a library that can selectively disrupt or facilitate PPI of interest can be screened by via use of positive and/or negative selection markers in a host cell.

In some embodiments, the molecule is small molecule. In some embodiments, the small molecule is peptidomimetic. The host cell can be made to become permeabilized to small molecules, for example by deletion of drug efflux pumps, such as PDR5, ERG6, or 12geneΔOHSR (*Chinen, 2011*), to

enable a small molecule screening approach. The host cell can additionally carry mutations to enable more efficient transformation with vectors and/or more efficient uptake small molecules.

In other embodiments, the molecule is peptide or protein. In some embodiments, the peptide or protein is derived from naturally occurring protein product. In another embodiment, the peptide or protein is synthesized protein product. In other embodiments, the peptide or protein is a product of recombinant genes.

In some embodiments, the molecule is introduced to the host cell exogenously. In other embodiments, the molecule is the expression product of test DNA inserted into the host cell, wherein the test DNA comprises of DNA sequences that encodes a polypeptide. Libraries can be formed by delivery of a plurality of test DNA molecules into host cells. In some embodiments, the peptide sequences of the polypeptides in the library are random. In some embodiments, the different peptide sequences are pre-enriched for binding to a target.

To screen for peptides that selectively disrupt or facilitate a PPI of interest, peptides from a randomized peptide library can be applied to the host cell. PLASMID 2 can be further used to express a randomized peptide library (such as a randomized NNK 60-mer sequences). PLASMID 2 can include a restriction site for integration of a randomized peptide library driven by a strong promoter (such as the ADH1 promoter) or an inducible promoter (such as the GAL1 promoter).

In some embodiments, the randomized peptide library is about 60-mer. In some embodiments, the randomized peptide library is from about 5-mer to 20-mer. In some embodiments, the randomized peptide library is less than 15-mer.

The library can also initiate with a fixed sequence of, for example, Methionine-Valine-Asparagine (MVN) for N-terminal stabilization and/or another combination of high-half-life N-end residues (see, for e.g., Varshavsky. *Proc. Natl. Acad. Sci. USA.* 93:12142-12149 (1996)) to maximize the half-life of the peptide, and terminate with the 3'UTR of a short protein (such as sORF1). The peptide can also be tagged with a protein tag such as Myc. In some embodiments, N-terminal residues of the peptide comprise Met, Gly, Ala, Ser, Thr, Val, or Pro or any combination thereof to minimize proteolysis.

The plurality of different short peptide sequences can be randomly generated by any method (e.g. NNK or NNN nucleotide randomization). The plurality of different short peptide sequences can also be preselected, either by previous experiments selecting for binding to a target, or from existing data sets in the scientific literature that have reported rationally-designed peptide libraries.

In some embodiments, the library comprises polypeptides about 60 amino acids or fewer in length. In another embodiment, the library comprises polypeptides about 30 or fewer amino acids in length. In another embodiment, the library comprises polypeptides about 20 or fewer amino acids in length.

Modification of Disrupting or Facilitating Peptides

The peptide that disrupts or facilitates PPI can also be a product of post-translational modification. The post-translational modification can include any one or combination of cleavage, cyclization, bi-cyclization, methylation, halogenation, glycosylation, acylation, phosphorylation, and acetylation. In some embodiments, the methylation comprises reacting with an N-methyltransferase. In some embodiments, the post-translational modification is done by naturally occurring enzymes. In some embodiments, the post-

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translational modification is done by synthetic enzymes. In some embodiments, the synthetic enzymes are chimeric.

The peptide can be ribosomally synthesized and post-translationally modified peptide (RiPP) whereby the core peptide is flanked by prepeptide sequence comprising a leader peptide and recognition sequences which signal for the recruitment of maturation, cleavage, and/or modifying enzymes such as excision or cyclization enzymes including, for example, lanthipeptides maturation enzymes from *Lactococcus lactis* (LanB, LanC, LanM, LanP) patellamide biosynthesis factors from cyanobacteria (PatD, PatG), butelase 1 from *Clitoria ternatea*, and POPB from *Galerina marginata*, *Lentinula edodes*, *Omphalotaca olearis*, *Dendrothele bisporea*, or *Amanita bisporigera*, or other species. In some embodiments, the cyclization or bicyclization enzymes are synthetic chimeras.

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In one example, as illustrated in FIG. 11, the variable peptide library region is embedded within the primary sequence of a modifying enzyme (e.g., the homolog of the omphalotin N-methyltransferase enzyme from *Dendrothele bisporea*, *Marasmius fiardii*, *Lentinula edodes*, *Fomitiporia mediterranea*, *Omphalotus olearius* or other) and contains random residues, some of which may be post-translationally decorated by additional modifications like hydroxylation, halogenation, glycosylation, acylation, phosphorylation, methylation, acetylation. This diversified variable region is excised and modified to form N-to-C cyclized, optionally backbone N-methylated macrocycles by the action of a prolyl endopeptidase belonging to the PopB family and N-methyltransferases belonging to the omphalotin methyltransferase family. An exemplary list of prolyl endopeptidases is shown in Table 2. An exemplary list of N-methyltransferases is shown in Table 3.

TABLE 2

Amino acid sequences of prolyl endopeptidase type cyclizing enzymes		
<i>Galerina marginata</i> CBS 339.88	SEQ ID NO.: 42 KDR68475.1 hypothetical protein GALMADRAFT_78538	MSSVTWAPGNYPSTRSDHVDITYQSASKGEVPPDPYQWLEESTDEVDKW TTAQADLAQSYLDQNADIQKLAEKFRASRYAKFSAPTLLDDGHWFYFN RGLQSQSVLYRSKEPALPDFSKGDDNVGDVFFDPNVLADGSGAMVLCFK SDGKFFAYAVSHLGGDYSTIYVRSTSSPLSQASVAQGVDRSLSDVKKWF KFSTIIWTKDSKGFLYQRYPARERHEGTRSDRNAMMCYHKVGTQTEEDII VYQDNEHPEWIIYGADTSEDGKYLILYQFKDTSKNLLWVAELDEDGVKSG IHWKVVNEYAADYNIITNHGSLVYIKTNLNPAPQYKVIITIDLSKDEPEIR DFIPEEKDAKLAQVNCANEEYFVAIYKRNKDEIYLYSKAGVQLTRLAP FVGAASIANRQKQTHFFLTLSGFNTPGTIARYDFTAPETQRFSLRITTKV NELDPDDFESTQVWYESKDGTKIPMFIVRHKSTKFDGTAAAIQYGYGGFA TSADPPFPSPILTLFLQTYGAIFAVPSIRGGGEGFEEWHKGGRRRETKVNT DDFIAAAQFLVKNKYAAPGKVAINGASNGLLVMGSIYRAPEGTFGAAVP EGGVADLLKFHKFTGGQAWISEYGNPSPIEEFDIYIYPLSPVHNVRTDKVM PATLITVNIQDGRVPMHSPKFIATLQHNVPQNPHPLLIKIDKSWLGHGM GKPTDKNVKDAADKWGFARALGLELKTVE
<i>Amanita bisporigera</i>	SEQ ID NO.: 43 ADN19205.1 prolyl oligopeptidase	MPPTPWAPHSYPSTRSDHVDVYQSASRGEVPPDPYQWLEENSNEVDEW TTAQTAFTQGYLDKNADRLQKLEEKFRASKDYVFKFSAPTLLDSGHWFYFN SGVQSQAVALYRSKPKVLPDFQGRTRKVGEVYFDPNVLSDAGTAIMGTCRF SPSGEYFAYAVSHLGGDYFTIYVRPTSSSLSQAPEAGGDRSLSDGVKWC KFTTIWTKDSKGFLYQRYPARESLVAKDRDKDAMVCYHRVGTQLEDII VQDQKENDPWTYGTDASEDGKYLILVYKDKASKQNLWVAEFDKGVKPE IPWRKVINEFGADYHVI TNHGLIYVKTNNVNPAPQYKVVITIDLTGEPEIR DFIPEQKDAKLQVCKVNGYFVAIYKRNKDEIYLYSKAGDQLSRLASD FIGVASITNREKQPHSFLTFSGFNTPGTISRVDFTAPDTQRLSILRTTKL NGLNADDFESTQVWYKSKDGTKVPMFIVRHKSTKFDGTAPAIQNGYGGFA ITADPPFPSPIMLTQMITYGAILAVPNIRGGGEGFEEWHKAGRRRETKNFT DDFIAAAQFLVKNKYAAPGKVAITGASNGGFLVCGSVVRAPEGTFGAAVS EGGVADLLKFHKFTGGMAWTSEYGNPFIKEDFDVQALSPVHNVPKDRVL PATLLMTNAGDDRVPVPMHSLKVFANLQYNVPQNPHPLLIRVDSKSWLGHGF GKTTDKHTKDAADKWSFVAQSLGLEWKTVD
<i>Hypsizygus marmoreus</i>	SEQ ID NO.: 44 KYQ30898.1 Prolyl endopeptidase	MAISPTPWTPTNYPSTRSSHVDIYKSATRGEVRVADPYQWLEENTEETD KWTTAQEEFTRSYLDKNTDRQLEDAFRTSTDYAKFSPTLYEDGRWYWF YNSGLQPPQLIYRSKGTLPDFSQDDNVVGEVFFDPNLLSDDGTAALSII DFSDCGYFAYGISFSGSDFSTIYVRSTESPLAKNSGSTDDRLSDEIK HVKFSAVTWTKDSKGFFYQRYPAHENAKEGIEGTGGVDAMIYYHVIQTSQ SEDILVHSDKSNPEWMSIDITEDGKYLILYTMKSSSRKNLMWIAELSKN EIGPNIQWNKI IDVFDAEYHLITNDGPILYVKTNADAPQYKLVMTDISGD KDISRDLIPEDKNANLVQVDCVNRDTFAVIYKRNKDEIYLYSKTGILQSL RLASDFVGAASISSREKQPHFFVTMTGFSTPGTVARYDFGAPEEQRWSII RSVKVNLNPDFFESKQVWYESKDGTKIPMFIVRHKATKFDGTAPAIQYGY YGGFISISINPFFSPTILTLFLQTYGAVLAVPNIRGGAEFGEDWHKAGTREK KGNVFDDFVAAATQYLVKNKYAGEGKVAINGGSSNGLLVGACINRAPEGTF GAFAEVGVMDLLKFSKFTTGKAWTSDYGDPPDKDFDFICPLSPLHNIP TDRVLPTTMLLTADHDDRVPVPMHSPKHAATLQYTLPHNPHPLVIRIDKKA GHGAGKSTEKIRIKESADKWGFVAQSLGLVWQPEA
<i>Conocybe apala</i>	SEQ ID NO.: 45 ACQ65797.1 prolyl oligopeptidase	MPPSTPNEYPPSTRSDVLTYSRSEKNGEVVPPDPYQWLEHNTEETDKWTT AQAAFTRAHLDKNPKRNAMEEAFANDYAKFSAPQLHDDGRWYWYNTG LQAQTCLEWRTRDDTIPDFSKQLEDVGEIYFDPNALS KDGTAAALSTYRFS RDGKYFAYAIQSGSDFNTIYVRPTDSPLTKRDESGRDPRLADEVKFVK FSGITWAPNSEGFFYQRYPHIDGATLEEGLIATRRDLHAMVYHVRVGTPO SEDILIHDPANPEWMPGVNVTDNGEYIELYISKSSSRKNMLNVANFAMN KIGEQFQWRKVINDFAAEYDVI TNHGPVYFRTDDGAPKHKILSINIDTN ERKLLVPESEDAALFSTVCVNKNYMALIYKRNKDEVHLYTLTEGKPVRRLL

TABLE 2-continued

Amino acid sequences of prolyl endopeptidase type cyclizing enzymes		
		AEDFVGACTISGKEKQPWFVVTMSGFTSPSTVGRYNFQIPEENRWSIFR AAKIKNLNPNDFEASQVWYKSKDGTNVPFIVRHKSTQFDGTAPALQYGY GGFSISIDPFFSASILTFLKVYGAILVVPISIRGGNEFGEEWHRGGMKQNK VNCFDDFIAATNHLVEHKYAPGKVAINGGSSNGGLLVAACINRAPEGTFG AAIAEVGVHMDMLKFHKFTIGKAWTSDYGNPDDPHDFDYIYPISPVHNVPT DKILPPTLLLTADHDDRVPVPMHTFKLAATLQHTLPHNPHPLLLRVDDKAG HGAGKPLQLKIREQADKWGFVAQSFQLVVRDGV
<i>Amanita bisporigera</i>	SEQ ID NO.: 46 GenBank HQ225841.1 POPB	MPPTPWAPHSPPTRRSDHVDVQASRGEVPVDPYQWLEENSNEVDEW TTAQTAFTQGYLDKNADRKLEEKFRASKDYVKFSAPTLLDSGHWYWFYN SGVQSQAVLYRSKKPVLPDFQGRGTRKVGVEVYFDPNVLSADGTAIMGTCRF SPSGEYFAYAVSHLGVDFYTIYVRPTSSSLSQAPEAEGGDRLSDGVKWC KFTTITWTKDSKGFYQRYPARESLVAKDRDKDAMVCYHRVGTQLEDII VQQDKENPDWYGTDAEDGKYIYLVVYKDAKQNLWVAEFDKDGVKPE IPWRKVINIEFGADYHVI TNHGSILYVKTNVNAPQYKVVTIDLSTGEPIR DFIPEQKDAKLTQVKCVNKGYPVAIYKRNVDKEIYLYSKAGDQLSRLASD FIGVASITNREKQPHSFLTFSGFNTPGTISRVDFTAPDTQRLSILRTTKL NGLNADDFESTQVWYKSKDGTKVPFIVRHKSTKFDGTAPAIQNGYGGFA ITADPFFSPIMLTFMQTYGAILAVPNIRGGGEFGGEWHKAGRRETKGNTF DDFIAAQFLVKNKYAAPGKVAITGASNGGFLVCGSVVRAPEGTFGAASV EGGVADLLKFNKFTGGMAWTSYGNPFIKEDDFVQALSPVHNVPKDRVL PATLLMTNAGDDRVPVPMHSLKFVANLQYNVPQNPHPHLLIRVDKSWLGHGF GKTTDKHTKDAADKWSFVAQSLGLEWKTVD
<i>Lentinula edodes</i>	SEQ ID NO.: 47 GenBank GAW09065.1 The DOE Joint Genome Institute (JGI) 011197; LENED_011197)	MFSATQESPTMSVPQWDPYPPVSRDETSAITYQSKLCGSVTVRDPYSALE VPFDDSEETKAPVHAQRKFARTYLDEIPDRETWLQTLKESWNYRRFTVPK RESDGYTYFEYNDGLQSQMSLRVKVSEEDTILTESGPGGELFFDPNLLS LDGNAALTGSMMSPCGKYWAYGVSEHGSWMTTYVRKTSPPHMPQSQEKKG DPGRMDDVIRYSRFFIVYSSDSKGFYSRYPPEDDEGKGNTPAQNCMVY YHRLGEKQEKDTLVYEDPEHPFWLWALQLSPSGRYALLTASRDASHTQLA KIADIGTSDIQNGIQWLTIHQWQARFVIIGDDSTIYFMTNLEAKNYLV ATLDIRHSEAGVKTLVAENPDALLISASILSTDKLVLVYLHNARHEIHVH DLNTGKPIRQIFDNLIGQFSLSGRRDDNDMFVHSGFTSPGTIYRFRINE DSNKGTLFRAVQVPGNLNSDFTTESVFYPSKDGTPIHMFITRLKDTVPDVG TAPVYIYGYGGFALAMLPTFSVSTLLFCKIYRAMYVVPNIRGGSEFGESW HREGMLDKKQNVFDDFNAATKWL VANKYANKYNVAIRGGSNGGVLTTACA NQAPELYRCVITIGGII DMLRFPKFTFGALWRSEYGDPEDEDFDIYKY SPYHNIPSGDVVLPAMLFFTAAYDDRVSPHLSFKHVAALQYNFPNGPNPV LMRIDLNTGHFAGKSTQKMLEETADEYRCDLLCCNLQL
<i>Omphalotaceae olearis</i>	SEQ ID NO.: 48 The DOE Joint Genome Institute (JGI) 2090; OMPOL1_2090	MSFPGWGPYPVERDETSAITYSSKLHGSVTVRDPYSQLEVPFEDSEETK APVHSQRKFARTYLDENPDREAWLETLLKSWNYRRFSALKPESDGHYYFE YNDGLQSQSLYRVRMGEEDTVLTESGPGGELFFPNPILLSLDGNAALTGF VMSPCGNYWAYGVSEHGSWMSIYVRKTSPPHLPQSQERKDPGRMNDKIR HVRFFIVSWTSDSKGFFYSRYPPEDDEGKGNAPAMNCMVYHYRIGEDQES DVLVHEDPEHPFWISSVQLTPSGRYILFAASRDASHTQLVKIADLHENDI GTNMKWKNLHDPWEARFTIVGDEGSKIYFMTNLKAKNYKVATFDANHPDE GLTTLIAEDPNAPFLVSASIIHAQDKLLLVYLRNASHEIHIRDLTTGKPLGR IFEDLLGQFMVSGRRQNDIFVLFSFLSPGTVYRYTFGEEKGHSSLFRA ISIPGLNLDDFMTESVFYPSKDGTSVHMFITRDKDVLDDGTSPLVLYQYGYG GFSLAMLPTFSLSTLLFCKIYRAIYAI PNIRGGSYGESWHREGMLDKKQ NVFDDFNAATEWLIANKYASKDRIAIRGGSNGGVLTTACANQAPGLYRCV ITIEGIDMLRFPKFTFGASWRSEYGDPEDEDFDIYKYSPYHNIPPPG DTIMPAMLFFTAAYDDRVSPHLSFKHVAALQHNFPKGNPNCLMRIDLNSG HFAGKSTQEMLEETADEYRLKVQ

TABLE 3

Amino acid sequences of N-methyltransferases		
<i>Lentinula edodes</i>	SEQ ID NO.: 49 GenBank GAW09067.1 The DOE Joint Genome Institute (JGI) 011194; LENED_011194)	METPTLNKSGSLTIVGTGIESIGQMTLQTLSTYIEAADKVFCVIDPATEAF ILTKNKDCVDLYQYYDNGKSRMDTYTQMSEVMLREVRKGLDVGVFYGHGPG VFVNP SLRALAIKASEGFKARMLPGVSAEDCLYADLCIDPSNPGCLTYEAS DFLIRERPTNIYSHFIFLQVGCVGIADEFNFTGFENSKFGLVDRLEKEYGA EHPVVHYIAAMLPHEDPVTQWTIGQLREPEFYKRVGGVSTFYIPPKERKE INVDIIRELKFLEPGKVPDTRTQIYPPNQWEPEVPTVPAYGSNEHAAIAQL DHTTPPEQYQPLATS KAMTDVMTKLALDPKALAEYKADHRAFAQSVPDLTA NERTALEIGDSWAFRCAMKEMPISLLDNAKQSMEEASEQGFPWIIIVGVGV VVGSVVSSA

TABLE 3-continued

Amino acid sequences of N-methyltransferases		
<i>Omphalotaceae</i>	SEQ ID	METSTQTAKAGSLTIVGTGIESIGQMTLQALSYLEAAKVFYCVIDPATEAF
<i>olearis</i>	NO.: 50	ILTKNKNVDLYQYYDNGKSRNTYTQMSSELMVREVRKGLDVGVFYGHGP
	The DOE	VFVNPSHRALAIKASEGYRARMPLPGVSAEDCLFADLCIDPSNPGCLTYEAS
	Joint	DFLIRDRPVSIIHSHLVLFQVGVCIADFNFTGFDNNKFGVLVDRLEQEYGA
	Genome	EHPVVHYIAAMMPHQDPVTDKYTVAQLREPEIAKRVGGVSTFYIPPKARKA
	Institute	SNLDIIRRLLELLPAGQVPDKKARIYPANQWEPDVPEVEPYRPSDQAAIAQL
	(JGI) 2087;	ADHAPPEYQPLATSKAMSDVMTKLALDPKALADYKADHRAFAQSVPDLT
	OMPOL1_2087	QERAALELGDSWAIRCAMKNMPSSLLDAARESGEASQNGFPWVIVGVIG
		VIGVSMSTE
<i>Dendrothele</i>	SEQ ID	MESSTQTKPGSLIVVGTGIESIGQMTLQALSYLEAAKVFYCVIDPATEAF
<i>bispora</i>	NO.: 51	ILTKNKNVDLYQYYDNGKSRMDTYTQMAELMLKEVRNGLDVGVFYGHGP
	The DOE	VFVNPSHRALAIARSEGYQARMPLPGVSAEDCLFADLCIDPSNPGCLTYEAS
	Joint	DELIRERPVNHSILHILFQVGVCIADFNFTSGFDNSKFTILVDRLEQEYGP
	Genome	DHTVVHYIAAMMPHQDPVTDKFTIGQLREPEIAKRVGGVSTFYIPPKARKD
	Institute	INTDIIIRLLEFLPAGKVPDKHTQIYPPNQWEPDVPTLPPYQNEQAAITRL
	(JGI)	EAHAPPEYQPLATSKAMTDVMTKLALDPKALAEYKADHRAFAQSVPDLT
	765759	QERAALELGDSWAIRCAMKNMPSSLLLEAASQSVVEEASMNFGFPWVIVTGIVG
		VIGSVVSSA
<i>Rhizopogon</i>	SEQ ID	MTTDTKRGTLTIAGSGIASIAHITLETLSYIKESDKLFYLVCDPVTEAFIQ
<i>vinicolor</i>	NO.: 52	DNATGDFDLSVFDKNSRYDSYIQMCEIMLRVRAGHVLGIFYGHGPV
	GenBank	FVSPSHRAIAVAREEGYKARMLPGVSAEDYMFADLEFDPSQSTCNTYEATE
	AM-OR11-	LLLRDRPLDPAIQNI IWQVGSVGVVDMFEFESKFHLLVDRLEQDFGPDHKV
	026;	VHYIGAVLPQSTTTMDIFTISDLRKENVAKQFGTISTLYIPPRDEGPVSS
	OAX31299.1	MTQAFDFKAGAMVYSPVKWAGPKLNIVSALSPYERDVISQIDTHVAPEGYK
		ILHTSAAMNKFMTDLSLKPFLFEEYKLYPEAVVESAEGLSNLEKFGLKFGS
		DGAVYILMKATESDIASGRQLTEDEIAKAHKSIVGFPTVLVLPTVIVVLIG
		RE
<i>Rhizopogon</i>	SEQ ID	MSTKRGTLTIAGSGIASVGHITLGTLSYIKESDKIFYLVCDPVTEAFIYDN
<i>vinicolor</i>	NO.: 53	STADCFDLSVFDKTKGRYDSYIQMCEVMLKAVRAGHDVLGVFYGHGPV
	GenBank	SPSHRAIAVARQEGYKAKMLPGISAEDYMFADLEFDPSVSGCKTCEATEIL
	AM-OR11-	LRDKPLDPTIQNI IWQVGSVGVVDMFEFSKSFQLLVDRLEKDFGPDHKVVH
	026;	YIGAVLPQSTTTMDTFTIADLRKEDVAKQFGTISTLYIPPRDEGHVNLMA
	OAX32862.1	KVFGPGASVKLNDISKWAGPKLNIVSANDPHERDVIAQVDTHVAPEGHKK
		LRVSAAMKKFMTDLALKPKFLFEEYKLDPVAVVESAEGLSNLERFGLKFARS
		GPADALMKATESDIASGRQLTEEEIAQGTGPVGLQTALALLVLLGLGVAIV
		TRPDD

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In other embodiments, the cyclization comprises reacting with beta-lactamase (FIG. 12). As shown in FIG. 12, a variable region is excised and end-to-end cyclized by the actions of an N-methyltransferase and a beta-lactamase family member. Table 4 shows an exemplary list of lacta-
mase and amino acid sequences of the processed cyclic

peptides. In some embodiments, some of the sidechains of the randomized residues are subsequently isomerized from the L- to D-configuration or decorated with additional modifications like hydroxylation, halogenation, glycosylation, acylation, phosphorylation, methylation, and acetylation.

TABLE 4

Amino acid sequences of the N-methyltransferase and beta-lactamase processed cyclic peptides	
<i>Rhizopogon</i>	SEQ ID
<i>vinicolor</i>	NO.: 54
	GenBank
	OAX32863.1
	hypothetical
	protein beta-
	lactamase
	(transpeptidase)
	MAKVFGLVLGFLSQTFTYPSQVWFSVPGANNGQVITPELSNSIQETLDVWN
	ITGLSVAIIIPKSGEPYHSWGDRTEDGESVTQDTLFHMASVSKAFCVSALG
	ILMDDFEHGRNVTPLPALTEFNWHTSIQDLLPGEWQLMDEWASRKANMKD
	ILSHVSGLPRHDFAFGPYESPKEAVSRLRYLRPAFELREQWSYNNQMFVMA
	GHIVETYSKTYTSFVEDRIFTPLGMSSSTESPAKAAKTGKFTQGWTS SGR
	LLPELFPEDMVMLMAGAGGVISSAVDMSKWVALWLNKGVDNVTVIPSSVY
	GNASQSYAVSISTPVDSEHSIQGYGLGWQNSYLGHNVVYHSGSIPGLSML
	VSFLPDDDVGFVVFANGGDKAAPVMNISNSIIDAALHLRSGPAPPIMPEKK
	AVTSPSEDIVNLELPLEEFSGTYTDPGYGTFTFCSPSSSSSYCQVMTDFT
	AVDSVHPSAPSPLQLLAAWPRMGSSHIRAVHQSGNKPLLLCTALFPPEGYGR
	DSTPFETAEIGTPGATAEFVVEDGKVVGEGLGLVDQVTERERTQTTVKDR
	AEVWFDKV

TABLE 4-continued

Amino acid sequences of the N-methyltransferase and beta-lactamase processed cyclic peptides	
<i>Rhizophogon</i> SEQ ID	MIMAKVFLVLGFLSQFTFTYPSQIRLSPVGVNNGQVITPELSNSIQETLDV
<i>vinicolor</i> NO.: 55	WNITGLSVAIIPKSGEPEYHSWGDRTEDGESVTQDTLPHMASVSKAFVSA
GenBank	LGILMDDFEHGRNVTPLPPALTEFNWHTSIQDLLPGEWQLMDEWASRKANV
OAX34183.1	KDILSHVSGLPSSHFAFGPYESPKEVVSRLRYLRPAFELREQWSYNNQMFT
hypothetical	VAGHIVETYSKITYTSFVEDRIFTPLGMFSSTFSPAKAVKTGKFTQGWTS
protein beta-	GRLLPEFFQEDMIMPMPAGPGGVISSAVDMSKWVALWLNKGVDNVTIIPSS
lactamase	VYGNASQSYAVSISTPVDSEHSILGYGLGWFRNSYLGHVYHSGSIPGLS
(transpeptidase)	TLVSFLPDDDVGVVFPANGDNKAAPVMNISNRIIDAALHLRSGPAPPIMPE
	KKAVTSPSEDIVNLELPLEEFSGTYTDPGYGTFTFCSPSSSSPYCQQVIAN
	FTTVDSVRPSAPSSLQLLAAWPRVGS SHIRTVHQSGNKFMLLPALTALFP
	GRDSTPFETAIGTRGAPVEFVVEDGRVVGFGLFGLVGQVTERERTQTTVK
	DRAGVWFDKV
<i>Rhizophogon</i> SEQ ID	MSTKRGTLTIAAGSGIASVGHITLGTLSYIKESDKIFYLVCDPVTEAFIYDN
<i>vinicolor</i> NO.: 56	STADCFDLVSFYDKTKGRYDSYIQMCEVMLKAVRAGHDVLGVFYGHPGVFV
GenBank	SPSHRAIAVARQEGYKAKMLPGISAEDYMFADLEFPDPSVSGCKTCEATEIL
OAX32862.1	LDRKPLDPTIQNI IWQVGSVGVVDMFESKSKFQLLVDRLEKDFGPDHKVVH
hypothetical	YIGAVLPQSTTTMDTFTIADLRKEDVAKQFGTISTLYIPRDEGHVNLMA
N-	KVFGGPGASVKLNDISKWAGPKLNIVSANDPHERDVIAQVDTHVAPEGHKK
methyltransferase	LRVSAAMKKFMTDLALKPKFLEEYKLDPVAVVESAEGLSNLERFGLKFARS
	GPADALMKATESDIASGRQLTEEEIAQGTGPVGLQ TALALLVLLGLGVAIV
	TRPDD
<i>Rhizophogon</i> SEQ ID	MTSDNLQPEVISANWLKSLEAASSTGDTASFVSHFLPDGWFDRDMLCFTWNF
<i>vinicolor</i> NO.: 57	RTLSGQEKIHGFISEVVDGQSRLSYSHLHDFKLLDHSVNAPSPFKLP
GenBank	IEGVQGAFTFSITKPAAYGRGFFRLTQDVHGNWKALTLFTNMQDLVGHEES
OAX34185.1	SADEYDPHEKANPTVVIVIKVGGGQSLICARLKGKLGIRALVIDKNARVG
hypothetical	DIWRQRYAEALPSFAVLSRQETQVPEPYAAYSQISKLLPYPSNFPKYLPGK
protein	KLANFLESYAINQELCIWLSSTVSPSPVYDSFSARWTEVEHENRKYILHP
FAD/NAD(P) -	KHLVLATGHGRPRIPTWNGMDDFQGTLYHSDFHRDAEKFRGKCVVIGAGN
dependent	ASGDICEDFVAQGAAEVTIVQRSATCVVSSATADAFVFKLPFSDKTPIEEL
oxidoreductase,	DFRHNSMPLAFVLQLMKSGGTQHMKAHDKEHHEGLRKAGFNLTWEPSPGSG
D-amino acid	EVGLLGFVFERAGSGTMDITGFGKLIVEGTVKVKQQNISHFDKEGIFTKD
dehydrogenase	GSKLPADVIVAATGNELTMDAIRAVLGDITAEQLPPKVGWGLDAEGELNQMY
	RPSGHPGLWFAVGS LGMTRFCSKHLGLQILAQEVGIA

In some embodiments, the cyclization comprises reacting with a prolyl endopeptidase, an N-methyltransferase, and a hydroxylase (FIG. 15). In some embodiments, the bicyclization comprises further modification of the indicated anchored residues on the cyclized peptide, forming an internal tryptathionine bridge (FIG. 16). FIG. 17 illustrates

a biochemical steps to create the tryptathionine bridge with hydroxylase and dehydratase. Step 1 involves hydroxylation of the 2-position of the indole ring of the tryptophan residue by a hydroxylase belonging to the cytochrome P450 family of oxygenases (FIG. 17). An example of such hydroxylase is shown in TABLE 5.

TABLE 5

Amino acid sequences of a hydroxylase		
<i>Galerina</i>	SEQ ID	MGKMAYHTVLDDIALYLLGSAALVIFYRSFFYPYFLSGRRLAPGPTKGELS
<i>marginata</i>	NO.: 58	KELKQFNNEINVHFLRHMVKEYGPIFRLVGAPMIPGGLVVCPTAQQRIL
CBS 339.88	KDR84981.1	VSNSINYGQPRLAFPRVWTGGLFTLPEREHRGMRKILDVPVFSFRNLISTTG
	hypothetical	VYNTVQSLITIFRSKIDGENGAKGDVILVYEWLARLAIDNVSEAILGFK
	protein	LDTLHDPNNELITTLDELSRIPTAAPELLVRVPGFLRLVTFDSVRHSTLWQ
	GALMADRAFT_260690	RRVPGRGLGVFFTFMRCLSTIRKNALAIKATILQEDSANRDLNVISVLQHMQ
		SSDETANADIAGNIIMLWMSGRATIATRISWLLWLLAKDQQCQQQLRDEIA
		PLFSRDPDPDYRSLDKLQWLDVIMESIRLFLFGPNIRVALNDYIDGVFV
		PKGTVVVIPLDLFTRGDIWGEDPDQFKPARWLDSTKRYKISPPFLSFLTGP
		HRCIAKGMAIMQTKIVIASLIANFEFKPAYEGQHVEGNPSIIGHGMLPHVK
		PIRPS

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Step 2 involves the formation of a tryptathionine bridge between the 2'-hydroxyl position on tryptophan and the thiol group from the cysteine residue. This

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condensation reaction is catalyzed by a novel family of dehydratases. Examples of the dehydratases are shown in TABLE 6.

TABLE 6

Amino acid sequences of dehydratases		
<i>Galerina marginata</i> CBS 339.88	SEQ ID NO.: 59 KDR80488.1 hypothetical protein GALMADRAFT_136963	MPYVDPKPYFEHREQSSGATLYYCLVCRDGRERQPHHIKTHEASQAHTAL SVFDSQAESSQQTHGNPTQPGYFDPVIDDAVRALLVSGSGDPHQPLYPAG HPNVYGEPNFTDSRRRTSPVTGIDWDQFEAQEDTHAVPSAQDQLRADICQA TLDWLNDDISDDDEREPSEVDSVSDAESDREPIDDQPRKARTNRDNP SEDWYPWQDKITCTLDILMHLPRSVFSRKQDLDFLWLLRVNNVDDVPTGKS MKMLNKILQGMCGIETIAYEGKLGHNHVHNNIAQILAQELCNPKVGPPIYF YPEDSGDNLAEARQAARWLHELPEETPMIHLPSGDYIYIEPAMLSNRSF CIPFRWFTRNGKFHARAWSLETGVVDNTLGWIVHKENEVEISEDDLKDF RFSSDCEAYNVPHPSRILGVSCADSGNLLPWNHTNPVLGNRWRQLAKGHRT LCLPLWMYCDDTSGNTSKKWNHNSFLFTLAGLPREHTAKEYNHFLCTSN LAPPLEMMDGVVSI EAAQQNGIWAWD CVRKEPVLIPTILALLGDNPMHS EFACHIGLRGKFFCRTCWVKGSDAQDDANI VTPGLHETPENS PAPS PAPS APSPAPSPAPSPALSMAPQSQPPTPSEPSMQVPAPPSTAAPT KARGKKKET MSAMLNRI TAFIKPGRLRNKSETQKTLQNFKEQAQTI GAKTKLKTARTETG IKDTVQEFFFEKLFSSYKNKRGPQAKQALDQAVNQLPSDITSPVWRLKGL DPHQDTPVEILHVLLGFIKYFWRDLVQNDQKQTLIQRNLNSFDVTGL GITQLGGETLVNYSAGSLTGRDFRAVAQVAPFVIYDMVPADVFDRAWLAKSKL VPLVWQPYIENVAQYLTLEHEIHVLLRTARWTTGWFNKSKFHIIHLHPS HIRRFGPAILFATEAFESFNAVIRAKSVHSNRQAPSRDIALAFAQGNIRH LLSGGHFLSADTHMVDPDQPGQYERLARGRWRVSGPGPGHLVSAEPIL PSYLGIPQSTTSAGLCKRKTTPQTFLQTLTGLKLPNVSRRPGARELWQT CSEVYLLNDKCLIGHHVI VQRQSEQASFVSPPIARI GEILQKVSANHA HDKPDGILVQTLKSS EVADKFMPLRVPQNEWSFVPLADILCTVNAQHDCD RNGCTASGFRYVYQERIQTNDQRPVVEHVNPQEDFILNTAQMRDALHLQKF RIRSRSLDEQTI IHESVARTINQRKAQDNSSSGTGAGVSGRGRGRGRGRG GGVEGPTSTRGRGGI EGRGASSSGNGRGRGRGARSQAQSVFP
<i>Galerina marginata</i> CBS 339.88	SEQ ID NO.: 60 KDR74877.1 hypothetical protein GALMADRAFT_99137	MPRKPAPECFETDEASKMIRCLICKENDTVQQGTWIKHGSASQHIETNAH KLAVARREQLLQVQEEERRLQETIYGNTTPLSGNAQLYPTYPANMYGNQ DAVDTMDNQNPPQAYMLCADIPDLGIKPIERPDPQSQRERLRQQVEQL LLQAEHEDEFGSPDDPDDLSTNIAQAFADLDLEEMLEDEEVEDYFNQVSP EHDYYPYPNKTMLLDILDNLPRLRMSNQLRLILWLKQTVGSNVPSFSG FRNMQTHLRNMC GTTPKQHVSSLGNI FYSNNIGESVMRD FANPEVAKHLHL YPEETEGPISEVWQAERWKEFAPSELTPMPSQGHROFFIDEVAQLQDGGYV IPRNWVRKGLTSDCHIVTVNVPVRSKLHGLVLVLKQCFQSGWTLLESET QIFHADDQFNFYDVVSRI RGPISWSEGTVPAMPNNLRELAGDDDLVVM VPLWCDDVSGNKSQYNKHINVYMANSNIPGRLLQQEYFVRVSTSPNATS PEQFSALKDQINETQKKPIQCYNHAHTNKKTRAILRVPGLPADNPQQSESC HMGGNANCKCRKCHVGGPHEKKESNEGYHEHYLTGIKRSABETRLLEKQI KLAMYGVEKPINETQNTGT KD KVAQHWIDILLAKSRELKSANPSRSVEEI AQELQTFWDEQPGDKINPLLSIAGLDPTQDTPVEILHTILLGIVKYAWHHL HSNWTEAQNLFTVRLQSTDIDGLSVPIRVAYMMQYRNLIGKHFKTLMQ TLFPFHVGTVSDAQFKLVKAIGELGSLVWVHEIGDMEKYLSDLEILIGNVL DAFAEIDPSTAMYARFIYEPMPVPSKIIIVKLKHLMLPHLIEDIKRFPAPAR NSTEVFEFCFNAIFRLCSILSNHQAASRDIALKFASMDRLKMLSGGYWLS VEEGKFEWIRAGENVRNIIQSEPTIQRHLGWAPSAKQSGRKRTPPTSWEN TKASQPMDS EETAAGFPNPRLLSWRKGVTTTAQSGDRCSGTGSWVVARNHK VCYLLASHYCSIAKNDQGESCIGRIHEIIGPDEKSASSTGIITLECFQLGK EHHPDFGLPTLQRPQADLPKYILKAWQDPLFIFSAHHDCHTASQATALQP QLQERQLTSRMNKLI AHNDSDHFIINLYGLHNAILLREPLPRELTAPQPLH QDRKAFHYEVAAKLRVQQA EKRKATNARRKATRAANKAKQVERQKQNP DHE QESQEMDERPNSENGSDIELGGDDDI EVETRRKRRRN
<i>Hypsizygus marmoreus</i>	SEQ ID NO.: 61 KYQ37095.1 hypothetical protein Hypma_08924	MGRRAEELPAYVELSEDGTLVRCNLCLMHNRLDYSKEWIKRKGWRSHKSGS IHDRSEAKQRVLD DAAMD LQEPASAEVEVTFNDILIINAPKTPTMNQSE EQAMWDHFDAGSFTLEAGEDPNHSSQRLYQDLARKADAYGAWDGEALPEY RDLDVVSQFLDEDEEEDLLSEILRGLGLEEEHEDSSDRNPAAEELNSPWYYP GSKLMFLLDITIDNLPRLRISGAMMRVFLWLLREVGVQVPSFDKLRKIQRK LREGSGVPTVHWMSPKGNAYSFNDPAVIVANDWASPI TRPHLRRYPV PKD GVITEVYHAEKWHREINRHFLT PMYDDGFRHYF IDELAQLKDGRYAVPVWR LEDVDGRIVADAWRVELEDNDRATIIDTATVRIHSQELALNFEEIIESNLM PEWSDTTTEAGHPSRMPNPDRA LAEGDPIYTSFIDIFGDDVSGNRKSKWNK HWNMYISHRNLPRKLLHQQYHTEFVSTSTFASIPEQFVGVEKAIESTHCKP VKVRDADTGKQIRLKIYCNCGPGDNPSQSETSGHIGGNGNYPCKCHTGGT QKSKETDEGYKMF TAGEARSSKETLAEVKSQVEAACTGAKTVADAQSDT GVKDAYTQYWDIAIEKARAMQKENPGMPTTTIQTALIKWVYDHEBAIYNS FTLDGFDASRDTPVEILHTILLGIVKYLWHRSHTSWNAQKKIYSTRLOQ TNTQGLSIHHIRANYIMQYANSLIGRLKTLAQVNVFHVYDLVDPLRFLFT KATGELCALLWFTEIRDL E EYLS D V D I A A A N V L D I A A V I D P S K I V S K I Y H LLSHLRDIIIRFGPLVGVA TEVFECPNAVERYCSILSNHLAPSRDIA YKLA AQETMKHFLSGGWWHVKDSVDLQGNPKWVQPGPSVRTFMASNPVLHTLCGW TRNNDSTPGTVKSEPRKRGDPKQTLPLVRLAWLETQGSRALNNTSPNNET QWQRCKVIAETQDQCNVGSWVFARSPLLENIPIGRIVEILQDTSASPSA

TABLE 6-continued

Amino acid sequences of dehydratases		
		FVVIDVFQVSATRDEVEGMPVLLRRFNECCLHVIPASSVIFDFNAQHDCRY AKCEATGEQPLIQERVPSGVTEFVVKAIIDRYLINIHALHNAHLIRATLP RDLTAPIYAPNREAHHSIAIAELRSAQDTKRAKTAAKTAANAAKAEAA LKDTTSGPAAKRRRVDDGSGEEDNRDVMVSV
Galerina	SEQ ID	MAKGRKLNPLPDFIEISNDGLQVRCTLCLEARQHNGSGWIKRGSVSNHLK
marginata	NO.: 62	SDNHTNSLEAHMKSAEKAEGRSVQEEIAMEEGMDFVILSSKIQPEITAP
CBS 339.88	KDR73903.1	ARAPRRSNEEQEMWDRTYTLGGEVFDAGVDHTLVEAEERKRLEREATDFDLW
	hypothetical	HGADFLPEEDPNDGELLLEDEQDDILSELLRNAHLNAPDAADVLTEEPRA
	protein	AADPRICDAWSPYESKMMFLDLDLNLRLRLISNSLMNVFLWILREGGARD
	GALMADRAFT_141673	VPSLYHLRQVQTTLRKSTGVPTTQHKSPKGNVYSMNDPRTLVMWDWNPVI CDHIRRYPVI PRNGVISEVYHAQKWKVDVPHTLSPMYDAGNCHYVIDEVA RLKNGTFIIPVRWLEDEDNRVCADAYVVFQDDQFIASVVDGETIIVQASDL QNNFLDLKDMGLLPTWGNQTIESGHPARMPNPDRLAEGDPLYTSWIDVFG DDVSGNRSKNWNKHWNIYI SHRNLPKLLQQEFHTHFVSTSPVASVTEQPH GIKQVIELTHKSPVKVRHGTSGAQIRFKINVNCGPGDNPQAQSEVCGHIGVN GNKLCRKCHTGGTHEVKESDEGFNSLFEPGDARSAQEIVADVESQVQLACL GIAQHVNQQTKNGIKDAYTQYWDYILINRARTLRKEQPRRTTADIQSELL VWVQEHKDEIYNPFLKLDGFDAAVDTPVEILHTILLGIVKYLWHGSHTSWT AIQKQTYSVRLQSTDTSGLSIAIRANYIMQYANSLIGRQFKTIAQVNVFH VYDLVDTTQFLLTAKAVGELTALLWIPEIANMEEYLLDVEAAANVLDLFLAL IDPSKMTNKLKLHLVHLKADILRFGPLVGVATETFECFNAIFRPFCSIYSN HLAPSRDIAFQLASQEVLYKRLTGGWWPASDGEWKRPGPSVRNFIHDHPTL QALLGWTKEEKLNVNGSFRLEPLKRDASQKIESRKHLPWLQTQGAKANVSSSE DNDSKWTACRFVANSNGDKCSVGSVWFATSPFNQSVTGRIVEVLAESG KRAVVLDIFEVCSRHKIFGMPMLARRHEEPVYAVIASTNIEFLYNVQHD CPLAKCTASGKQPLIQERVESGLFKTYIEHKPIEREVINTHAFHNAHLRA VLQRLVVPPIPLYPEIRKTKHAEFAHNLQATQVKLEARAQAQKAKEIITP ADKTDSTIPKKTRSEMETETDDTAIATQADVFFNAQGC

Step 3 describes S-oxygenation of the tryptathionine thiol by a flavin-monooxygenase enzyme that converts it to a sulfinyl form. Examples of such monooxygenase are shown in TABLE 7. Step 4 describes potential future modification steps such as hydroxylation of side chains on the peptide such as the hydroxylation of position 6 on the indole ring of the tryptathionine-forming tryptophan residue by a P450 family monooxygenase.

TABLE 7

Amino acid sequences of monooxygenases		
Galerina	SEQ ID	MVQIKRLLGLSSPSQTPLESNHGVPVPSKSIIVGAGSAGLAMLRLTVEL
marginata	NO.: 63	EAFSRNNWEVVLVEERESVGGIWLPDNDVFPPEIPKTPLYPLRLTNPVP
CBS 339.88	KDR68385.1	SMTYPGFPFPSTPLYPRHDHVEAYHLRYARRHLLDFIKFDTMVEKAFWN
	hypothetical	GTPEEGYWNLTLSKKEGRMYKTEDHLVATGNNHIPIPHVWKQEDWLAS
	protein	PANHRSKI IHSVYYRGPEAFSNQTVLIVNGGSGRDAATQILGYASQTFMS
	GALMADRAFT_104945	IRRSYGPVDDGVIKPDISHFTEAGVVEVDGTILDPDVILLGTGYEMQKPL LSEGGELSFDPATAKDNSSVRGILVTNGHYIFPLHRRHIFSLSPRYPPNALAF IGLLSFIAACPSDIAQSLFAAHAILDPSILPRLHLLLELASYEDKARRQG LDPYLGKPIMLNNTSNDYQDELVEYLKQKNAPDDGKGFVEEWRREILAYH YLQRGWSRIEKLGMGPAWTEGVKTEAQWFDLMTRVNEWQKNWETENGIAFR VDLDLTG

A gene organization within two exemplary loci in *Rhizopogon vinicolor* that encode for methyltransferase, beta-lactamase, hopene cyclase, beta-lactamase 2, dehydrogenase, and glycine oxidase is shown in FIG. 13.

The sequence which flanks the encoded random peptide library can be, for example, as shown in FIG. 14, by using N-term and C-term flanks from the MSDIN family genes (toxin preproprotein sequences) identified in the genomes of *Amanita bisporigera* and *Amanita phalloides*. The low consensus central regions indicate areas where a random peptide library could be inserted to facilitate post-translational processing into a cyclic peptide. See Pulman, Jane A., et al. "Expansion and Diversification of the MSDIN Family of Cyclic Peptide Genes in the Poisonous Agarics *Amanita*

Phalloides and *A. Bisporigera*." *BMC Genomics*, vol. 17, no. 1, 15 Dec. 2016, p. 1038., doi: 10.1186/s12864-016-3378-7.

The enzymes can additionally be targeted to a specific cellular compartment to increase peptide synthesis efficiency and increase yield for peptide production purposes.

Disclosed herein, in certain embodiments, is a method of detecting interaction between a first test protein and a second

test protein in a host cell, comprising: expressing in the host cell a first fusion protein comprising the first test protein and a second fusion protein comprising the second test protein; delivering a first molecule to the host cell; modifying the first molecule while in the host cell via a modifying enzyme; and allowing the first molecule to modulate the interaction between the first test protein and the second test protein, wherein the first molecule is a product of an encoded DNA sequence, wherein the first molecule comprises a randomized polypeptide library and one or more modifying enzymes, wherein the one or more modifying enzymes modify the randomized polypeptide library.

Host Cells

In some embodiments, the host cell is a eukaryote or a prokaryote. In some embodiments, the host cell is from

animal, plant, a fungus, or bacteria. In some embodiments, the fungus is *Aspergillus*, *S. cerevisiae*, or *Pichia pastoris*. In some embodiments, the host cell is a haploid yeast cell. In other embodiments, the host cell is a diploid yeast cell. In some embodiments, the diploid yeast cell is produced by mating a first host cell comprising DNA sequences encoding the first chimeric gene, the second chimeric gene, and the third chimeric gene, to a second host cell comprising DNA sequences encoding the death agent, positive selection reporter, and the mRNA comprising a nucleotide sequence encoding a polypeptide. In some embodiments, the plant is *Nicotiana tabacum* or *Physcomitrella patens*. In some embodiments, the host cell is a sf9 (*Spodoptera frugiperda*) insect cell.

Disclosed herein, in certain embodiments, is a host cell configured to express a first fusion protein comprising a DNA-binding moiety; a second fusion protein comprising a gene activating moiety; a third fusion protein comprising a different DNA-binding moiety; a death agent, wherein the expression of the death agent is under control of a promoter DNA sequence specific for one of the DNA-binding moiety; a positive selection reporter, wherein the expression of the positive reporter is under control of a promoter DNA sequence specific for the other DNA-binding moiety; and a polypeptide of 60 or fewer amino acids, wherein the polypeptide modulates an interaction between the first test protein and the second test protein; wherein the host cell optionally has a mutant background enabling uptake of small molecules; and wherein the host cell optionally has a mutant background enabling increased transformation efficiency.

Disclosed herein, in certain embodiments, is a host cell comprising a plasmid vector which comprises the components of PLASMID 1, or any combination of the components of PLASMID 1; or the plasmid vector, wherein a DNA sequence encoding a first polypeptide is inserted in frame with Gal4-DBD, a second polypeptide is inserted in frame with LexA-DBD, and wherein a DNA sequence encoding a third polypeptide is inserted in frame with VP64-AD.

Disclosed herein, in certain embodiments, is a kit comprising PLASMID 1, PLASMID 2, and PLASMID 3; and transfectable host cells compatible with PLASMIDS 1-3, or any combination thereof. In some embodiments, the provided host cells are already transfected with PLASMID 1 or 2. In some embodiments, the kit includes selectable agents for use with host cells transfected with PLASMIDS 1-3. In some embodiments a library of variants of PLASMID 1 are provided, wherein more than a single pair of Y2H interactors are represented. Such a library can be used to, for example, screen for protein-protein interactions that are inhibited by a defined agent. In some embodiments a library of variants of PLASMID 2 are provided, wherein a plurality of different short test polypeptide sequences for screening are represented. The plurality of different short peptide sequences can be randomly generated by any method (e.g. NNK or NNN nucleotide randomization). The plurality of different short peptide sequences can also be preselected, either by previous experiments selecting for binding to a target, or from existing data sets in the scientific literature that have reported rationally-designed peptide libraries.

The host cell can additionally be made to be permeable to small molecules, for example by deletion of drug efflux pumps, such as PDR5, ERG6, or 12geneΔ0HSR (*Chinen*, 2011), to enable a small molecule screening approach.

The host cell can additionally carry mutations to enable more efficient transformation with vectors and/or more efficient uptake small molecules.

PLASMIDS 1, 2, and 3 can be used in various permutations. In some embodiments, integration of PLASMID 1 into the genome of the host cell (as confirmed using PLASMID 3) is followed by transformation of a library of PLASMID 2 with randomly encoded peptides using, for example, NNK or NNN codons.

In some embodiments, to perform a screen to identify a peptide that can disrupt a PPI, the host cell is propagated in selection media to ensure the presence of PLASMID 1 and a proper PPI (e.g. on media lacking the positive selection marker for yeast, or in media containing antibiotic for human or bacterial cells). This host cell is then be transformed with PLASMID 2, and immediately transferred to selection media to ensure all components are present (i.e. on media lacking both plasmid selection markers for yeast, or antibiotics for bacterial or mammalian cells), and are inducing expression of any inducible component (e.g. with Gal, doxycycline, etc).

In other embodiments, the plasmids are used as a 'plug and play platform' utilizing the yeast mating type system, where the 2 or more plasmids (or the genetic elements therein) are introduced into the same cell by cell fusion or cell fusion followed by meiosis instead of transfection. This cell fusion involves two different yeast host cells bearing different genetic elements. In this embodiment, yeast host cell 1 is one of MATa or MATalpha and includes an integration of PLASMID 1. In this embodiment, yeast host cell 1 strain can be propagated on positive selection media to ensure a proper PPI is present. In this embodiment, the yeast host cell 2 can be the opposite mating type. This strain carries (or has integrated) the randomized peptide library and 'death agent' (e.g. cytotoxic reporter) plasmid (PLASMID 2). Yeast host cell 2 can be generated via large batch high efficiency transformation protocols which ensure a highly diversified library variation within the cell culture. Aliquots of this library batch can then be frozen to maintain consistency. In this embodiment, the strains are mated in batch to result in a diploid strain that carries all the markers, the PPI, positive selection, 'death agents' and peptide. This batch culture then can be propagated on solid medium that enables selection of all the system components (i.e. media lacking both positive selection markers), and inducing expression of any inducible component (i.e. with Gal).

Surviving colonies from limiting dilution experiments performed on host cells bearing both the Y2/3H and library/cytotoxic constructs (either introduced to the cell by transfection or mating) can constitute colonies with a specific PPI that has been disrupted by a peptide and no longer triggers the death cascade triggered by the encoded 'death agents' (e.g. cytotoxic reporters) while maintaining a differential PPI driving a positive selection marker. The peptide sequence can be obtained by DNA sequencing the peptide-encoding region of PLASMID 2 in each surviving colony.

To ensure that survival is due to inhibition of the PPI rather than stochastic chance or faulty gene expression, an inducible promoter can be used to inactivate the production of either the PPI or the peptide and confirm specificity. In some embodiments, cell survival is observed only on media with galactose wherein all the components are expressed; and no survival is observed on media without galactose when expression of the peptide is lost.

The plasmids can also be isolated and re-transformed into a fresh host cell to confirm specificity. Biochemical fractionation of the viable host cells which contain the PPI, peptide, positive selection and 'death agent' followed by pull-down experiments can confirm an interaction between the peptide sequence and either PPI partner using the

encoded tags (e.g. Myc-tag, HA-tag, His-tag). This is also helpful to perform SAR to determine the binding interface.

The peptides to be used in screening assay can be derived from a complex library that involves post-translational modifying enzymes. The modified peptides can be analyzed by methods such as mass spectrometry, in addition being sequenced to ID the primary sequence. The peptides can also be tested for inherent membrane permeability by reapplying them onto the host cells exogenously (from a lysate) and observing for reporter inactivation or activation.

Once enough surviving host cell colonies are sequenced, highly conserved sequence patterns can emerge and can be readily identified using a multiple-sequence alignment. Any such pattern can be used to 'anchor' residues within the library peptide insert sequence and permute the variable residues to generate diversity and achieve tighter binding. In some embodiments, this can also be done using an algorithm developed for pattern recognition and library design. Upon convergence, the disrupting peptide pattern, as identified through sequencing, can be used to define a peptide disruptor sequence. Convergence is defined by the lack of retrieval of any new sequences in the last iteration relative to the penultimate one.

EXAMPLES

Example 1: Method for Identifying a Disruptor of a Protein-Protein Interaction

This is an example of a system that uses two variants of one protein, fused to different DBDs to identify inhibitors against a specific PPI. The PPI integration plasmid (PLASMID 1; FIG. 8) is used to integrate into *Saccharomyces cerevisiae* three proteins that constitute the binary PPIs of interest. The plasmid encodes for the fusion of an AD (VP16) with c-Raf, a DBD (LexA) with KRas(G12D), and a DBD (Gal4BD) with KRas, driven by a strong promoter and terminator ADH1. Fusion proteins VP16-c-Raf, LexA-KRas(G12D), and Gal4BD-KRas are tagged with MYC, FLAG, and HA, respectively. The plasmid further includes bacterial selection and propagation markers (ori and AmpR), and yeast replication and selection markers (TRP1 and CEN). The plasmid also has sites for integration into the genome at a specified locus.

Saccharomyces cerevisiae is co-transformed with the selection and library plasmid (PLASMID 2; FIG. 9) for the expression of a randomized peptide library (NNK 20-mer sequences). The plasmid is driven by a strong promoter, ADH1. The initiation sequence of the selection and library plasmid is a fixed sequence of methionine-valine-asparagine (MVN) to maximize the half-life of the peptide, and terminated with an untranslated region (UTR) of a short protein (sORF1). The selection and library plasmid also comprises a sequence that encodes a His-tag. The translated protein has the N-terminus of methionine to minimize proteolysis.

The selection and library plasmid additionally comprises a LexAop sequence, which induces 'death agents' (cytotoxic reporter expression) when bound by a functional transcriptional factor that is formed by LexA-KRas(G12D) fusion protein and VP64-c-Raf fusion protein, unless interrupted by a disrupter peptide. The selection and library plasmid also contains a positive selection marker, ADE2 which is under control of Gal4BD-KRas. The plasmid further includes bacterial selection and propagation markers (ori and AmpR), and yeast replication and selection markers (TRP1 and CEN).

To confirm expression of the reporters and the successful construction of the strains, *Saccharomyces cerevisiae* is transformed with a confirmation plasmid (PLASMID 3; FIG. 10). The confirmation plasmid includes a direct fusion between the AD fusion protein (VP64-c-Raf) and DBD fusion proteins (LexA-KRas(G12D) or Gal4BD-KRas). The confirmation plasmid includes yeast replication and selection markers (TRP1 and CEN). The confirmation plasmid is used to confirm integration of LexAop promoter before the ADE2 gene. Transformation with the confirmation plasmid allows for the activation of ADE2 only if the promoter is properly integrated. Also, confirmation of death agents is enabled by inducible expression of the confirmation fusion construct.

The screen is performed by mating the strains in a batch to result in a diploid strain, which carries all the markers, the PPIs, the positive selection, the death agents, and the peptide. This batch culture is then propagated on solid medium, which enables selection of all the system components (media lacking two nutritional components) and induces expression of any inducible component with Gal.

Surviving colonies constitute colonies with a specific PPI(KRas-c-Raf) that have been disrupted by a peptide and no longer trigger the death cascade induced by the encoded death agents and maintain positive selection of the remaining PPI.

The peptide sequence that disrupts the death agent-driving PPI is obtained by DNA sequencing the peptide-encoding region of the selection and library plasmid in each surviving colony.

To confirm specificity, the inducible marker is used to inactivate the production of the PPI and confirm specificity. The plasmid is then isolated and re-transformed into a fresh parental strain to confirm specificity.

Biochemical fractionation of the viable strain that contained the PPI, peptide, selection marker, and death agent, followed by pull-down experiments is done to confirm an interaction between the peptide sequence and either PPI partner using the encoded tags.

An alternative example can be made by switching LexA with TetR. In another alternative example, fusion proteins in the PPI integration plasmid and the randomized peptide library in selection and library plasmid are driven by an inducible promoter, GAL1, instead of ADH1 used alternatively in another example. In yet another example, the N-terminus of the peptide translated from the selection and library plasmid can alternatively be glycine, alanine, serine, threonine, valine, or proline. In other examples, the genetic reporter in the confirmation plasmid is HIS3 or URA3, in place of ADE2. Either mating types of *Saccharomyces cerevisiae* haploid state can be used as background strain in alternative examples. In other examples, background strains also express the enzymes for the cyclization and methylation of peptides like lanthipeptides maturation enzymes from *Lactococcus lactis* (LanB, LanC, LanM, LanP), patellamide biosynthesis factors from cyanobacteria (PatD, PatG), butelase 1 from *Clitoria ternatea*, and GmPOPB from *Galerina marginata* or other species.

Example 2: Method for Identifying Protein-Protein Interaction Disruptor

This is an example of system that uses two different proteins, fused to different DBDs to identify inhibitors against a specific PPI. The PPI integration plasmid (PLASMID 1; FIG. 8) is used to integrate into *Saccharomyces cerevisiae* three proteins that constitute the binary PPIs of

interest. The plasmid encode for the fusion of an AD (VP64) with TEAD, a DBD (LexA) with YAP, and a DBD (Gal4BD) with VGLL4, each associated with ADH1 promoter. Three protein fusion sequences are tagged with either FLAG, MYC or HA. The plasmid further includes yeast replication and selection markers (TRP1 and CEN). The plasmid also has sites for integration into the genome at a specified locus.

The *Saccharomyces cerevisiae* is co-transformed with the selection and library plasmid (PLASMID 2; FIG. 9) for the expression of a randomized peptide library (NNK 20-mer sequences). The selection plasmid is driven by a strong promoter, ADH1. The selection and library plasmid also comprises a sequence that encodes a His-tag.

The selection and library plasmid additionally comprises a LexAop sequence, which induces 'death agent' (cytotoxic reporter expression) when bound by a functional transcriptional factor that is formed by LexA-YAP fusion protein and VP64-TEAD fusion protein, unless interrupted by a disrupter peptide. The selection and library plasmid also contains a positive selection marker, ADE2 which is under control of Gal4BD-VGLL4 and VP64-TEAD. The plasmid further includes yeast replication and selection markers (TRP1 and CEN).

To confirm expression of the reporters and the successful construction of the strains, *Saccharomyces cerevisiae* is transformed with a confirmation plasmid (PLASMID 3; FIG. 10). The confirmation plasmid includes a direct fusion between the AD fusion protein (VP64-TEAD) and DBD fusion proteins (LexA-YAP or Gal4BD-VGLL4). The confirmation plasmid includes yeast replication and selection markers (TRP1 and CEN). The confirmation plasmid is used to confirm integration of LexAop promoter before the ADE2 gene. Transformation with the confirmation plasmid allows for the activation of ADE2 only if the promoter is properly integrated. Also, confirmation of death agents is enabled by inducible expression of the confirmation fusion construct.

The screen is performed by propagating the parental strain on selection media to ensure the presence of the PPI Integration plasmid, and that a proper PPI has occurred, which is confirmed via use of the confirmation plasmid. The strain is cultured on media lacking nutrient markers against positive selection markers to ensure selection of colonies where the desired interaction occurred. The strain is then transformed with the selection and library plasmid, and is immediately plated on selection media to ensure all components are present (on media lacking the two nutritional markers) and is induced expression of any inducible component (with Gal).

Surviving colonies constitute colonies with a specific PPI(YAP-TEAD) that has been disrupted by a peptide and no longer triggers the death cascade induced by the encoded death agents and maintain positive selection of the remaining PPI.

The peptide sequence that disrupts the death agent-driving PPI is obtained by DNA sequencing the peptide-encoding region of the selection and library plasmid in each surviving colony.

To confirm specificity, the inducible marker is used to inactivate the production of the PPI and confirm specificity. The plasmid is then isolated and re-transformed into a fresh parental strain to confirm specificity.

Biochemical fractionation of the viable strain that contained the PPI, peptide, selection marker, and death agent is followed by pull-down experiments to confirm an interaction between the peptide sequence and either PPI partner using the encoded tags.

An alternative example can be made by switching LexA with TetR. In another alternative example, fusion proteins in the PPI integration plasmid and the randomized peptide library in selection and library plasmid are driven by an inducible promoter, GAL1, instead of ADH1 used alternatively in another example. In yet another example, the N-terminus of the peptide translated from the selection and library plasmid can alternatively be glycine, alanine, serine, threonine, valine, or proline. In other examples, the genetic reporter in the confirmation plasmid is HIS3 or URA3, in place of ADE2. Either mating types of *Saccharomyces cerevisiae* haploid state can be used as background strain in alternative examples. In other examples, background strains also express the enzymes for the cyclization and methylation of peptides like lanthipeptides maturation enzymes from *Lactococcus lactis* (LanB, LanC, LanM, LanP), patellamide biosynthesis factors from cyanobacteria (PatD, PatG), butelase 1 from *Clitoria ternatea*, and GmPOPB from *Galerina marginata* or other species.

Example 3: Method for Identifying Protein-Protein Interaction Facilitator

This is an example of system that uses two variants of one protein, fused to different DBDs to identify facilitator for a specific PPI. The PPI integration plasmid (PLASMID 1; FIG. 8) is used to integrate into *Saccharomyces cerevisiae* three proteins that constitute the binary PPIs of interest. The plasmid encodes for the fusion of an AD (VP64) with Mdm2, a DBDs (LexA) with KRas, and a DBD (Gal4BD) with KRas(G12D). DBD-KRas, each driven by ADH1 promoter. Three protein fusion sequences are tagged with either FLAG, MYC or HA. The plasmid further includes yeast replication and selection markers (TRP1 and CEN). The plasmid also has sites for integration into the genome at a specified locus.

The *Saccharomyces cerevisiae* is co-transformed with the selection and library plasmid (PLASMID 2; FIG. 9) for the expression of a randomized peptide library, NNK 20-mer sequences. The selection plasmid is driven by a strong promoter, ADH1. The selection and library plasmid also comprises a sequence that encodes a HIS tag.

The selection and library plasmid additionally comprises a LexAop sequence, which induces 'death agents' (cytotoxic reporter expression) when bound by a functional transcriptional factor that is formed by LexA-KRas fusion protein and VP64-Mdm2 fusion protein, when mediated by a facilitator peptide. The selection and library plasmid also contains a positive selection marker, ADE2 which is under control of Gal4BD-KRas(G12D) fusion protein and VP64-Mdm2 fusion protein and leading to expression of the positive selection marker when the fusion proteins are mediated by a facilitator. The plasmid further includes yeast replication and selection markers (TRP1 and CEN).

To confirm expression of the reporters and the successful construction of the strains, *Saccharomyces cerevisiae* is transformed with confirmation plasmid (PLASMID 3; FIG. 10). The confirmation plasmid includes a direct fusion between the AD fusion protein (VP64-Mdm2) and DBD fusion proteins (LexA-KRas or Gal4BD-KRas(G12D)). The confirmation plasmid includes yeast replication and selection markers (TRP1 and CEN). The confirmation plasmid is used to confirm integration of LexAop promoter before the ADE2 gene. Transformation with the confirmation plasmid allows for the activation of ADE2 only if the promoter is

properly integrated. Also, confirmation of death agents is enabled by inducible expression of the confirmation fusion construct.

The screen is performed by mating the strains in a batch to result in a diploid strain, which carries all the markers, the PPIs, the positive selection, the death agents, and the peptide. This batch culture is then propagated on solid medium, which enable selection of all the system components (media lacking two nutritional components) and induce expression of any inducible component with Gal.

Surviving colonies constitute colonies with a specific PPI(KRas(G12D)-Mdm2) that has been facilitated by a peptide and do not have nonspecific PPI(KRas-Mdm2) that can trigger the death cascade induced by the encoded death agents and maintain positive selection of the remaining PPI.

The peptide sequence that is able to facilitate a specific PPI is obtained by DNA sequencing the peptide-encoding region of the selection and library plasmid in each surviving colony.

To confirm specificity, the inducible marker is used to inactivate the production of the PPI and confirm specificity. The plasmid is then isolated and re-transformed into a fresh parental strain to confirm specificity.

Biochemical fractionation of the viable strain that contained the PPI, peptide, selection marker, and death agent is followed by pull-down experiments to confirm an interaction between the peptide sequence and either PPI partner using the encoded tags.

An alternative example can be made by switching LexA with TetR. In another alternative example, fusion proteins in the PPI integration plasmid and the randomized peptide library in selection and library plasmid are driven by an inducible promoter, GAL1, instead of ADH1 used alternatively in another example. In yet another example, the N-terminus of the peptide translated from the selection and library plasmid can alternatively be glycine, alanine, serine, threonine, valine, or proline. In other examples, the genetic reporter in the confirmation plasmid is HIS3 or URA3, in place of ADE2. Either mating types of *Saccharomyces cerevisiae* haploid state can be used as background strain in alternative examples. In other examples, background strains also express the enzymes for the cyclization and methylation of peptides like lanthipeptides maturation enzymes from *Lactococcus lactis* (LanB, LanC, LanM, LanP), patellamide biosynthesis factors from cyanobacteria (PatD, PatG), butelase 1 from *Clitoria ternatea*, and GmPOPb from *Galerina marginata* or other species.

Example 4: Reversible Induction of a Nutritional Reporter by Protein-Protein Interaction

This is an example of two platforms that either used two variants or two different proteins, fused to different DBDs to identify inhibitor by nutrient based selection. In the first platform, KRas and KRas(G12D) fused to DBDs and c-Raf fused to AD were expressed in *Saccharomyces cerevisiae* cells using an integration plasmid (FIG. 6). In the absence of any inhibitors, the DBD fusion protein and AD fusion protein pairs maintained their interaction to drive expression of Nutritional reporters 1 and 2. A 5-fold dilution series starting at 10^4 cells were spotted onto selective media with or without inhibitor and visualized after 2 days of growth at 30° C. The results showed that the cells grown on media that selected for Nutritional reporter 2 had poor survival rate when added the inhibitor, illustrating the specificity of the inhibitor for KRas(G12D) and c-Raf interaction.

In the second platform, VGLL4 or YAP fused to DBD and TEAD fused to AD were expressed in *Saccharomyces cerevisiae* cells with an integration plasmid (FIG. 6). In the absence of any inhibitors, the DBD fusion protein and AD fusion protein pairs maintained their interaction to drive expression of Nutritional reporters 1 and 2. A 5-fold dilution series starting at 10^4 cells were spotted onto selective media with or without inhibitor and visualized after 2 days of growth at 30 C. The results showed that the cells grown on media that selected for Nutritional reporter 2 had particularly poor survival rate when added the inhibitor, illustrating the specificity of the inhibitor for YAP and TEAD.

Example 5: Reversible Induction of a Cytotoxic Reporter by Protein-Protein Interaction

This is an example of two platforms that used either two variants or two different proteins, fused to different DBDs to identify inhibitor by toxicity based selection. In the first platform, KRas and KRas(G12D) fused to DBDs and c-Raf fused to AD were expressed in *Saccharomyces cerevisiae* cells with an integration plasmid. In the absence of inhibitors, the KRas(G12D) and c-Raf maintained an interaction to drive expression of cytotoxic reporter. A nutritional reporter was controlled by KRas and c-Raf interaction. The cells were patched onto selective media for a nutritional marker with or without inhibitor and visualized after 4 days of growth at 30° C. In cell populations with KRas(G12D) and c-Raf interaction, only those with the inhibitor showed enhanced cell viability, illustrating the specificity of the inhibitor to KRas(G12D) and c-Raf interaction. In the second platform, VGLL4 or YAP fused to DBD and TEAD fused to AD were expressed in *Saccharomyces cerevisiae* cells with an integration plasmid. In the absence of inhibitors, the YAP and TEAD maintained their interaction to drive expression of cytotoxic reporter. A nutritional reporter was controlled by VGLL4 and TEAD interaction. The cells were patched onto selective media for a nutritional marker with or without inhibitor and visualized after 4 days of growth at 30 C. In cell populations with YAP and TEAD interaction, only those with the inhibitor showed enhanced cell viability, illustrating the specificity of the inhibitor to YAP and TEAD interaction.

Example 6: Cyclization of Peptides

This is an example of a system that induces cyclization of randomized peptides in a complex to achieve enhanced cell permeability and peptide stability. The *Saccharomyces cerevisiae* is transformed with the selection and library plasmid (PLASMID 2; FIG. 9) for the expression of a randomized peptide library, NNK 60-mer sequences. The selection plasmid is driven by a strong promoter, ADH1. The initiation sequence of the selection and library plasmid is a fixed sequence that encodes Methionine Valine Asparagine (MVN) at N-terminus of peptide to maximize the half-life of the peptide, and terminate with sequence that encodes the UTR of a short protein (sORF1). The peptide is also tagged with a HIS tag.

The variable peptide library region of the selection and library plasmid is embedded within primary sequence of modifying enzyme, a homolog of N-methyltransferase from *Rhizophogun vinicolor* and contains randomized residues. The diversified variable region is excised and end-to-end cyclized by the action of a beta-lactamase DD-transpeptidase from *R. vinicolor*. Some of the side chains of the

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randomized residues are subsequently post-translationally isomerized from L- to D-configuration and hydroxylated.

Example 7: Cyclization of Peptides

This is an example of a system that induces cyclization of randomized peptides in a complex to achieve enhanced cell permeability and peptide stability. The *Saccharomyces cerevisiae* is transformed with the selection and library plasmid (PLASMID 2; FIG. 9) for the expression of a randomized peptide library, NNK 60-mer sequences. The selection plasmid is driven by a strong promoter, ADH1. The initiation sequence of the selection and library plasmid is a fixed sequence that encodes Methionine Valine Asparagine (MVN) at N-terminus of peptide to maximize the half-life of the peptide, and terminate with sequence that encodes the UTR of a short protein (sORF1). The peptide is also tagged with a HIS tag.

The variable peptide library region of the selection and library plasmid is embedded within primary sequence of modifying enzyme, a homolog of prolyl endopeptidases belonging to the PopB family and contains randomized residues. A post-translational processing of the variable peptide by the co-expressed prolyl endopeptidases leads to the generation of N-to-C cyclized macrocycles.

Example 8: Bicyclization of Peptides

This is an example of a system that induces cyclization of randomized peptides in a complex to achieve enhanced cell permeability and peptide stability. The *Saccharomyces cerevisiae* is transformed with the selection and library plasmid (PLASMID 2; FIG. 9) for the expression of a randomized peptide library, NNK 60-mer sequences. The selection plasmid is driven by a strong promoter, ADH1. The initiation

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sequence of the selection and library plasmid is a fixed sequence that encodes Methionine Valine Asparagine (MVN) at N-terminus of peptide to maximize the half-life of the peptide, and terminated with sequence that encodes the UTR of a short protein (sORF1). The peptide is also tagged with a HIS tag.

The variable peptide library region of the selection and library plasmid is embedded within primary sequence of modifying enzyme, a homolog of prolyl endopeptidases belonging to the PopB family and contains randomized residues. A post-translational processing of the variable peptide by the co-expressed prolyl endopeptidases leads to the generation of N-to-C cyclized macrocycles.

The macrocycles is then hydroxylated at the 2-position of the indole ring of the tryptophan residue by a hydroxylase belonging to the Cytochrome P450 family of oxygenases. A condensation reaction is followed catalyzed by dehydratase to form a tryptathionine bridge between the 2'-hydroxyl position on tryptophan and the thiol group from the cysteine residue. A flavin-monooxygenase enzyme converts the intermediate product to a sulfinyl form by S-oxygenation of the tryptathionine thiol. Some of the side chains of the peptide are subsequently hydroxylated at position 6 on the indole ring of the tryptathionine-forming tryptophan residue by a P450 family monooxygenase. The resulting bicyclized macrocycles comprises a tryptathionine bridge.

While preferred embodiments of the present disclosure have been shown and described herein, it will be obvious to those skilled in the art that such embodiments are provided by way of example only. Numerous variations, changes, and substitutions will now occur to those skilled in the art without departing from the disclosure. It should be understood that various alternatives to the embodiments of the disclosure described herein may be employed in practicing the disclosure.

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GNGGDDYCR L VYNWTHLRN LQKDGWLWR VESKYLDGGY PSLRFKIKMT SNKDNVCRIV 480
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DTFPLNQKEL GNPKFIAIN EVNQKKEGKG VNAKADEGR DLLFGKELKK NLEHGQLVDL 180
DLISGNLSEF QHVFASFAF YYPEPHYKEAL KSYAPALFNY MLELDQMRFK EISDDVKEKN 240
KNVLDPKWYT RKAESWGVQT FKNWKENLTI SEKDIITGYT GSKYDPINEY LRKYDGEIIP 300
NIGGDLKKS KKALEKIEHQ IKNLDAALQK SKITENLIVY RRVSELQFGK KYEDYNLRQN 360
GIINEEKVME LESNFKGQTF IQHNYMSTSL VQDPHQSYSN DRYPILLEIT IPEGVHGAYI 420
ADMSEYPGQY EMLNIRGYTF KYDKFSIVKP TREEDKGKEY LKVNLSIYLG NLNREK 476

SEQ ID NO: 8      moltype = AA length = 487
FEATURE          Location/Qualifiers
REGION          1..487
note            = strain V583
source          1..487
                 mol_type = protein
                 organism = Enterococcus faecalis

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SEQUENCE: 8

MSQLNKWQKE	LQALQKANYQ	ETDNQLFNQV	RQSLIDIKKR	LKVYTENAES	LSFSTRLEVE	60
RLFSVADEIN	AILQLNSPKV	EKTIKGYSAK	QAEQGYGLW	YTLQSQSNI	LSMPLINHDI	120
IMNLVNPVPA	GKRLSKRLYK	YRDELAQNV	NNIITGLFEG	KSVAEIRAWI	NEETEASYKQ	180
ALRIARTEAG	RTQSVTTQKG	YEEAKELGIN	IKKKWLATID	KHTRRTHQEL	DGKEVDVDEE	240
FTIRGHSAGK	PRMFGVASED	VNCRCTTIEV	VDGISPRLRK	DNESKEMSEF	KSYDEWYADR	300
IRQNESKPKP	NFTELDFFGQ	SDLQDDSDKW	VAGLKPEQVN	AMKDYTSDAF	AKMNKILRNE	360
KYNPREKPYL	VNIQNLDDA	ISKFKLKHDI	ITYRGVSANE	YDAILNGNVF	KEFKSTSINK	420
KVAEDFLNFT	SANKDGRVVK	FLIPKGTQGA	YIGTNSMMKK	ESEFLLNRNL	KYTVEIVDNI	480
LEVITILG						487

SEQ ID NO: 9 moltype = AA length = 457
FEATURE Location/Qualifiers
source 1..457
 mol_type = protein
 organism = *Pseudomonas aeruginosa*

SEQUENCE: 9

MHIQSSQQNP	SFVAELSQA	AGRLGQVEAR	QVATPREAQ	LAQRQEAPKG	EGLLSRLGAA	60
LARPFVAIIE	WLGKLLGSRA	HAATQAPLSR	QDAPPAASLS	AAEIKQMLLQ	KALPLTLGGL	120
GKASELATLT	AERLAKDHTR	LASGDGALRS	LATALVGIRD	GSLIEASRTQ	AARLLEQSVG	180
GIALQQWGTG	GGAASQHVLS	ASPEQLREIA	VQLHAVMDKV	ALLRHAVESE	VKGEPVDKAL	240
ADGLVEHFGL	EAEQYLGEHP	DGPYSDAEVM	ALGLYTNGEY	QHLNRSRLRQ	RELDAGQALI	300
DRGMSAAFEK	SGPAEQVVK	FRGTQGRDAF	EAVKEGQVGH	DAGYLSTSRD	PSVARSFAGL	360
GTITTLFGRS	GIDVSEISIE	GDEQEILYDK	GTDMRVLLSA	KDGQGVTRRV	LEEATLGERS	420
GHSEGLLDAL	DLATGTDRSG	KPQEQLRLR	MRGLDLA			457

SEQ ID NO: 10 moltype = AA length = 265
FEATURE Location/Qualifiers
REGION 1..265
 note = Description of Unknown: CdtB toxin sequence
source 1..265
 mol_type = protein
 organism = unidentified

SEQUENCE: 10

MKKIICLFLS	FNLAFFANLEN	FNVTGTWNLQ	SSAATESKWS	VSVRQLVSGA	NPLDILMIQE	60
AGTLPRATAT	TGRHVQQGGT	PIDEYEWNLG	TLSPRDRVFI	YYSRVVDVGN	RVNLAIVSRM	120
QAEVIVLPLP	PTTVSRPIIG	IRNGNDAFFN	IHALANGGTD	VGAITAVDA	HFANMPQVNW	180
MIAGDFNRDP	STITSTVDRE	LANRIRVVFP	TSATQASGGT	LDYAITGNSN	RQQTYTPPLL	240
AAILMLASLR	SHIVSDHFPV	NFRKF				265

SEQ ID NO: 11 moltype = AA length = 560
FEATURE Location/Qualifiers
source 1..560
 mol_type = protein
 organism = *Corynebacterium diphtheriae*

SEQUENCE: 11

MSRKLFAFAS	IGALLGIGAP	PSAHAGADDV	VDSSKSFVME	NFSSYHGTHK	GYVDSIQKGI	60
QPKSGTQGN	YDDWKGFFYS	TDNKYDAAGY	SVDNENPLSG	KAGGVVKVTY	PGLTKVLALK	120
VDAETIKKE	LGLSLTEPLM	EQVGTEEFIK	RFGDGASRVV	LSLPFAEGSS	SVEYINNWEQ	180
AKALSVELEI	NFETRGRKGQ	DAMYEMAQA	CAGNRVRRSV	GSSLSCLNLD	WDVIRDKTKT	240
KTESLKEHGP	IKNMSESPN	KTVSEKAKQ	YLEEFHQAL	EHPSELSELT	VTGTNPVFAG	300
ANYAAWAVNV	AQVDSSETAD	NLEKTTAALS	ILPGIGSVMG	IADGAVHHNT	EEIVAQSIAL	360
SSLMVAQAIP	LVGELVDIGF	AAYNFVESII	NLFQVVHNSY	NRPAYSPGHK	TQPFLLHDGYA	420
VSWNTVEDSI	IRTFGQGESG	HDIKITAENT	PLPIAGVLLP	TIPGKLDVKN	SKTHISVNGR	480
KIRMRCAID	GDVTFCRPKS	PVVVGNVSHA	NLHVAFHRSS	SEKIHSENEIS	SDSIGVLGYQ	540
KTVDHTKVNS	KLSLFFFEIKS					560

SEQ ID NO: 12 moltype = AA length = 621
FEATURE Location/Qualifiers
REGION 1..621
 note = Description of Unknown: ExoU/VipB toxin sequence
source 1..621
 mol_type = protein
 organism = unidentified

SEQUENCE: 12

MKLAEIMTKS	RKLKRNLEI	SKTEAGQYSV	SAPCHKGLVL	SGGGAKGISY	LGMIALQER	60
GKIKNLTHVS	GASAGAMTAS	ILAVGMDIKD	IKKLIIEGLDI	TKLLDNSGVG	FRARGDRFRN	120
ILDVIYMMQM	KKHLESVQOP	IPPEQQMNYG	ILKQKIALYE	DKLSRAGIVI	NNVDDIINLT	180
KSVKDLEKLD	KALNSIPTEL	KGAKGEQLEN	PRLTLGDLGR	LRELLPENK	HLIKNLSVVV	240
TNQTKEHLER	YSEDTPPQOS	IAQVVQWGA	HPVLFVPGRN	AKGEYIADGG	ILDNMPEIEG	300
LDREEVLCVK	AEAGTAFEDR	VNKAKQSAME	AISWFKARMD	SLVEATIGGK	WLHATSSVLN	360
REKVVYNIDN	MIYINTGEVT	TTNTSPTPEQ	RARAVKNGYD	QTMQLLDSHK	QTFDHPLEMAI	420
LYIGHDKLKD	ALIDEKSEKE	IFEASAHQA	ILHLQEQIVK	EMNDGDYSSV	QNYLDQIEDI	480
LTVDKAMDDI	QKEKAFALCI	KQVNFLESEG	LETYLNKVEA	EAKAAEPSW	ATKILNLLWA	540
PIEWVVSFLK	GPAQDFKVEV	QPEPVKVSTS	ENQETVSNQK	DINPAVEYRK	IIAEVRREHT	600
DPSPSLQKEK	RVGLSTTFGG	H				621

SEQ ID NO: 13 moltype = AA length = 211
FEATURE Location/Qualifiers

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source                1..211
                      mol_type = protein
                      organism = Pseudomonas syringae

SEQUENCE: 13
MNRVSGSSSA TWQAVNDLVE QVSERTTLST TGYQTAMGRL NKPEKSDADA LMTMRAQQY 60
TDSAKRTYIS ETLMLNLADLQ QRKIYRTNSG NLRGAIENTP TQLTDCVQKC REEGFSNCDI 120
QALEIGLHLR HKLGISDFTI YSNRKLSHNY VVIHPSNAFP KGAIVDSWTG QGVVELDFKT 180
RLKFKHREEN YAVNANMHEW IERYGQAHVI D 211

SEQ ID NO: 14          moltype = AA length = 204
FEATURE               Location/Qualifiers
source                1..204
                      mol_type = protein
                      organism = Pseudomonas syringae

SEQUENCE: 14
MGNICGTSGS RHVYSPSHTQ RITSAPSTST HVGGDTLTSI HQLSHSQREQ FLNMHDPMRV 60
MGLDHDTELF RTTDSRYIKN DKLAGNPQSM ASILMHEELR PNRFASHTGA QPHEARAYVP 120
KRIKATDLGV PSLNVMTGSL ARDGIRAYDH MSDNQVSVKM RLGDFLERGG KVVADASSVA 180
DDGETSQALI VTLPGQKQVP VERV 204

SEQ ID NO: 15          moltype = AA length = 493
FEATURE               Location/Qualifiers
source                1..493
                      mol_type = protein
                      organism = Pseudomonas syringae

SEQUENCE: 15
MQIKNSHLYS ASRMVQNTFN ASPKMEVTNA IAKNNEPAL SATQTAKTHE GDSKGQSSNN 60
SKLPFRAMRY AAYLAGSAYL YDKTANNFFL STTSLHDGKG GFTSDARLND AQDKARKRYQ 120
NNHSSTLENK NSLLSPLRLC GENQFLTMID YRAATKIYLS DLVDTEQAHT SILKNIMCLK 180
GELTNEEAIK KLNPEKTPKD YDLTNSEAYI SKNKYSLTGV KNEETGSTGY TSRSITKPFV 240
EKGLKHFIKA THGEKALTPK QCMETLDNLL RKSITLNSDS QFAAGQALLV FRQVYAGEDA 300
WGDAERVILK SHYNRGTVLQ DEADKIELSR PFSEQDLAKN MFKRNTSIAG PVLHYHAYIYI 360
QEKIFKLPPD KIEDLKHKSM ADLKNLPLTH VKLSNSGVGF EDASGLGDSF TALNATSCVN 420
HARIMSGEPP LSKDDVILI GCLNAVYDNS SGIRHSLREI ARGCFVGAGF TVQDGDFFYK 480
QICKNASKQF YNG 493

SEQ ID NO: 16          moltype = AA length = 530
FEATURE               Location/Qualifiers
source                1..530
                      mol_type = protein
                      organism = Vibrio cholerae

SEQUENCE: 16
MPKISVSQQA NVMSTSDTAQ RSSLKISIKS ICNKSLSKKL HTLAEKCRFF SQELKEHTAS 60
KKQIVEQATT TVRESSLTKS DSELGSSRSL LTSDVLSSSS SHEDLTAVNL EDNDSVFVTI 120
ESSSELIVQK DGSIPPAPPL PGNIPPAPPL PSAGNIPTAP GLPKQKATTE SVAQTSNDRS 180
KLMEETRQGV KLRATPKSSS TEKSASDPHS KLMKELINHG AKLKKVSTD IPVPPPLPAA 240
FASKPTDGRS ALLSEIAGFS KDRLRKAGSS ETLNVSQPTV AESSIPEAYD LLLSDEMFINL 300
SPKLSETELN TLADSLADYL FKAADIDWMQ VIAEQTKGST QATSLKSQLE QAPEYVKAFK 360
DEILKFPDCY KSADVASPES PKAGPSSVID VALKRLQAGR NRLFSTIDAK GTNELKKGEA 420
ILESAINAAR SVMTAEQKSA LLSSNVKSAT FKVFSSEPCM EGFABQNGKA AFNALRLAFY 480
SSIQSGDTAQ QDIARFMKEN LATGFSGYSY LGLTSRVAQL EAQLAALTTK 530

SEQ ID NO: 17          moltype = AA length = 288
FEATURE               Location/Qualifiers
REGION               1..288
                      note = Description of Unknown: YopJ toxin sequence
source                1..288
                      mol_type = protein
                      organism = unidentified

SEQUENCE: 17
MIGPISQINI SGGELSEKETS SLISNEELKN IITQLETDIS DGSWFHKNYS RMDVEVMPAL 60
VIQANNKYPE MNLNLVTSPL DLSIEIKNVI ENGVRSSRFI INMGEGGIHF SVIDYKHING 120
KTSILILFEP NFNSMGPAFL AIRTKTAIER YQLPDCHFMS VEMDIQRSSS ECGIFSFLA 180
KKLYIERDSL LKIHEDNIKG ILSDGENPLP HDKLDPLYLPV TFYKHTQGGK RLNEYLNTNP 240
QGVGTVVNKK NETIVNRFDN NKSIVDGKEL SVSVHKKRIA EYKTLCLKV 288

SEQ ID NO: 18          moltype = AA length = 579
FEATURE               Location/Qualifiers
source                1..579
                      mol_type = protein
                      organism = Pseudomonas sp.

SEQUENCE: 18
MAGINGAGPS GAYFVGHTDP EPASGGAGHS SSGASSNSP RLPAPPDAPA SQARDREML 60
LRARPLSRQT REWVAQGMPP TAEAGVPIRP QESAEEAAPQ ARAEERTHPE ADAAASHVRT 120
EGGRTPQALA GTSRPTGAV PHANRIVQQL VDAGADLAGI NTMIDNAMRR HAIALPSRTV 180
QSILIEHFPF LLAGELISGS ELATAFRAAL RREVRQOEAS APPRTAARSS VRTPERSTVP 240
PTSTESSSGS NQRTLLGRFA GLMTPNQRRP SSASNASASQ RPPVDRSPPRV NQVPTGANRV 300
VMRNHGNNEA DAALQGLAQQ GVDMEDLRRA LERHILHRPP IPMDIAYALQ GVGIAPSIDT 360
GESLMENPLM NLSVALHRAL GPRPARAQAP RPAVPVAPAT VSRRPDSARA TRLOQVIPARE 420

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DYENNVAYGV	RLLSLNPGAG	VRETVAAFPV	NRYERQAVVA	DIRAALNLK	QFNKLRTVSK	480
ADAASNKPGF	KDLADHPDDA	TQCLFGEELS	LTSSVQQVIG	LAGKATDMSE	SYSREANKDL	540
VFMDMKLLAQ	FLAGKPEHPM	TRETLNAENI	AKYAFRIVP			579

SEQ ID NO: 19 moltype = AA length = 1116
 FEATURE Location/Qualifiers
 REGION 1..1116
 note = Description of Unknown: SdbA toxin sequence
 source 1..1116
 mol_type = protein
 organism = unidentified

SEQUENCE: 19

MHKKYNYYSL	EKEKKTWFQH	ILDILKAPFR	LPGWVVSFFL	ARNITHVALN	PNNIPQQRLI	60
HLTKTSNRPE	DDIVVINFKK	RPPHKWFNDT	LIKIAINTIAA	LPFVTPRLRT	RLHYDNENDI	120
NHVNKLLAEI	DALVQGSQKQ	KYCKGRAFDW	SKIHLLKGLEF	LDPKMRGVVY	EQLHEKYGYV	180
SYTTKRKPNI	EFFTLKTPDG	SELDVQVTG	EDEEKKPMGE	RKFITCIAR	DQNFINWIKD	240
LNYYAKNLGA	TAISFNRYGV	DYSRGLVWTE	NNLVDDILAQ	VQRLISLGAD	PKNICLDGMC	300
IGGAVATIAA	AKLHEKGMKV	KLNNERSFTS	LSSLVFGFIV	PELQTANWWS	PLTYGRFLLA	360
GVVYALLTPL	IWLAWGPVDV	TKAWNRIPAQ	DKMYSVVRDK	DNGLYDGVVIH	DHFCSTIASLV	420
DSQINSILYK	LSTDQPLTEE	EKQILCDDQF	SHHFKPSQSV	LKNPKYKGP	FISRQDLVAE	480
LGHREEYTNH	DYFLDRLREK	FQLDRATRPV	ALAEDGEKDI	DGISSQLSNN	KERPLIIASS	540
GGTGHSATH	GIINDLQSKT	DNVVITQHHA	ELYKNKPFSI	TSVLIRIGVW	FTSLPILEDI	600
LKGVMRFIGY	PVLPSSSIFW	DQMSKIQQSE	TKKENGIETG	RTRPYVDMML	DIYPEGYEYT	660
AFNNATHLTS	SIEDIQTMS	FKGHVEEDNR	NIVYQNILQR	LMHAAKQNT	YTRLISTQAL	720
SLGAICDAVK	YNTVFLPVY	NAERGTSYQP	IAIDQYMTDL	PSLGCIFHMN	NLEELTSEQR	780
QLMEIHAVNM	SEPFKEAHFG	KEQGFKAVHN	IDPRNNPMIR	NAFKDPSLTK	YLDKTSQFSL	840
HPNVYKKEKQ	NALPVLNGKE	KITIKPHAKI	ASIMIGSLAA	NASADYAKYL	LNQGYEHIFL	900
FGGLNDSIAA	RIDQIINSYP	APTRDEIRKK	IILLGNQSDV	EMAPIMTRSN	CVVIRGGGLS	960
VMEQMAMPIM	DDKIVLLHHE	DNEEGPLTSG	LSWEDGNSDK	LIEYLSEKGA	YAKKTSPLGLC	1020
SGHLHEAEKS	FEKKYHGQLK	STETKKKVDL	TIPQOQETYS	LKKEWDRKTGY	TESGHILSHQ	1080
HRFFNTIPEV	REPFCSEKDL	HHNELSSQSL	VSVSAG			1116

SEQ ID NO: 20 moltype = AA length = 965
 FEATURE Location/Qualifiers
 source 1..965
 mol_type = protein
 organism = Legionella pneumophila

SEQUENCE: 20

MRSKDEVLE	ANDSLFGITV	QTWGTNDRPS	NGMMNFADQQ	FFGGDVGHAS	INMKLPVTDK	60
TKQWIEKYCY	SQTYDQFKKV	KGNEDKTYEE	YLKTAKRLLP	VELKTQVTRK	AQYDSNGNLV	120
TTHEKAYEQI	YFDIDWSWWP	GRLQNTEDDM	VWEREGKHFE	YDEKWKEYLQ	PEQVRVHRGKL	180
GRKMDYAPT	SIHQDRIPT	SELEKITRDH	KIHTIEEKL	VVKLLQSKID	EMPHTKMSPS	240
MELMPKNLGI	NVEKLLDETK	DNGVDPTNLE	AMREYLTNRL	TERKLELETE	LSEAKKEVDS	300
TQVKNKVEDV	YYDFEYKLNQ	VRKKMEEVNS	QLEKMDSLH	KLEGNTSGPI	PYTAIEDLM	360
SVLPFLKEEL	ELENGTSPK	SIENLIDHID	ELKNELASKQ	EKKNERNLNL	IKKYEELCEQ	420
YKDEEGLEE	ALWEEGIDVE	EVNSAKKDIS	KPAETIQKLT	DLQEQLRNHK	ESGVKLSSEL	480
EETLNSSVKM	WTKKIDSPCQ	VISESSVKAL	VSKINSTRPE	LVKEKEQLPE	QEESSLSKEAK	540
KAQELIKIQ	EFQFYSSENS	SAYMVIPLPP	HHQVSLPLAV	NGKRGLHPEA	MLKKMHLEVA	600
GPEKKEFNLH	TNNCSLTSIE	VLSAGAQHDP	LLHSIMGTRA	LGFFGTPQQV	LENAKLTSKT	660
INEGKKSNI	TPLVTASPLD	RALGYAMSIY	MDPEASKAKQ	NAGLALGVLV	GLAKTPGIII	720
GSLNPKQGF	NDILNTLNLV	YSRNTGLKV	GLTLMALPAM	IVLAPLAAIQ	KGVEVIAETI	780
AKPFKLIANL	FKQKPESTDF	ITVSVGSKKV	AEKEGSYSNT	ALAGLVNSKI	SKSIDENTIT	840
VEFQKSPQKM	IEEFESQLKE	NPGKVVLSE	KAHNAVLKPV	SKSDDEALKQ	KFYDCCNQSV	900
ARSQKFAPKT	RDEIDELVEE	VTSTDKTELT	TSPRQEPSMS	STIDEENID	SEHQIETGTE	960
STMRI						965

SEQ ID NO: 21 moltype = AA length = 665
 FEATURE Location/Qualifiers
 source 1..665
 mol_type = protein
 organism = Legionella pneumophila

SEQUENCE: 21

MKTQEVSSQ	DKLKDSKST	PLQTKETWFI	SDALNITFDP	YDFSISVTEQ	APMPYRIVFS	60
GGGSRILAH	GALDELTRHG	LKFTFSGSS	AGAMVAFAF	LGYNCSSEIK	IISWFNEDKL	120
LDSPLIFNPN	NIKQIFNKG	LSSAKLMRQA	ANYVILKKVM	DIISDEKFKT	RFQKQNFLE	180
ENIYRCPENI	TFQTLARIKE	ICPECELGEK	LFITGTNLST	QKHEVFSIDT	TPSMALADAI	240
IISANLPIAF	BRICQKNVY	SDGGISNNLP	AHCFSKGGHK	TTFLKHKDDV	DFSVLALQFD	300
NGLLENALYS	QNPPIKQSWL	SNTFYSLITG	HPNVTENWYE	DLQILRRHAH	QSILIKPTPTI	360
ALTNLTIQD	TKKALVESGR	TAAKTYLELH	EFYTDYGN	RHNECLHEKF	QKPEELLDYC	420
VLHSHFELK	KIKQAISCSQ	YLEKGYKHYL	CELCNLLP	QLKCPNEGSG	TEQPEIKLEK	480
DTIICEKN	SGLTFSMTFF	GVSPPLVKTL	NQDSPCLKIK	LFTGLYPILI	QNWQNLCPVS	540
GISGILNSIR	MSFVEISSTD	TCIKTLIDKL	NEIEIGHFLI	PVFKAALKNY	DKHDFILLK	600
NLKLHHSIE	LIRNKPFSHD	DRFYGQWSFE	GHPDKRILEF	IKSDDISGLM	TILEDKKALP	660
NNKPN						665

SEQ ID NO: 22 moltype = AA length = 246
 FEATURE Location/Qualifiers
 source 1..246
 mol_type = protein

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organism = Legionella pneumophila

SEQUENCE: 22
MVSLEHIQKL ISECRKLGKD GLDNGTNGLI PELEIDVVPV SAFLGVGNPN AIFVNSKTYK 60
LMRTTHEKVV ENKTIVFKSY LLSQPAIKII GAIVHETGHA FNVAAKIPNT EANACIFEIE 120
VLMRLFQVKS PLLLGCTELD MQSYFKSRLT DYNKCVKDCQ CLAEMVEFIT HQFKLDEVSI 180
SEKENQIPLL SISNKWPGLF AKKQIAPDMD KLLTSPVTIT PEVKILFYQL VKEHFHSPET 240
EIKLDI 246

SEQ ID NO: 23      moltype = AA length = 644
FEATURE            Location/Qualifiers
source             1..644
                   mol_type = protein
                   organism = Legionella pneumophila

SEQUENCE: 23
MYKIYSYLGW RIDMKTENLP QAGQEAQIDK KIHFIWVGHI MPQKNIQVVS EWAEKNPGYE 60
TIIWVDKKIA PAKELDLFIL DMKSKGITVK DINEEGVCRD SIRHELDQES PNYGMVSDML 120
RLNIIAAEGG IYLDSDILCS APFPDEIYAP FGFLSPWSQ GANNTLCNDI ILCSKGNQII 180
QQADAIEQS YIARDSFEFT HEYASMKETK GERIAKTLGV TGPGLFPHQL KKMGIILNKS 240
EMEAIHWELQ DQRYLIDGSV KEPDYFVVPQ NNTNDASWVP SIKRPGIENM SFQERLENAV 300
QLIAFDIQT GLFNLDHYAN ELKVKQNSWC IAAETSPELK PDSYLLIRPR DKTGEWTLYY 360
VDEKKLNPV TLPVIKGAIK LSEVSDPLRK FHTLLSQVSD PVNPTAHELK QIGRALIELK 420
PRQDEWHCKN KWSGAEBIAQ ELWQRTSNE TLRAQIKQCF TQFESLKPRV AELGLTRASG 480
AGTEVEAHES TVKEQEIIISQ NTVGEEGTKE KNSVQLASEN SSDEKIKTAH DLIDEIIQDV 540
IQLDGKLGLL GGNTRQLEDG RVINIPNGAA MIFDDYKKYK QGELTAESAL ESMIKIAKLS 600
NQLNRHTFFN QRQPETGQFY KKVAAIDLQT TIAAEYDNNH GLRI 644

SEQ ID NO: 24      moltype = AA length = 219
FEATURE            Location/Qualifiers
source             1..219
                   mol_type = protein
                   organism = Yersinia sp.

SEQUENCE: 24
MKISSFISTS LPLPTSVSGS SSVGEMSGRS VSQQTSDQYA NNLAGRTESE QGSSLASRII 60
ERLSSVAHSV IGFQIRMFSE GSHKPVVTPA PTPAQMPSPS SFSDSIKQLA AETLPKYMQQ 120
LNSLDAEMLQ KNHDQFATGS GPLRGSITQC QGLMQFCGGE LQAEASAILN TPVCGIPFSSQ 180
WGTIGGAASA YVASGVDLTQ AANEIKGLAQ QMQKLLSLM 219

SEQ ID NO: 25      moltype = AA length = 543
FEATURE            Location/Qualifiers
source             1..543
                   mol_type = protein
                   organism = Salmonella sp.

SEQUENCE: 25
MLKYEERKLN NLTLSFSKSV GVSNDARLYI AKENTDKAYV APEKFSSKVL TWLGKMPLFK 60
NTEVVQKHTE NIRVDQKIL QTFPLHALTEK YGETAVNDAL LMSRINMNKP LTQRLAVQIT 120
ECVKAADGEF INLIKSKDNV GVRNAALVIK GGDTKVAEKN NDVGAESKQP LLDIALKGLK 180
RTLPLQLEQMD GNSLRENFQE MASGNGPLRS LMTNLQNLNK IPEAKQLNDY VTTLTNIQVG 240
VARFSQWGTG GGEVERWVDK ASTHELTQAV KKIHVIAKEL KNVTALEKI EAGAPMPQTM 300
SGPTLGLARF AVSSIPINQQ TQVKLSDGMP VPVNTLTFDG KPVALAGSYP KNTPDALAEH 360
MKMLLEKECS CLVVLTSEDQ MQAKQLPPYP RGSYTFGEVH TNSQKVSSAS QGEAIDQYNM 420
QLSCGEKRYT IPVLVHKNWP DHQPLPSTDQ LEYLADRVKN SNQNGAPGRS SSDKHLPMIH 480
CLGGVGRTGT MAAALVLKDN PHSNLEQVRA DFRDSRNNRM LEDASQFVQL KAMQAQLLMT 540
TAS 543

SEQ ID NO: 26      moltype = AA length = 240
FEATURE            Location/Qualifiers
source             1..240
                   mol_type = protein
                   organism = unidentified
                   note = Salmonella typhimurium

SEQUENCE: 26
MTNITLSTQH YRIHRSDVEP VKEKTEKDI FAKSITAVRN SFISLSTSLS DRFSLHQQTD 60
IPTTHFHRGN ASEGRAVLTS KTVKDFMLQK LNSLDIKGNA SKDPAYARQT CEAILSAVYS 120
NNKDQCKLLK ISKGVSTPFP LKEIGEAAQN AGLPGEIKNG VFTPGGAGAN PFVVPPIASA 180
SIKYPHMFN HNQQVSFKAY AEKIVMKEVT PLFNKGTMPPT PQQFQLTIEN IANKYLQNAS 240

SEQ ID NO: 27      moltype = AA length = 561
FEATURE            Location/Qualifiers
source             1..561
                   mol_type = protein
                   organism = unidentified
                   note = Salmonella typhimurium

SEQUENCE: 27
MQIQSFYHSA SLKTQEAFKS LQKTLYNGMQ ILSGQKGAPA KAPDARPEII VLREPGATWG 60
NYLQHQKASN HSLHNLNLYQ RDLLTVAATV LGKQDPVLTS MANQMELAKV KADRPATKQE 120
EAAAKALKKN LIELIAARTQ QQDGLPAKEA HRFAAVAFRD AQVKQLNNQP WQTIKNTLTH 180
NGHHYNTQQL PAAEMKIGAK DIFPSAYEGK GVCSWDTKNI HHANNLWMST VSVHEDGKDK 240
TLFCGIRHGV LSPVHEKDPL LRHVGAENKA KEVLTAALFS KPELLNKALA GEAVSLKLVS 300
VGLLTASNIF GKEGTMVEDQ MRAWQSLTQP GKMIHLKIRN KDGLDQTVKI KPDVAAPNVG 360

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VNELALKLGF	GLKASDSYNA	EALHQLLGND	LRPEARPGGW	VGEWLAQYPD	NYEVVNTLAR	420
QIKDIWKNNQ	HHKDGGEFYK	LAQRLAMLAH	EIDAVPAWNC	KSGKDRTGMM	DSEIKREIIS	480
LHQTHMLSAP	GSLPDSGGQK	IFQKVLLNSG	NLEIQKQNTG	GAGNKVMKNL	SPEVLNLSYQ	540
KRVGDENIWQ	SVKGISSLIT	S				561

SEQ ID NO: 28 moltype = AA length = 685
 FEATURE Location/Qualifiers
 source 1..685
 mol_type = protein
 organism = unidentified
 note = Salmonella typhimurium

SEQUENCE: 28

MVTSVRTQPP	VIMPGMQTEI	KTOATNLAAN	LSAVRESATT	TLSGEIKGPQ	LEDFFALIKQ	60
ASLDALFKCG	KDAEALKEVF	TNSNNVAGKK	AIMEFAGLFR	SALNATSDSP	EAKTLLMKVG	120
AEYTAQIIKD	GLKEKSAFGP	WLPETKKAEA	KLENLEKQLL	DIKNNNTGGE	LSKLSTNLVM	180
QEVMPYIASC	IEHNFGCTLD	PLTRSNLTHL	VDKAAAKAVE	ALDMCHQKLT	QEQGTSGVRE	240
ARHLEMQTLI	PLLLRNVFAQ	IPADKLDPDPK	IPEPAAGPVP	DGGKKAEPGT	INININIDSS	300
NHSVDNSKHI	NNSRSHVDNS	QRHIDNSNHD	NSRKTIDNSR	TFIDNSQRNG	ESHHSTNSSN	360
VSHSHSRVDS	TTHQTETAHS	ASTGAIDHGI	AGKIDVTAHA	TAEAVTNASS	ESKDGKVVTS	420
EKGTTGETTS	FDEVDGVTSK	SIIGKPVQAT	VHGVDNKKQ	SQTAEIVNVK	PLASQLAGVE	480
NVKTDTLQSD	TTVITGNKAG	TTDNDNSQTD	KTGPFSGLK	KQNSFLSTVP	SVTNMHSMHF	540
DARETFGLVI	RKALEPDTST	PFPVRRAFDG	LRAEILPNDT	IKSAALKAQC	SDIDKHPELK	600
AKMETLKEVI	THHPQKEKLA	EIALQFAREA	GLTRLKGETD	YVLSNVLDGL	IGDGSWRAGP	660
AYESYLNKPG	VDRVITTVDG	LHMQR				685

SEQ ID NO: 29 moltype = AA length = 732
 FEATURE Location/Qualifiers
 source 1..732
 mol_type = protein
 organism = Yersinia pseudotuberculosis

SEQUENCE: 29

MKSVKIMGTM	PPSISLAKAH	ERISQHWQNP	VGELNIGGKR	YRIIDNQVLR	LNPMSGFSLF	60
REGVGKIFSG	KMFNFSIARN	LTDTLHAAQK	TTSQELRSDI	PNALSNLFGA	KPQTELP LGW	120
KGEPLSGAPD	LEGMRVAETD	KFAEGESHIS	IIETKDKQRL	VAKIERSIAE	GHLFAELEAY	180
KHIYKTAGKH	PNLANVHGMA	VVPYGNRKEE	ALLMDEV DGW	RCSDTLRTLA	DSWKQGGKINS	240
EAYWGTIKFI	AHRLLDVTNH	LAKAGVVHND	IKPGNVV FDR	ASGEPVVIDL	GLHSRSGEQP	300
KGFTESFKAP	ELGVGNLGAS	EKSDVFLVVS	TLLHCIEGFE	KNPEIKPNQG	LRFITSEPAH	360
VMDENGYPHI	RPGIAGVETA	YTRFITDILG	VSADSRPDSN	EARLHEFLSD	GTIDEESAKQ	420
ILKDTLTGEM	SPLSTDV RRI	TPKKLRELS	LLRTHLSSAA	TKQLDMGGVL	SDLDTMLVAL	480
DKAEREGGVD	KDQLKSFNSL	ILKTYRVIED	YVKGREGDTK	NSSTEVSPYH	RSNFMLSIVE	540
PSLQRIQKHL	DQTHSFSDIG	SLVRAHKHLE	TLLVLVLTLS	QQGQPVSSST	YGFLNRLAEA	600
KITLSQQLNT	LQQQQESAKA	QLSILINRSG	SWADVARQSL	QRFDSTRPVV	KFGTEQYTAI	660
HRQMMAAHAA	ITLQEVSEFT	DDMRNFTVDS	IPLLIQLGRS	SLMDEHLVEQ	REKLRELTTI	720
AERLNRLERE	WM					732

SEQ ID NO: 30 moltype = AA length = 529
 FEATURE Location/Qualifiers
 source 1..529
 mol_type = protein
 organism = Yersinia sp.

SEQUENCE: 30

MFINPRNVSN	TFLQEPLRHS	SNLTEMPVEA	ENVKSKEYY	NAWSEWERN	PPGNGEQREM	60
AVSRLRDCLD	RQAHELELNN	LGLSSLPPEL	PHLESIVASC	NSLTPELPEL	QSLKSLQVEN	120
NNLKALPDLP	PSLKKLHVRE	NDLTDLPPEL	QSLESRLVDN	NNLKALSDLP	PSLEYLTASS	180
NKLEELPELQ	NLPFLAAIYA	DDNNLETLDP	LPPSLKKLHV	RENDLTDLPE	LPQSLESLOV	240
DDNNLKALSD	LPPSLEYLTA	SSNKLEELPE	LQNLPLFLAAI	YADNNLLET	PDLPPHLEIL	300
VASYNSTEL	PELPQSLKSL	RVDNNNLKAL	SDLPPSLEYL	TASSNKLEEL	PELQNLPLFLA	360
AIYADNNLLE	TLPLDPPSLK	KLHVRENDLT	DLPELPQSLT	FLDVSDNNIS	GLSELPPNLY	420
YLDASSNEIR	SLCDLPPLSV	DLNVKSNQLS	ELPALPPHLE	RLIASFNLYA	EVPELPQNLK	480
QLHVEQNALR	EFPDIPESLE	ELEMDSERVV	DPYEFAHETT	DKLEDDVFE		529

SEQ ID NO: 31 moltype = AA length = 35
 FEATURE Location/Qualifiers
 REGION 1..35
 note = Description of Unknown: Amatoxin sequence
 source 1..35
 mol_type = protein
 organism = unidentified

SEQUENCE: 31

MSDINATRLP	IWGIGCNPCV	GDDVTTLTTR	GEALC	35
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SEQ ID NO: 32 moltype = AA length = 34
 FEATURE Location/Qualifiers
 source 1..34
 mol_type = protein
 organism = unidentified
 note = Amanita phalloides

SEQUENCE: 32

MSDINATRLP	AWLVDCPCVG	DDVNRLLTRG	ESLC	34
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SEQ ID NO: 33 moltype = AA length = 291
 FEATURE Location/Qualifiers
 source 1..291
 mol_type = protein
 organism = Ustilago maydis

SEQUENCE: 33

MIKPERSILT	ILIGILCLLA	YVLANGEPHD	GDNEWSSYCS	DQGFRRSDDG	LVTTPDVQGE	60
SIGKNSINGS	ELVDYLQCLK	VRLNGQKQVV	SNDGWLLLLV	QEPSVNVVTK	AMSECNYNVS	120
SGHKAGSYIQ	VTNTPADYKV	ISRRGSYEGD	QLPEDVKPYF	GVQKTSYRYP	ISKRINPNLT	180
LRQLAYNFAA	LNMCSLWCNS	CISRSCPYYI	AELTVHVNMI	HHGTVWLHHF	CRNASPQGGN	240
LYSTLTISHK	DTAYYVGTGW	WKVRSTAATT	NDVAGDWYPA	SWNQYWCAPH	Y	291

SEQ ID NO: 34 moltype = AA length = 219
 FEATURE Location/Qualifiers
 source 1..219
 mol_type = protein
 organism = Ustilago maydis

SEQUENCE: 34

MLIFSVLMYL	GLLLAGASAL	PNGLSPRNNA	FCAGFGLSCK	WECWCTAHGT	GNELRYATAA	60
GCGDHLISKAY	YDARAGHCLF	SDDLNRNQFYS	HCSSLNNNMS	CRSLSKRTIQ	DSATDVTDLG	120
AELHRDDPPP	TASDIGRKGK	RPRPVMCQCV	DTTNGGVRLD	AVTRAACSID	SFIDGYYTEK	180
DGFCRAKYSW	DLFTSGQFYQ	ACLRYSHAGT	NCQPDQPQYE			219

SEQ ID NO: 35 moltype = AA length = 316
 FEATURE Location/Qualifiers
 source 1..316
 mol_type = protein
 organism = Saccharomyces cerevisiae

SEQUENCE: 35

MTKPTQVLVR	SVSILFFITL	LHLVVALNDV	AGPAETAPVS	LLPREAPWYD	KIWEVKDWLL	60
QRATDGNWGK	SITWGSFVAS	DAGVVIFGIN	VCKNCVGERK	DDISTDCGKQ	TLALLVSIFV	120
AVTSGHHLIW	GGNRFPVSQSD	PNGATVARRD	ISTVADGDIP	LDFSALNDIL	NEHGISILPA	180
NASQVVKRSD	TAEHTTSFVV	TNNYTSLHTD	LIHHGNGTYT	TFTTPHIPAV	AKRYVYPMCE	240
HGIKASYCMA	LNDAMVSANG	NLYGLAEKLF	SEDEGQWETN	YYKLYWSTGQ	WIMSMKFIEE	300
SIDNANNDFE	GCDTGH					316

SEQ ID NO: 36 moltype = AA length = 296
 FEATURE Location/Qualifiers
 source 1..296
 mol_type = protein
 organism = Saccharomyces cerevisiae

SEQUENCE: 36

MGHAILLFSI	IAVLNIATAV	ASSDSIYLKG	HRVGQDIDSL	YRVYDNGTMY	PVTFNEWLND	60
LTMNDLATN	NATILKRDS	DVSCVTETCQ	YVDYHVDDEG	VITIDISTYR	IPVEWDSGSA	120
GNASVGVSKR	TKYETETCKK	KICGINVSGF	CNAYDFAVHA	FDFGGSVYNP	VSGITDRIKE	180
ATKRDKTECL	GYELDHVRID	PAVDWSISIS	TWKQGSANCD	TQASADSLKC	AAQKALESEH	240
NHQKTAFCIH	LDNGGSFNLD	IRLISELSFS	KYNPWALPCP	KYKGSNSWQV	VSDCFQ	296

SEQ ID NO: 37 moltype = AA length = 708
 FEATURE Location/Qualifiers
 source 1..708
 mol_type = protein
 organism = Saccharomyces cerevisiae

SEQUENCE: 37

MPRFAIFAL	LIAYSLFLST	LFTGSIPDRA	NTVTSNAPCQ	VVIWDWIRTR	RICNCCSRLC	60
YSLGGRSNLS	RTAKRGVCTI	AGAVLATAAV	IVA AVLVGKS	SGSATKRGLT	KTISVLNHTI	120
PPTDHILNGQ	TLNNGTGSNF	VTIGFSGYAV	HATIKRASTT	DIISWVIPES	MEPTLARVAS	180
YVSSSSINLA	AVPDTGGNAS	ALSFQNAVQE	FATSWVSMY	DQSYGDLRNV	ANDEGGEEIL	240
ILMRKRSYRI	SFQVIETGST	ALLLRTRRVV	SQLITMTYLV	TVQARVGIQI	GDIFQHYGGI	300
DNYVMTSISV	LRTLEDKAFH	ENKLLIVREP	PNKSNQDANQ	SYRLRPFSAN	DLIQNLKSVD	360
IGFLAFCSFF	DKYAHYPEII	MMKITIFISK	GNLWSIIYVI	QARYVRKRV	KVRGQMPGGL	420
LTNMESLLNI	VSTPNLNISE	FHIQTHSMSQ	SKPMYFQKQC	YSSQNNIIYI	YNSIHITCGA	480
VVVIHVDVRT	PSVFVLIELR	NCKPLKNSWC	ETTKTSPRDT	KIKKNEYNET	VCRRAGALLD	540
GRVRTIRFLM	MRTHWSRVKG	VSCNTANRLS	RPCNHVVSYY	PSQNATIHLL	PTSLRAESLE	600
QQYTTRPLSS	SNNRFCLKS	IFINNCKKAC	ESPSLVSCNL	QQTAEELLMVY	YLYICEACYV	660
SRNHDLLSKQ	CMSTVRAYV	ARMRLPKFRS	TFPCMPRLCW	LVNGVVVV		708

SEQ ID NO: 38 moltype = AA length = 768
 FEATURE Location/Qualifiers
 source 1..768
 mol_type = protein
 organism = Bacillus anthracis

SEQUENCE: 38

MHVKEKEKNK	DENKRKDEER	NKTQEEHLKE	IMKHIVKIEV	KGEEAVKKEA	AEKLEKVP	60
DVLEMYKAIG	GKIYIVDGD	TKHISLEALS	EDKKIKDIY	GKDALLHEHY	VYAKEGYEPV	120
LVIQSSSEYV	ENTEKALNVY	YEIGKILSRD	ILSKINQPYQ	KFLDVLNTIK	NASDSGQDL	180
LFTNQLKEHP	TDFSVEFLEQ	NSNEVQEVFA	KAFAYYIEPQ	HRDLVLQYAP	EAFNYMDKFN	240
EQEINLSLEE	LKDQRMLSR	EKWEKIKQHY	QHWSDSLSEE	GRGLLKLQI	PIEPKKDDII	300

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HLSQEEKEFL	LKRIQIDSSD	FLSTEEKEFL	KKLQIDIRDS	LSEEEKELLN	RIQVDSSNPL	360
SEKEKEFLKK	LKLDIQPYDI	NQRLQDTGGL	IDSPSINLDV	RKQYKRDIQN	IDALLHQSIG	420
STLYNKIYLY	ENMNINNLTA	TLGADLVDS	DNTKINRGIF	NEFKKNFKYS	ISSNYMIVDI	480
NERPALDNER	LKWRIQLSPD	TRAGYLENGK	LILQRNIGLE	IKDVQIIKQS	EKEYIRIDAK	540
VVPKSKIDTK	IQEAQLNINQ	EWNKALGLPK	YTKLITFNVH	NRYSNIVES	AYLILNEWKN	600
NIQSDLIKKV	TNYLVDNGNR	FVFTDITLPN	IABQYTHQDE	IYEQVHSGKL	YVPESRSILL	660
HGPSKGVLEL	NDSGFIHEF	GHAVDDYAGY	LLDKNQSDLV	TNSKKFIDIF	KEEGSNLTSY	720
GRTNEAEFFA	EAFRLMHSTD	HAERLKVQKN	APKTFQFIND	QIKFIINS		768
SEQ ID NO: 39	moltype = AA length = 319					
FEATURE	Location/Qualifiers					
REGION	1..319					
	note = Description of Unknown: Shiga toxin sequence					
source	1..319					
	mol_type = protein					
	organism = unidentified					
SEQUENCE: 39						
MKCIILLKWVL	CLLLGFSSVS	YSREFTIDFS	TQSSYVSSLN	SIRTEISTPL	EHISQGTTSV	60
SVINHTPPGS	YFAYDIRGLD	VYQARFDHLR	LIIEQNNLVY	AGFVNTATNT	FYRFSDFAHI	120
SVPGVTTVSM	TTDSSYTTLQ	RVAALERSGM	QISRHSVLSS	YLALMEFSGN	TMTRDASRAV	180
LRFVTVTAEA	LRFRQIQREF	RQALSETAPV	YTMTPGDVDL	TLNWGRISNV	LPEYRGEDGV	240
RVGRISFNMI	SAILGTVAVI	LNCHHQGARS	VRAVNEESQP	ECQITGDRPV	IKINNTLWES	300
NTAAAFNLNRK	SQSLYTTGE					319
SEQ ID NO: 40	moltype = AA length = 299					
FEATURE	Location/Qualifiers					
source	1..299					
	mol_type = protein					
	organism = Saponaria officinalis					
SEQUENCE: 40						
MKSWIMLVVT	WLIILOTTVT	AVIIEYELNLQ	GTTKAQYSTF	LKQLRDDIKD	PNLHYGGTNL	60
PVIKRPVGGP	KFLRVNLKAS	TGTVSLAVQR	SNLYVAAYLA	KNNKQFRAY	YFKGFQITTN	120
QLNNLFPEAT	GVSNQQLGY	GESYPQIQNA	AGVTRQQAGL	GIKKLAESMT	KVNGVARVEK	180
DEALPLLIIV	QMVGEAARFK	YIENLVNNF	DTAKEVEVPV	DRVIILENW	GLLSRAAKTA	240
NNGVPQTPLV	LTSYAVPGVE	WRVTTVAEVE	IGIFLNVDMN	GLPSIIYNNI	ISGAPGDTY	299
SEQ ID NO: 41	moltype = AA length = 565					
FEATURE	Location/Qualifiers					
source	1..565					
	mol_type = protein					
	organism = Saponaria officinalis					
SEQUENCE: 41						
MYAVATWLCF	GSTSGWSFTL	EDNNIFPKQY	PIINFTTAGA	TVQSYTNFIR	AVRGRLTGTA	60
DVRHDIPVLP	NRVGLPINQR	FILVELSNHA	ELSVTLALDV	TNAYVVGYRA	GNSAYFFHPD	120
NQEDAEATH	LFTDVQNYRT	FAPGGNYDRL	EQLAGNLREN	IELGNGLPEE	AISALYYYST	180
GGTQLPTLAR	SFIICQMIS	EAARFQYIEG	EMRTRIRYNR	RSAPDPSVIT	LENSWGRLST	240
AIQESNQGAF	ASPIQLQRRN	GSKFSVYDVS	ILIPIIALMV	YRCAPPPSSQ	FSLIRPVVP	300
NFNADVCMDD	EPIVRIVGRN	GLCVDVDRGR	PHNGNAIQLW	PCKSNTDANQ	LWTLKRDNTI	360
RSNGKCLTTY	GYSYGVVYMI	YDCNTAATDA	TRWQIWDNGT	IINPRSSLVL	AATSGNSGTT	420
LTVQTNIIAV	SQGWLPNTNT	QPFVTTIVGL	YGLCLQANSQ	QVWIEDCSSE	KAEQQWALYA	480
DGSIRPQQNR	DNCLTSDSNI	RETUVKILSC	GPASSGQWRM	FKNDGTILNL	YSGLVLDVRR	540
SDPSLKQIIL	YPLHGDPNQI	WLPLF				565
SEQ ID NO: 42	moltype = AA length = 730					
FEATURE	Location/Qualifiers					
source	1..730					
	mol_type = protein					
	organism = unidentified					
	note = Galerina marginata					
SEQUENCE: 42						
MSSVTWAPGN	YPSTRRSDHV	DTYQSASKGE	VPVPDPYQWL	EESTDEVKDW	TTAQADLAQS	60
YLDQNADIQK	LAEKFRASRN	YAKFSAPTLL	DDGHWYWFYN	RGLQSQSVLY	RSKEPALPDF	120
SKGDDNVGVD	FFDPNVLAAD	GSAGMVLCKF	SPDGKFFAYA	VSHLGGDYST	IYVRSTSSPL	180
SQASVAQGVV	GRLSDEVKWF	KFSTIIWTKD	SKGFLYQRYP	ARERHEGTRS	DRNAMMCYHK	240
VGTTQEEDII	VYQDNEHPEW	IYGADTSEDG	KYLYLYQFKD	TSKKNLLWVA	ELDEDGVKSG	300
IHWKVVNEY	AADYNIITNH	GSLVYIKTNL	NAPQYKVTI	DLSKDEPEIR	DFIPEEKDAK	360
LAQVNCANEE	YFVAIYKRN	KDEIYLYSKA	GVQLTRLAPD	FVGAASIANR	QKQTHFFLTL	420
SGFNTPGTIA	RYDFTAPETQ	RFSILRTTKV	NELDPDDFES	TQVWYESKDG	TKIPMFIVRH	480
KSTKFDGTAA	AIQYGYGGFA	TSADPPFSPI	ILTFLLQTYGA	IFAVPSIRGG	GEFGEEWHKG	540
GRRETQVNTF	DDFIAAAQFL	VKNKYAAPGK	VAINGASNGG	LLVMGSIVRA	PEGTFGAAPV	600
EGGVADLLKF	HKFTGGQAWI	SEYGNPSIPE	EPDYIYPLSP	VHNVRTDKVM	PATLITVNI	660
DGRVVPMSHF	KFIATLQHN	PQNPHPLLIK	IDKSWLGHGM	GKPTDKNVKD	AADKWGFIAR	720
ALGLELKTVE						730
SEQ ID NO: 43	moltype = AA length = 730					
FEATURE	Location/Qualifiers					
source	1..730					
	mol_type = protein					
	organism = unidentified					

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note = *Amanita bisporigera*

SEQUENCE: 43

MPPTPWAPHS	YPPTRRSDHV	DVYQSASRGE	VPVPDPYQWL	EENSNEVDEW	TTAQTAFQTG	60
YLDKNADRQK	LEEKFRASKD	YVKFSAPTLL	DSGHWYWFYN	SGVQSQAVLY	RSKKPVLPDF	120
QRGTRKVGGEV	YFDPNVLSAD	GTAIMGTCRF	SPSGEYFAYA	VSHLGVDFYFT	IYVRPTSSSL	180
SQAPEAEGGD	GRLSDGVKWC	KFTTITWTKD	SKGFLYQRYP	ARESLVAKDR	DKDAMVCYHR	240
VGTTQLEDII	VQQDKENPDW	TYGTDASEDG	KYIYLVVYKD	ASKQNLLWVA	EFDKDGVKPE	300
IPWRKVINEF	GADYHVITNH	GSLIYVKTNV	NAPQYKVVTI	DLSTGEPEIR	DFIPEQKDAK	360
LTQVKCVNKG	YFVAIYKRN	KDEIYLYSKA	GDQLSRLASD	FIGVASITNR	EKQPHSFLTF	420
SGFNTPGTIS	RYDFTAPDTQ	RLSILRTTKL	NGLNADDDES	TQVWYKSKDG	TKVPMFIVRH	480
KSTKFDGTAP	AIQNGYGGFA	ITADPPFFSPI	MLTFMQTYGA	ILAVPNIRGG	GEFGGEWHKA	540
GRRETGKNTF	DDFIAAAQFL	VKNKYAAPGK	VAITGASNGG	FLVCGSVVRA	PEGTFGAAVS	600
EGGVADLLKF	NKFTGGMAWT	SEYGNPFIKE	DFDFVQALSP	VHNVPKDRVL	PATLLMTNAG	660
DDRVVPMHSL	KFVANLQYNV	PQNPHELLIR	VDKSWLGHGF	GKTTDKHTKD	AADKWSFVAQ	720
SLGLEWKTVD						730

SEQ ID NO: 44 moltype = AA length = 734

FEATURE Location/Qualifiers

source 1..734

 mol_type = protein

 organism = unidentified

 note = *Hypsizygus marmoreus*

SEQUENCE: 44

MAISPTPWTP	NTYPTTRSS	HVDIYKSATR	GEVVRADPYQ	WLEENTEETD	KWTTAQEETF	60
RSYLDKNTDR	QRLEDAFRTS	TDYAKFSSPT	LYEDGRWYWF	YNSGLQPQPL	IYRSKGKTLF	120
DPSQDDNVVG	EVFFDPNLLS	DDGTAALSIY	DFSDCGKYFA	YGISFSGSDF	STIYVRSTES	180
PLAKNKSST	DDRLSDEIK	HVKFSAVTWT	KDSKGFFYQR	YPAHENAKEG	IETGGDVTAM	240
IYHVHVTGQ	SEDLVHSDK	SNPEWMNSID	ITEDGKYLIL	YTMKDSSRKN	LMWIALSKN	300
EIGPNIQWNK	IIDVFDAEYH	LITNDGPILY	VKTNADAPQY	KLVMTDISGD	KDISRDLIPE	360
DKNANLVQVD	CVNRDTFAVI	YKRNVDDEIY	LYSKTGIQLS	RLASDFVGAA	SISSREKQPH	420
FFVTMTGFTS	PGTVARYDFG	APEEQRWSIY	RSVKVGNLNP	DDFESKQVWY	ESKDGTKIMP	480
FIVRHKATKF	DGTAPAIQYG	YGGFSISINP	FFSPTILTFL	QTYGAVLAVP	NIRGGAEFGE	540
DWHKAGTREK	KGNVDFDFA	ATQYLKKNKY	AGEGKVAING	GSNGGLLVGA	CINRAPEGTF	600
GAAVAEGVM	DLLEKSKFTI	GKAWTSYDYG	PDDPKDFDFI	CPLSPLHNIP	TDRLVPPTML	660
LTADHDDRVP	PMHSFKHAAT	LQYTLPHNPH	PLVIRIDKKA	GHGAGKSTEK	RIKESADKWG	720
FVAQSLGLVW	QEPA					734

SEQ ID NO: 45 moltype = AA length = 733

FEATURE Location/Qualifiers

source 1..733

 mol_type = protein

 organism = unidentified

 note = *Conocybe apala*

SEQUENCE: 45

MPPSTPNEYF	PTRRSDDVLT	YRSEKNGEVV	VPDPYQWLEH	NTEETDKWTT	AQAAFTRAHL	60
DKNPKRNALE	EAFTAANDYA	KFSAPQLHDD	GRWYWYNTG	LQAQTCLEWRT	RDDTIPDFSK	120
QLDEDVGEIF	FDPNALSCKG	TAALSTYRFS	RDGKYFAYAI	AQSGSDFNTI	YVRPTDSPLT	180
KRDESGRDP	RLADEVKFVK	FSGITWAPNS	EGFFYQRYPH	IDGATLEEGL	IATRRDLHAM	240
VYVHRVGTQ	SEDLIHRDP	ANPEWMFQVN	VTNDGEYIEL	YISKDSSRKN	MLWVANFAMN	300
KIGEQQWRK	VINDFAAEYD	VITNHGPPVY	FRTDDGAPKH	KILSINIDTN	ERKLLVPESE	360
DAALFSTVCV	NKNYMALIYK	RNVKDEVHLY	TLEGKPVRL	AEDFVGACTI	SGKEKQPWF	420
VTMSGFTSP	TVGRYNFQIP	EEENRWSIFR	AAKIKNLNPN	DFEASQVWYK	SKDGTNVPMF	480
IVRHKSTQFD	GTAPALQYGY	GGFSISIDPF	FSASILTFKL	VYGAILVVP	IRGGNEFGEE	540
WHRGGMKQNK	VNCFDDFIAA	TNHLVEHKYA	APGKVAINGG	SNGGLLVAAC	INRAPEGTFG	600
AAIAEVGVHD	MLKFHKFTIG	KAWTSYDGNP	DDPHDFDIYI	PISPVHNVPT	DKILPPTLLL	660
TADHDDRVP	MHTFKLAATL	QHTLPHNPH	LLLVRDKKAG	HGAGKPLQLK	IREQADKWGF	720
VQSQQLVWR	DGV					733

SEQ ID NO: 46 moltype = AA length = 730

FEATURE Location/Qualifiers

source 1..730

 mol_type = protein

 organism = unidentified

 note = *Amanita bisporigera*

SEQUENCE: 46

MPPTPWAPHS	YPPTRRSDHV	DVYQSASRGE	VPVPDPYQWL	EENSNEVDEW	TTAQTAFQTG	60
YLDKNADRQK	LEEKFRASKD	YVKFSAPTLL	DSGHWYWFYN	SGVQSQAVLY	RSKKPVLPDF	120
QRGTRKVGGEV	YFDPNVLSAD	GTAIMGTCRF	SPSGEYFAYA	VSHLGVDFYFT	IYVRPTSSSL	180
SQAPEAEGGD	GRLSDGVKWC	KFTTITWTKD	SKGFLYQRYP	ARESLVAKDR	DKDAMVCYHR	240
VGTTQLEDII	VQQDKENPDW	TYGTDASEDG	KYIYLVVYKD	ASKQNLLWVA	EFDKDGVKPE	300
IPWRKVINEF	GADYHVITNH	GSLIYVKTNV	NAPQYKVVTI	DLSTGEPEIR	DFIPEQKDAK	360
LTQVKCVNKG	YFVAIYKRN	KDEIYLYSKA	GDQLSRLASD	FIGVASITNR	EKQPHSFLTF	420
SGFNTPGTIS	RYDFTAPDTQ	RLSILRTTKL	NGLNADDDES	TQVWYKSKDG	TKVPMFIVRH	480
KSTKFDGTAP	AIQNGYGGFA	ITADPPFFSPI	MLTFMQTYGA	ILAVPNIRGG	GEFGGEWHKA	540
GRRETGKNTF	DDFIAAAQFL	VKNKYAAPGK	VAITGASNGG	FLVCGSVVRA	PEGTFGAAVS	600
EGGVADLLKF	NKFTGGMAWT	SEYGNPFIKE	DFDFVQALSP	VHNVPKDRVL	PATLLMTNAG	660
DDRVVPMHSL	KFVANLQYNV	PQNPHELLIR	VDKSWLGHGF	GKTTDKHTKD	AADKWSFVAQ	720
SLGLEWKTVD						730

-continued

SEQ ID NO: 47 moltype = AA length = 738
 FEATURE Location/Qualifiers
 source 1..738
 mol_type = protein
 organism = unidentified
 note = Lentinula edodes

SEQUENCE: 47

MFSATQESPT	MSVPQWDPPY	PVSRDETS	AI	TYQSKLCGSV	TVRDPYSALE	VPFDDSEETK	60
AFVHAQRKFA	RTYLDEIPDR	ETWLQTLKES		WNYRRFTVPK	RESDGYTYFE	YNDGLQSQMS	120
LRRVKVSEED	TILTESGPGG	ELFFDPNLLS		LDGNAALTGS	MMSPCGYWA	YGVSEHGS	180
MTTYVRKTSS	PHMPSQEKKG	DPGRMDDVIR		YSRFFIVYWS	SDSKGFFYSR	YPPEDDEGKG	240
NTPAQNCMVY	YHRLGEKQEK	DTLVYEDPEH		PFWLWALQLS	PSGRYALLTA	SRDASHTQLA	300
KIADIGTSDI	QNGIQWLTIH	DQWQARFVII		GDDDDSTIYFM	TNLEAKNYLV	ATLDIRHSEA	360
GVKTLVAENP	DALLISASIL	STDKLVLVYL		HNARHEIHVH	DLNTGKPIRQ	IFDNLIGQFS	420
LSGRRDDNDM	FVFHSGFTSP	GTIYRFRLLNE		DSNKGTLFRA	VQVPGLNLSD	FTTESVFYPS	480
KDGTPIHMF	TRLKDTFVDG	TAPVYIYGYG		GFALAMLPTF	SVSTLLFCKI	YRAMYVVPNI	540
RGGSEFGESW	HREGMLDKKQ	NVFDDFNAAT		KWLVANKYAN	KYNVAIRGGS	NGGVLTTACA	600
NQAPELYRCV	ITIGGIIDML	RFPKFTFGAL		WRSEYGDPE	PEDFDFIYKY	SPYHNIPSGD	660
VVLPAMLFFT	AAyddrvsP	HSPKHVAALQ		YNFPNGPNFV	LMRIDLNTGH	PAGKSTQKML	720
EETADEYRCD	LLCCNLQL						738

SEQ ID NO: 48 moltype = AA length = 723
 FEATURE Location/Qualifiers
 source 1..723
 mol_type = protein
 organism = unidentified
 note = Omphalotacae olearis

SEQUENCE: 48

MSFPGWGPYP	PVERDETS	AI	TYSSKLHGSV	TVRDPYSQLE	VPFEDSEETK	AFVHSQRKFA	60
RTYLDENPDR	EAWLETLLKS		WNYRRFSALK	PESDGHYYFE	YNDGLQSQLS	LYRVRMGEE	120
TVLTESGPGG	ELFFNPPLL		LDGNAALTGF	VMSPCGNYWA	YGVSEHGS	MSIYVRKTSS	180
PHLPSQERKG	DPGRMNDKIR		HVRFFIVSWT	SDSKGFFYSR	YPPEDDEGKG	NAPAMNCMVY	240
YHRIGEDQES	DVLVHEDPEH		PFWISSVQLT	PSGRYILFAA	SRDASHTQLV	KIADLHENDI	300
GTNMLKWNLH	DPWEARFTIV		GDEGSKIYFM	TNLKAKNYKV	ATFDANHPDE	GLTTLIAEDP	360
NAFLVSASIH	AQDKLLLVYL		RNASHEIHIR	DLTTGKPLGR	IFEDLLGQFM	VSGRRQDNDI	420
FVLFSFLSP	GTVYRYTFGE		EKGHSSLFRA	ISIPGLNLDD	FMTESEVFP	KGTSVHMF	480
TRPKDVLDDG	TSPVLQYGYG		GFSLAMLPTF	SLSTLLFCKI	YRAIYAI	PNI	540
HREGMLDKKQ	NVFDDFNAAT		EWLIANKYAS	KDRIAIRGGS	NGGVLTTACA	NQAPGLYRCV	600
ITIEGIIDML	RFPKFTFGAS		WRSEYGDPE	PEDFDFIYKY	SPYHNIPPPG	DTIMPAMLFF	660
TAAYDDRVS	LHTFKHVAAL		QHNFPKGNP	CLMRIDLNSG	HPAGKSTQEM	LEETADEYRL	720
KVQ							723

SEQ ID NO: 49 moltype = AA length = 417
 FEATURE Location/Qualifiers
 source 1..417
 mol_type = protein
 organism = unidentified
 note = Lentinula edodes

SEQUENCE: 49

METPTLNKSG	SLTIVGTGIE		SIGQMTLQTL	SYIEAADKVF	YCVIDPATEA	FILTKNKDCV	60
DLYQYYDNKG	SRMDTYTQMS		EVMLREVRKG	LDVVGVFYGH	PGVFNPNPSLR	ALAIKSEGF	120
KARMLPGVSA	EDCLYADLCI		DPSNPGCLTY	EASDFLIRER	PTNIYSHFIL	FQVGCVGIAD	180
FNFTGFENSK	FGILVDRLEK		EYGAHPVVH	YIAAMLPHE	PVTDQWTIGQ	LREPEFYKRV	240
GGVSTFYIPP	KERKEINVDI		IRELKFLPEG	KVPDTRTQIY	PPNQWEPEVP	TVPAYGSNEH	300
AAIAQLDTH	PPEQYQPLAT		SKAMTDVMTK	LALDPKALAE	YKADHRAFAQ	SVPDLTANER	360
TALEIGDSWA	PRCAMKEMPI		SLLDNAQSQM	EEASEQGFPW	IIVVGUVGVV	GSVVSSA	417

SEQ ID NO: 50 moltype = AA length = 417
 FEATURE Location/Qualifiers
 source 1..417
 mol_type = protein
 organism = unidentified
 note = Omphalotacae olearis

SEQUENCE: 50

METSTQTKAG	SLTIVGTGIE		SIGQMTLQAL	SYIEAAAKVF	YCVIDPATEA	FILTKNKNVC	60
DLYQYYDNKG	SRLNTYTQMS		ELMVRVVRKG	LDVVGVFYGH	PGVFNPNPSHR	ALAIKSEGY	120
RARMLPGVSA	EDCLFADLCI		DPSNPGCLTY	EASDFLIRDR	PVSIHSHLV	FQVGCVGIAD	180
FNFTGFDNNK	FGVLVDRLEQ		EYGAHPVVH	YIAAMMPHQD	PVTDKYTVAQ	LREPEIAKRV	240
GGVSTFYIPP	KARKASNLDI		IRRLELLPAG	QVPDKKARIY	PANQWEPDVP	EVEPYRPSDQ	300
AAIAQLADHA	PPEQYQPLAT		SKAMSDVMTK	LALDPKALAD	YKADHRAFAQ	SVPDLTPQER	360
AALELGDSWA	IRCAMKNMPS		SLLDAARESG	EEASQNGFPW	VIVVGIVGVI	GSMVSTE	417

SEQ ID NO: 51 moltype = AA length = 417
 FEATURE Location/Qualifiers
 source 1..417
 mol_type = protein
 organism = unidentified
 note = Dendrothele bispora

SEQUENCE: 51

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MESSTQTKPG SLIVVGTGIE SIGQMTLQAL SYIEAASKVF YCVIDPATEA FILTKNKNV 60
DLQYYDNGK SRMDTYTQMA ELMLEKVRNG LDVGVVFYGH PGVFNPSHR ALAIARSEGY 120
QARMLPGVSA EDCLFADLCI DPSNPGCLTY EASDFLIRER PVNVHSHLIL FQVGCVGAD 180
FNFSGFDNSK FTILVDRLEQ EYGPDHTVVH YIAAMMPHQD PVTDKFTIGQ LREPEIAKRV 240
GGVSTFYIPP KARKDINTDI IRLLEFLPAG KVPDKHTQIY PPNQWEPDVP TLPYQNEQ 300
AAITRLEAHA PPEEYQPLAT SKAMTDVMTK LALDPKALAE YKADHRAFAQ SVPDLTPQER 360
AALELGDSWA IRCAMKNMPS SLLEAASQSV EEASMNPFV VIVTGIVGVI GSVVSSA 417

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SEQ ID NO: 52      moltype = AA length = 410
FEATURE           Location/Qualifiers
source            1..410
                  mol_type = protein
                  organism = unidentified
                  note = Rhizopogon vinicolor

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SEQUENCE: 52
MTTDTKRGTL AGSGIASI AHITLETLSY IKESDKLFYL VCDPVTEAFI QDNATGDFFD 60
LSVFYDKNKS RYDSYIQMCE IMLRAVRAGH SVLGIFYGHP GVFVSPSHRA IAVAREEGYK 120
ARMLPGVSAE DYMFADLEFD PSQSTCNTYE ATELLLRDRP LDPAIQNIW QVGSVGVDM 180
EPEKSKFHL RLEKDFGPDH KVVHYIGAVL VLPQSTTMD IFTISDLRKE NVAKQPGTIS 240
TLYIPPRDEG PVSSSMTQAF DFKAGAMVYS PVKWAGPKLN IVSALSPYER DVISQIDTHV 300
APEGYKILHT SAAMNKFM TD LSLKPKFLEE YKLYPEAVVE SAEGLSNLEK FGLKFGSDGA 360
VYILMKATES DIASGRQLTE DEIAKAHKS VGPFTVLVILP TVIVVLIGRE 410

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SEQ ID NO: 53      moltype = AA length = 413
FEATURE           Location/Qualifiers
source            1..413
                  mol_type = protein
                  organism = unidentified
                  note = Rhizopogon vinicolor

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SEQUENCE: 53
MSTKRGTLTI AGSGIASVGH ITLGTLSYIK ESDKIFYLVC DPVTEAFIYD NSTADCFDL 60
VFYDKTKGRY DSYIQMCEVM LKAVRAGHDV LGVFYGHGPGV FVSPSHRAIA VARQEGYKAK 120
MLPGISAEDY MFADLEFDPS VSGCKTCEAT EILLRDKPLD PTIQNIWQV GSVGVVDM 180
SKSKFQLLV RLEKDFGPDH KVVHYIGAVL PQSTTMDTIF TIADLRKEDV AKQFGTISTL 240
YIPPRDEGHV NLSMAKVFPG PGASVKLNDK IKWAGPKLNI VSANDPHERD VIAQVDTHVA 300
PEGHKKLRVS AAMKFKMTDL ALKPKFLEEY KLDPVAVVES AEGLSNLERF GLKFARSGPA 360
DALMKATESD IASGRQLTEE EIAQGTGPVG LQTALALLVL LGLGVAIVTR PDD 413

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SEQ ID NO: 54      moltype = AA length = 569
FEATURE           Location/Qualifiers
source            1..569
                  mol_type = protein
                  organism = unidentified
                  note = Rhizopogon vinicolor

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SEQUENCE: 54
MAKVFGVLV LGFLSQTFTYPS QVWFSPVGAN NGQVITPELS NSIQETLDVW NITGLSVAII 60
PKSGEPEYHS WGDRTEDGES VTQDTLFHMA SVSKAFCSVA LGILMDDFEH GRNVTPPLPA 120
LTFENWHTSI QDLLPGEWQL MDEWASRKAN MKDILSHVSG LPRHDFAFGP YESPKEAVSR 180
LRYLRPAFEL REQWSYNNQM FMVAGHIVET YSGKTYTSFV EDRIFTPLGM SSSTFSPA 240
AKTGKFTQGW TSSGRLLPEL FPEDMVLMMA GAGGVISSAV DMSKWVALWL NKGVDNVTV 300
IPSSVYGNAS QSYAVSISTP VDSEHSIQGY GLGWFGNSYL GHNVVYHSGS IPGLSMLVSF 360
LPDDDVGFVV FANGGDKAAP VMNISNSIID AALHLRSGPA PPIMPEKKAV TSPSEDIVNL 420
ELPLEEFSGT YTDPGYGTFT FCSPSSSSSY CQQVMTDFTA VDSVHPSAPS PLQLLAAPWR 480
MGSSHIRAVH QSGNKFLLLC TALFPEGYGR DSTPFETA EI GTPGATAEFV VEDGKVVVGF 540
LFLGLVQVTE RERTQTTVKD RAEVWFDDK 569

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SEQ ID NO: 55      moltype = AA length = 571
FEATURE           Location/Qualifiers
source            1..571
                  mol_type = protein
                  organism = unidentified
                  note = Rhizopogon vinicolor

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SEQUENCE: 55
MIMAKVFGLV LGFLSQTFTY PSQIRLSPVG VNNGQVITPE LSNSIQETLD VWNITGLSVA 60
IIPKSGEPEY HSWGDRTEG ESVTQDTLFH MASVSKAFCV SALGILMDDF EHGRNVTPLP 120
PALTEFNWHT SIQDLLPGEW QLMDEWASRK ANVKDILSHV SGLPSHHFAF GPYESPKEV 180
SRLRYLRPAF ELREQWSYNN QMFTVAGHIV ETYSGKTYTS FVEDRIFTPL GMFSSTFSPA 240
KAVKTGKFTQ GTYTDPGYGT EFPQEDMIMP MAGPGGVISS AVDMSKWVAL WLNKGVDNV 300
TIIPSSVYGN ASQSYAVSIS TPVDSEHSIL GYGLGWFRNS YLGHVYVHGS GSIPGLSTLV 360
SFLPDDDVGF VVFANGDNKA APVMNISNRI IDAALHLRSG PAPPIMPEKK AVTSPSEDIV 420
NLELPLEEFS GTYTDPGYGT FTFCSPSSSS PYCQQVIANF TTVDVSRPSA PSSQLLAAP 480
PRVGSSHIRT VHQSNGKFM LPTALFPEGY GRDSTPFETA EIGTRGAPVE FVVEDGRVVG 540
FGLFLVGVQV TERERTQTTV KDRAGVWFDK 571

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SEQ ID NO: 56      moltype = AA length = 413
FEATURE           Location/Qualifiers
source            1..413
                  mol_type = protein
                  organism = unidentified

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-continued

note = *Rhizophogon vinicolor*

SEQUENCE: 56

MSTKRGTLLTI	AGSGIASVGH	ITLGTLSYIK	ESDKIFYLVC	DPVTEAFIYD	NSTADCFDLS	60
VFYDKTKGRY	DSYIQMCEVM	LKAVRAGHDV	LGVFYGHPGV	FVSPSHRAIA	VARQEGYKAK	120
MLPGISAEDY	MFADLEFDPS	VSGCKTCEAT	EILLRDKPLD	PTIQNIWQV	GSVGVVDMEF	180
SKSKFQLLVD	RLEKDFGPDH	KVVHYIGAVL	PQSTTTMDTF	TIADLRKEDV	AKQFGTISTL	240
YIPPRDEGHV	NLSMAKVFPG	PGASVKLND	IKWAGPKLNI	VSANDPHERD	VIAQVDTHVA	300
PEGHKKLRVS	AAMKKFMTDL	ALKPKFLEEV	KLDPVAVVES	AEGLSNLERF	GLKFARSGPA	360
DALMKATESD	IASGRQLTEE	EIAQGTGPVG	LQTALALLVL	LGLGVAIVTR	PDD	413

SEQ ID NO: 57 moltype = AA length = 598

FEATURE Location/Qualifiers

source 1..598

 mol_type = protein

 organism = unidentified

 note = *Rhizophogon vinicolor*

SEQUENCE: 57

MTSDNLQPEV	ISANWLKSL	AASSTGDTAS	FVSHFLPDGW	FRDMLCFTWN	FRTLSGQEKI	60
HGFISEVVDG	QSRLSYSHLH	DFKLDHDSVN	APSPFKLPGP	PDIEGVQGA	TFSITKPAAY	120
GRGFFRLTQD	VHGNWKALTL	FTNMQDLVGH	EESSADEYDP	HEKANPTVVI	VIKVGGSQSG	180
LICARLGLKL	GIRALVIDKN	ARVGDIDWRQ	YAEALPSFAV	LSRQETQVPE	PYAAYSQISK	240
LLPYPSNFPK	YLPKGLANF	LESYAINQEL	CIWLSSTVSP	SPVYDSFSAR	WTVEVEHENR	300
KVILHPKHLV	LATGHGRPRI	PTWNGMDDFQ	GTLYHSDPFR	DAEKFRGKCV	VVIGAGNASG	360
DICEDFVAGQ	AAEVTIVQRS	ATCVVSSATA	DAFVFKLPFS	DKTPIEELDF	RHNSMPLAFV	420
LQLMKSGGTQ	HMKAHDKHH	EGLRKAGFNL	TWEPSPGSGE	VGLLGFVFER	AGSGTMIDTG	480
FGKLIVGTV	KVKQGGNISH	FDKEGITFKD	GSKLPADVIV	AATGNELTMD	AIRAVLGDIT	540
ABQLPPKVVW	LDAEAGELNQM	YRPSGHFGLW	FAVGLGMTR	FCSKHLGLQI	LAQEVGIA	598

SEQ ID NO: 58 moltype = AA length = 515

FEATURE Location/Qualifiers

source 1..515

 mol_type = protein

 organism = unidentified

 note = *Galerina marginata*

SEQUENCE: 58

MGKMAYHTVL	DDIALYLLGS	AALVIFYRSF	FYPYFLSGRR	LAPGPTKGEL	SKELKQFNNE	60
INVHLRHMV	KEYGPIFRLV	GAPMIPGPGL	VVCTPTAQQR	ILVNSINSYG	QPRLAFFRWV	120
TGGLFTLPER	EHRGMRKILD	PVFSFRNLIS	TTGVYYNTVQ	SLITIFRSKI	DGENGAKDGD	180
VILVYEWLLR	LAINDVSEAI	LGFKLDTLHD	PNNELITLTD	ELSRIPATAF	ELLVRVPGL	240
RLVTDFSVRH	STLWQRRVR	RLGVFFTFMR	CLSTIRKNAL	AIKATILQED	SANRDLNVIS	300
VLQHMQSDE	TANADIAGNI	IMLWMSGRAT	IATRISWLLW	LLAKDQQCQQ	QLRDEIAPLF	360
SRDPRPDYRS	LDKLQWLDV	IMESIRLFLF	GPNIIRVALND	DYIDGVFVPK	GTVVVIPLDL	420
FTRGDIWGED	PDQFKPARWL	DSTKRYKISP	PFLSPLTGP	RCIAKGMAIM	QTKIVIASLI	480
ANFEFKPAYE	GQHVGEVNSI	IGHGMPLHVK	PIRPS			515

SEQ ID NO: 59 moltype = AA length = 1318

FEATURE Location/Qualifiers

source 1..1318

 mol_type = protein

 organism = unidentified

 note = *Galerina marginata*

SEQUENCE: 59

MPYVDPKPYF	EHREQSSGAT	LYYCLVCRDG	RERQPHHIKT	HEASQAHRTA	LSVFDSQAES	60
SSQQTGHNPT	QPGYFDPVID	DAVRALLVSG	SGDPHQPLYP	AGHPNVYGE	NFTDSRRRTS	120
PVTGIDWDQF	EAQEDTHAVP	SAQDQLRADI	CQATLDWLND	DISDDDEREP	SEVDSVSDSA	180
ESDREPIPDD	QPRKRARTNR	DNPISDWYP	WQDKITCTLD	ILMHLPRSVF	SRKQLDLFLW	240
LLRVNNVDV	PTGKSMKMLN	KILQGMCGIE	TIAYEGKLG	NYHVNNIAQI	LAQELCNPKV	300
GPHIYFYPED	SGDNLAEARQ	AARWLHELRL	EETTPMIHLP	SGDYIYIYEP	MLSNRSFCIP	360
FRWFRTRNGK	HARAWSLETG	VVDNTLWIV	HKENEVEISE	DDLKDFTRF	SSDCEAYNVP	420
HPSRILGVSC	ADSGNLLPWN	HTNPVLGNRW	RQLAKGHRTL	CLPLWMYCDD	TSGNTSKKWN	480
EHNSFLFTLA	GLPREHTAKE	YNIHFLCTSN	LAPPLEMMDG	VVSQIEAAQQ	NGIWANDCVR	540
KEPVLIFPTI	LALLGDNPMH	SEFACHIGLR	GKFFPCRTCW	KGSDAQDDAN	IVTPGLHETP	600
ENSPAPSPAP	SPAPSPAPSP	APSPALSMAP	QSQPPTPSEP	SMQVPAPPST	AAPTAKARGKK	660
KETMSAMLRN	ITAFIKPGR	RNKSETQKTL	QNFKEQAQTI	GAKTKLKTAR	TETGIKDTVQ	720
EFFFEKLFSS	YKNKRGPOAK	QEALDQAVNQ	LPSDITSPVW	RLKGLDPHQD	TPVEILHVVL	780
LGFIKYFWRD	LVQIQINDQ	KQTLIQRLNS	FDVTGLGITQ	LGGETLVNYA	GSLTGRDFRA	840
VAQVAPFVIY	DMVPADVDDA	WLALSKLVPL	VWQPYIENVA	QYLTTLEHEI	HVLLRTARW	900
TTGWPNKSKF	HIIHLPSHI	RRFGPAILFA	TEAFESFNAV	IRAKSVHSNR	QAPSRDIALA	960
FAQGNRIRHL	LSGGHFLSAD	THMVDPDQ	QLGQYERLAR	GRWRSVGP	GHLVSAEPIL	1020
PSYLGIQQS	TTSSAGLCKR	TKTPPQTFLQ	TLTGLKLPNV	SRPGARELWQ	TCSEVYLLND	1080
DKCLIGHHVI	VQRQSEQASF	VSPFFIARIG	EILQKVGSA	HAHDKPDGIL	VQTLKSSEVA	1140
DKFQMPRLVP	QNEWSFVPLA	DILCTVNAQH	DCDRNGCTAS	GFRYVYQERI	QTNDQRPVVE	1200
HVNQPEDFTL	NTAQMRDALH	LQKFIRSR	LDEQTIHES	VARTINQRKA	QDNSSSGTGG	1260
AGVSGRGRGR	GRGRGGGVEG	PSTSRGRGGG	IEGRGASSSS	GNGRGRGRGA	RSQSVFPF	1318

SEQ ID NO: 60 moltype = AA length = 1262

FEATURE Location/Qualifiers

source 1..1262

 mol_type = protein

-continued

organism = unidentified
note = *Galerina marginata*

SEQUENCE: 60

MPRKKPAPEC	PETDEASKMI	RCLICKENDT	VQQGTWIKHG	SASQHIETNA	HKLAVARREQ	60
LLQVQQEEER	RLQEIYGGNT	IPLSGNAQLY	PTYPRANMYG	NQDAVDTMD	NQNSPPQAYM	120
LCDADIPDLG	IKPIERPDP	QERERLRQQV	EQLLLQAEHE	DEFGSPDDPD	DLTSTNIAQA	180
FADLDLEEML	DEEEVDFYFN	QVSPHEDYYP	YPNKTTMLLD	ILDNLPLRLM	SSNQLRLILW	240
LLKQTGVSNV	PSFSGFRNMQ	THLRNMC GTT	PKQHVSSLGN	IFYSNNIGES	VMRDFANPEV	300
AKHLHLYPEE	TEGPISVWQ	AERWKEFAPS	ELTPMFSQGH	RQFFIDEVAQ	LQDGGYVIPR	360
NWVMRKGLKT	SDCHIVTVNP	VRFKSLHGSL	VLVLKQCFQS	GWTLTSETQI	FHADDQFQNY	420
FDVVSRIIRG	ISWSEGTEVP	AMPNNLRELA	GDDDLVVMV	PLWCDDVSGN	KSKQYNKHIN	480
VYMANSNIPG	RLLQQEYFVR	FVSTSPNATS	PEQFSALKDQ	INETQKKPIQ	CYNAHTNKKT	540
RAILRVGGLP	ADNPQQSEES	CHMGGNANCK	CRKCHVGGPH	EKKESNEGYH	EHYLTGIKRS	600
AEETRLELEK	QIKLAMYGVE	KPINETQTNT	GTKDKVAQHW	IDILLAKSRE	LKSANPSRSV	660
EEIAQELQTW	FDEQPGDKIN	PLLSIAGLDP	TQDTPVEILH	TILLGIVKYA	WHHLHSNWTE	720
AEQNLFITVRL	QSTDIDGLSV	PPIRVAYMMQ	YRNLGIGKHF	KTLMQTLFFH	VHGTVSDAQF	780
KLVKAIGELG	SVLWVHEIGD	MEKYLSDLEI	LIGNVLDAFA	EIDPSTAMYA	RFIYEPMPVP	840
SKIIIVKLKHL	MLPHLIEDIK	RFGPAIRNST	EVFECFNAIF	RLCSILSNHQ	AASRDIALKF	900
ASMDRLKML	SGGWLSSEVE	EGKFEWIRAG	ENVRNILQSE	PTIQRHLGWA	PSAKFQSGRK	960
RTPPTSWENT	KASQFMDSEE	TAAIGFPNPR	LLSWRKGVTT	TAQSGDRCST	GSWVVARNHK	1020
VCIYLASHYC	SIKANDQGES	CIGRIHEIIG	PDEKSSASTG	IITLECFQLG	KEHHPDFGLP	1080
TLQRPOADLP	KYILKAWDQP	LFIKSAHHC	HTASCQATAL	QPQLQERQLT	SRMNKLIHNA	1140
DSDFIINLY	GLHNAILLRE	FLPRELTAPQ	PLHQDRKAPH	YEVAAKLRVQ	QAEKRAKTNA	1200
RRKATRAANK	AKQVERQKQN	PDHEQESEQE	MDERPNSENG	SDIELGGDDD	IEVETRRKRR	1260
RN						1262

SEQ ID NO: 61 moltype = AA length = 1206
FEATURE Location/Qualifiers
source 1..1206
 mol_type = protein
 organism = unidentified
 note = *Hypsizygus marmoreus*

SEQUENCE: 61

MGRRAEELPA	YVELSEDGTL	VRCNLCMLHN	RLDYSKEWIK	RKGWRSHKGS	GIHDRSEAKQ	60
RVLDDAAMD	QEPASAEVEV	VTFNDILIIN	APKTPGTGNM	SEEQAMWDHF	DAGSFTLEAG	120
EDPNHSSQRL	YQDLARKADA	YGAWDGTEAL	PEYRDLDDVS	QFLDEDEEED	LLSEILRGLG	180
LEEEHEDSSD	RNPAAEELNSP	WYPYGSKLMF	LLDTIDNLPR	LRISGAMMRV	FLWLLREVGV	240
RQVPSFDKLR	KIQRKLREGS	GVPTVHWMSP	KGNAYSFNDP	AVIVANDWAS	PITRPHLRRY	300
PVIPKDGUIT	EVYHAEKWHR	EINRHFLTPM	YDDGFRHYFI	DELAQLKDG	YAVPVWRLED	360
VDGRIVADAW	RVELEDGNRA	TIIDTATVRI	HSQELALNFE	EIESNLMP	WSDTTTEAGH	420
PSRMPNPDR	LAEGDPIYTS	FIDIFGDDVS	GNSRSKSNKH	WNMYISHRNL	PRKLLHQQYH	480
THFVSTSTFA	SIPEQFVGVK	EATSTHSESK	VKVRDADTGK	QIRLKIYCN	GPNDNPSQSE	540
TSGHIGNGN	YPCRKCHTGG	TQSKETDEG	FYKMFATGEA	RSSKETLAEV	KSQVEAACTG	600
VAKTVADAQS	DTGVKDAYTQ	YWIDAIEKA	RAMQKENPGM	PTTTIQATLI	KWYVDHEEAI	660
YNSFLTLDGF	DASRDTPVEI	LHTILLGIVK	YLWHSHTSW	NAQKKIYST	RLQGTNTQGL	720
SIHHRANYI	MQYANSILGR	QKTLAQOVN	PHVYDLVDPL	RFLFTKATGE	LCALLWFTEI	780
RDLEEYLSDV	DIAAANVLDI	AAVIDPSKIV	SKIYHLLSH	LREDIIRFGP	LVGVATEVFE	840
CNAVFRYCS	ILSNHLAPSR	DIAYKLAAQE	TMKHLFSGGW	WHVKDSVDLQ	GNPKWVQPGP	900
SVRTFMASNP	VLHTLCGWNR	NNDSTPGTVK	SEPRKRGPDK	QTLPLVRLA	WLETQGSRAL	960
NNTSPNNETQ	WQRCKYVIAE	TQDQCNVGSW	VFARSPLEN	IPIPGRIVEI	LQDTSASPSA	1020
FVVIDVQVVS	ATREDEVGMP	VLLRRFNECC	LHVIPASSVI	PDFNAQHDCR	YAKCEATGEQ	1080
PLIQERVPSSG	VTFENFVPHKA	IDRYLINIHA	LHNAHLIRAT	LPRDLTAPIP	YAPNREAHHS	1140
AIAAELRSAQ	DTKRAKTAAK	TAANAANKKA	EAALKDTTSG	PAAKRRRVDD	EGSGEEDNRD	1200
VDMSV						1206

SEQ ID NO: 62 moltype = AA length = 1213
FEATURE Location/Qualifiers
source 1..1213
 mol_type = protein
 organism = unidentified
 note = *Galerina marginata*

SEQUENCE: 62

MAKGRKLNNP	LPDFIEISND	GLQVRCTLCL	AARQHNGSGW	IKRGSVSNHL	KSDNHTNSLE	60
AHEMKKSAEK	AEGRSVQEEI	AMEEGMDFVI	LSSKIQPEIT	APARAPRRSN	EEQEMWDRTY	120
LGGEVFDAGV	DHTLVEAEER	KRLEREATDF	DLWHGADFLP	EEDPNDGELL	LDELEQDDIL	180
SELLRNAHLN	APDAADVLT	EPRAAADPRI	CDAWSPYESK	MMFLDLTLDN	LPRLRISNSL	240
MNVFLWLIRE	GGARDVPSLY	HLRQVQTTLR	KSTGVPTTQH	KSPKGNVYSM	NDPRTLVA	300
WANPVICDHI	RRYPVIRP	VISEVYHAQK	WRKDVDPHTL	SPMYDAGNCH	YYIDEVARLK	360
NGFTIIPVRW	LEDEDRNVCA	DAYVVQFDDQ	FIASVVDGET	IIVQASDLQN	NFLDLKDMGL	420
LPTWNGQTIE	SGHPARMPNP	DRALAEGDPL	YTSWIDVFGD	DVSGNRSKNW	NKHWNIIYISH	480
RNLPRKLLQ	EFHTHFVSTS	PVASVTEQFH	GIKQVIELTH	KSPVKVRHGT	SGAQIRFKIN	540
VNCGPGDNPA	QSEVCGHIGV	NGNKLCKKCH	TGGTHEVKES	DEGFNSLFEP	GDARSAQEIV	600
ADVESQVQLA	CLGIAQHVQN	QQTNGIKDA	YQYWIWIDYI	NRARTLRKEQ	PRRTTADIQS	660
ELLVWVQEHK	DEIYNFLIKL	GGDAADVTP	VEILHTILG	IVKYLWHGSH	TSWTAIQKQT	720
YSVRLQSTDT	SGLSIHAIIRA	NYIMQYANSL	IGRQFKTIAQ	VNVFHVYDLV	DTTQFLLTKA	780
VGELTALLWI	PEIANMEEYL	LDVEAAAANV	LDLALIDPS	KMTNKLKLHL	LVHLKADILR	840
FGPLVGVATE	TFECFNAIFR	FCSIYSNHLA	PSRDIAFQLA	SQEVLYKRLT	GGWWPASDGE	900
WKRPGSPSVN	FIIHDHPTLQA	LLGWTKEEKL	VNGSPRLLEPL	KRDASQKIES	RKHLPLWLQTQ	960
GAKAVNSSSD	NDSKWTACRF	AVANSGBKCS	VGSWFATSP	FNSNQSVTGR	IVEVLAESEG	1020

-continued

KRAVVVDIF	EVCSTRHKIF	GMPMLARRHE	EPVYAVIAST	NIEFLYNVQH	DCPLAKCTAS	1080
GKQPLIQERV	ESGLFKTYIE	HKPIERFVIN	THAFHNAHRL	RAVLQRLVV	PIPLYPPEIR	1140
KTKHAEFAHN	LQATQKVLE	ARAAQKAKEI	ITPADKTDST	IPKKRTRSEM	ETETDDTAIA	1200
TQADVFFNAQ	GCP					1213

SEQ ID NO: 63 moltype = AA length = 517
 FEATURE Location/Qualifiers
 source 1..517
 mol_type = protein
 organism = unidentified
 note = *Galerina marginata*

SEQUENCE: 63

MVQIKRLLLG	FLSSPSQTPL	ESNHGVPVPSK	SIADVAGASA	GLAMLRITVE	LEAFSRNNWE	60
VVLYEERESV	GGIWLDPDND	VFPPEIPKTP	LYPLLRTNTP	VPSMTYPGFP	FPPSTPLYPR	120
HDHVEAYHLR	YARRHNLDF	IKFDTMVEKA	FWNGTPEEGY	WNLTLSKEG	RMRYKTFDHL	180
VVATGNHHP	HIPVWKGQED	WLASPNHSHR	KIIHSVYYRG	PEAFSNQTVL	IVNGGSGRD	240
AATQILGYAS	QTFMSIRRSY	GPVDDGVIVK	PDISHFTEAG	VVFVDGTILD	PDVILLGTGY	300
EMQKPLLSSEG	GELSFDPYAK	DNSSVRGTLV	TNGHYIFPLH	RHIFSLSPRY	PPNALAFIGL	360
LSFIASCPSP	IAQSLFAAHA	ILDPSILPPR	HLLLEELASY	EDKARRQGLD	PYLKGPIMLN	420
NTSNDYQDEL	VEYLKQKNAI	PDDGKKFVEE	WRREILAYHY	LQRGWSRIEK	LGMGPAWTEG	480
VKTEAQWFDL	MTRVNEWQKN	WETENGIAFR	VLDLDTG			517

What is claimed is:

1. A method of screening for a molecule that disrupts an interaction between a first test protein and a second test protein in a host cell, while not disrupting an interaction between a third test protein and a second test protein in the host cell, the method comprising:

expressing in the host cell a first fusion protein comprising the first test protein and a first DNA-binding moiety;

expressing in the host cell a second fusion protein comprising the second test protein and a gene activating moiety;

expressing in the host cell a third fusion protein comprising a third test protein and a second DNA-binding moiety, wherein the second DNA-binding moiety is different from the first DNA-binding moiety; and

delivering a molecule to the host cell to test for disruption of the interaction of the first test protein and the second test protein without disrupting the interaction between the third test protein and the second protein;

wherein a plurality of death agent genes are disposed within the host cell, each operably linked to a promoter DNA sequence specific for the first DNA-binding moiety,

wherein a positive selection reporter is disposed within the host cell and operably linked to a promoter DNA sequence specific for the second DNA binding moiety,

wherein, when the molecule does not disrupt the interaction between the first test protein and the second test protein and does not disrupt the interaction between the second test protein and the third test protein, an interaction between the first test protein and the second test protein causes the gene activating moiety to activate expression of the death agent genes, while an interaction between the second test protein and the third test protein causes the gene activating moiety to activate expression of the positive selection reporter;

wherein, when the molecule disrupts the interaction between the first test protein and the second test protein but not the third test protein and the second test protein, (1) an interaction between the second test protein and the third test protein causes the gene activating moiety to activate expression of the positive selection reporter; and (2) disruption of the interaction between the first

test protein and the second test moiety does not activate expression of the death agents, and wherein the host cell is an animal cell.

2. The method of claim 1, wherein the host cell comprises an integrated DNA encoding the first fusion protein, an integrated DNA encoding the second fusion protein, an integrated DNA encoding the third fusion protein, a plasmid DNA encoding death agents produced by the death agent genes, and a plasmid DNA encoding the positive selection reporter.

3. The method of claim 1, wherein the first test protein is a Ras variant, the second test protein is a Raf kinase, and the third test protein is Ras.

4. The method of claim 1, wherein the first DNA-binding moiety, the second DNA-binding moiety, or both, is a LexA, cI, Gli-1, YY1, Glucocorticoid receptor, TetR, or Ume6 DNA binding moiety.

5. The method of claim 1, wherein the gene activating moiety is a VP16, GAL4, NF-κB, B42, BP64, VP64, or p65 gene activating moiety.

6. The method of claim 1, wherein death agent(s) encoded by the death agent genes are selected from the group consisting of: a ribosomally encoded xenobiotic agent, a ribosomally encoded poison, a ribosomally encoded endogenous or exogenous gene that results in severe growth defects upon mild overexpression, a ribosomally encoded recombinase that excises an essential gene for viability, a limiting factor involved in the synthesis of a toxic secondary metabolite, and any combination thereof.

7. The method of claim 1, wherein death agent(s) encoded by the death agent genes are selected from the group consisting of: Cholera toxin, SpvB toxin, CARDS toxin, SpyA Toxin, HopU1, Chelt toxin, Certhrax toxin, EFV toxin, ExoT, CdtB, Diphtheria toxin, ExoU/VipB, HopPtoE, HopPtoF, HopPtoG, VopF, YopJ, AvrPtoB, SdbA, SidG, VpDA, Lpg0969, Lpg1978, YopE, SptP, SopE2, SopB/SigD, SipA, YpkA, YopM, Amatoxin, Phallacidin, Killer toxin KP1, Killer toxin KP6, Killer Toxin K1, Killer Toxin K28 (KHR), Killer Toxin K28 (KHS), Anthrax lethal factor endopeptidase, Shiga Toxin, Saporin Toxin, Ricin Toxin, and any combination thereof.

8. The method of claim 1, wherein the molecule is a small molecule.

9. The method of claim 8, wherein the small molecule is a peptidomimetic.

10. The method of claim 1, wherein the molecule is a peptide or a protein.

11. The method of claim 10, wherein the peptide or the protein is a naturally occurring protein product, is a synthesized protein product, or is produced by expression of a recombinant gene. 5

12. The method of claim 10, wherein the peptide or protein is expressed from a library comprising test DNA molecules comprising DNA sequences that encode polypeptides, and wherein the peptide or the protein is expressed from the test DNA molecules. 10

13. The method of claim 12, wherein the polypeptides are 60 or fewer amino acids in length.

14. The method of claim 12, wherein the polypeptides are processed into cyclic or bicyclic peptides in the host cell. 15

15. The method of claim 1, wherein the first test protein is YAP or TAZ, the second test protein is TEAD, and the third test protein is VGLL4.

16. The method of claim 1, wherein the animal cell is a mammalian cell. 20

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